

Super Multi System

Service Manual
Air Conditioner - Multi Split Type System

HFC R407C

TOSHIBA
AIR CONDITIONING

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Introduction

Precautions

-  Please read these instructions carefully before starting the installation.
-  This equipment should only be installed by suitably trained operatives.
-  In all cases ensure safe working practice: Observe precautions for persons in the vicinity of the works.
-  Ensure that all local, national and international regulations are satisfied.
-  Check that the electrical specifications of the unit meet the requirements of the site.
-  Carefully unpack the equipment, check for damage or shortages. Please report any damage immediately.

Model MAR-F105HTM8-PE complies with the following EU Directives:

73/23/EEC (Low Voltage Directive), 89/336/EEC (Electromagnetic Compatibility) and 97/23/EC (Pressure Equipment Directive). Accordingly, they are designated for use in commercial and industrial environments.

97/23/EC Pressure Equipment Directive information

Conformity assessment procedure: Module D1

Pressure equipment:

Compressor, category II, Module A1

Liquid receiver, category II

Accumulator, category I

High pressure switch, category II, Module A1

Notified body for inspection and quality assurance systems: BSI, Maylands Avenue, Hemel Hempstead, WP2 4SQ, UK.

Avoid installation in the following locations:

-  Where there is danger of flammable gas leakages.
-  Where there are high concentrations of oil.
-  Where the atmosphere contains an excess of salt (as in coastal areas). The air conditioner is prone to failure when used under this condition unless special maintenance is provided.
-  Where the airflow from the outdoor unit may cause annoyance.
-  Where the operating noise of the outdoor unit may cause annoyance.
-  Where the foundation is not strong enough to fully withstand the weight of the outdoor unit.
-  Where the water drainage may cause a nuisance or a hazard when frozen.
-  Where strong winds may blow against the air outlet of the outdoor unit.

Precautions for R407C outdoor units

-  R407C outdoor units use synthetic oils which are extremely hygroscopic. Therefore ensure that the refrigerant system is NEVER exposed to air or any form of moisture.
-  Mineral oils are unsuitable for use in these units and may lead to premature system failure.
-  Use only equipment which is suitable for use with R407C. Never use equipment which has been used with R22.
-  R407C should only be charged from the service cylinder in the liquid phase. It is advisable to use a gauge manifold set equipped with a liquid sight glass fitted in the centre (entry) port.

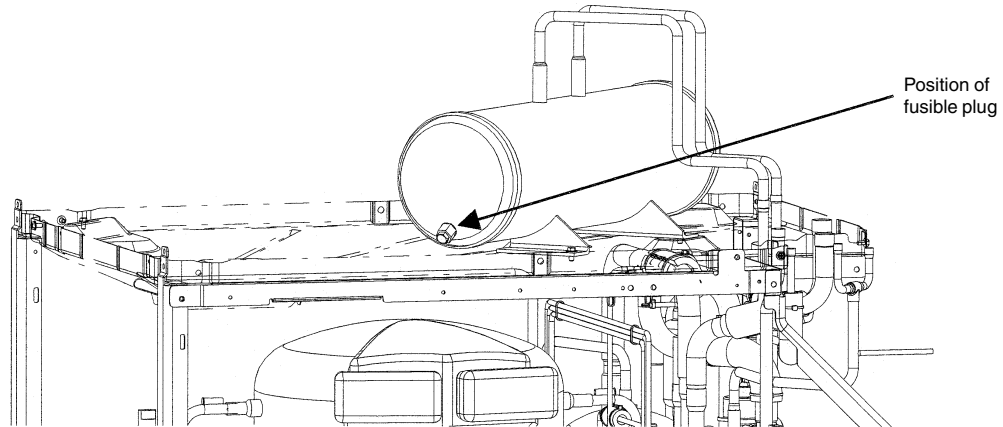
Introduction

Precautions

Precautions for R-407C outdoor units

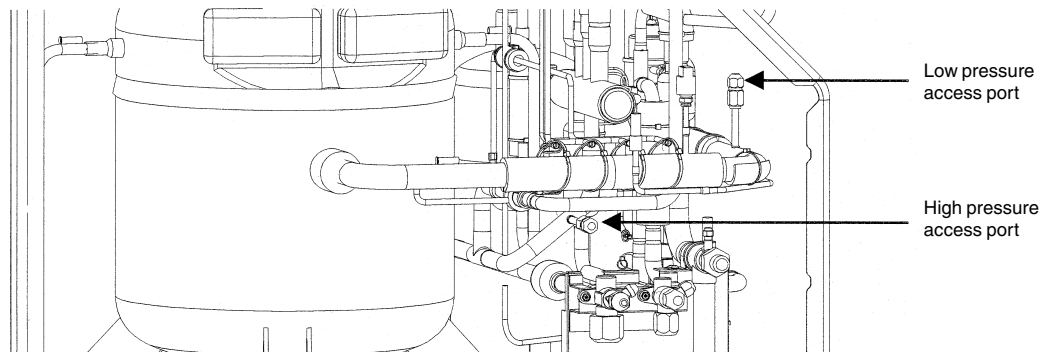
Liquid receiver fusible plug

In the event of the system being subjected to abnormal conditions it is protected by a fusible plug, positioned on the liquid receiver, within the outdoor unit. It is rated to fail at 70°C.



System pressure measurement

To measure the system's high and low pressures, connect a gauge manifold to the corresponding access port as indicated below.



Explanation of Toshiba serial number

A serial label is attached to all Toshiba air conditioning units. Located on the label is an 8-digit number, which represents the month, year and batch number of the manufactured unit. A breakdown of the 8-digit number is defined below.

2	4	4	8	0	0	0	1
↓	↓	↓		}			
Year of manufacture	Month of manufacture	Site of manufacture		Model batch serial number			
2001 = 1 2002 = 2 2003 = 3	41 = Jan. 42 = Feb. 43 = Mar. 44 = Apr. 45 = May 46 = Jun. 47 = Jul. 48 = Aug. 49 = Sep. 50 = Oct. 51 = Nov. 52 = Dec.	8 = Plymouth					

Introduction

Components

Components - 3 pipe system

1. Outdoor unit

Model name	Inverter unit
MAR-F105HTM8-PE	10 HP

2. Multi Controllers

Model name	No. of indoor units connectable
RBM-Y1034F-PE	3
RBM-Y1044F-PE	4

3. Interface control kit

Model name	Requirement
RBC-16DIF1-PE	3 or 4 Multi Controllers used on a system

Combination of Multi Controllers, indoor units and interface kits

No. of indoor units	No. of 3-way Multi Controllers	No. of 4-way Multi Controllers	No. of interface kits
1-8	1-2 Multi Controllers		0
9	3	0	1
10	2	1	1
11	1	2	1
12	0	3	1
13	3	1	2
14	2	2	2
15	1	3	2
16	0	4	2

Introduction

Operating conditions

- Operating conditions of the unit are as follows:

Outdoor temperature	-5 ~ 43°C	Cooling
	-15 ~ 21°C	Heating
Room temperature	18 ~ 32°C	Cooling
	15 ~ 29°C	Heating
Room humidity	<80%	Cooling

Note 1: Cooling capacity is rated at the following temperature conditions:

Indoor air inlet temperature	27°C DB, 19°C WB.
Outdoor air inlet temperature	35°C DB.

Note 2: Heating capacity is rated at the following temperature conditions:

Indoor air inlet temperature	20°C DB.
Outdoor air inlet temperature	7°C DB, 6°C WB.

Note 3: For details about the outdoor unit, indoor units or remote controller installation refer to the relevant literature, i.e. Installation Instructions supplied with the units.

Note 4: Operatives handling refrigerants must be suitably qualified in accordance with local and national codes of practice and statutory requirements.

Note 5: Legislation may regulate the removal of waste refrigerant from the systems. We advise awareness of any regulations and duty of care. Waste refrigerant must NEVER be discharged to atmosphere.

Note 6: Electrical work should be in accordance with all relevant codes of practice and should be carried out by suitably qualified personnel.

Note 7: Metric/Imperial pipe conversion.

Diameter (mm)	6.4	9.5	12.7	15.9	19.0	22.0	28.6
Nominal diameter (inch)	1/4	3/8	1/2	5/8	3/4	7/8	1-1/8

Note 8: Within this manual:

ODU	= Outdoor Unit	IDU	= Indoor Unit
R/C	= Remote Controller	D.O.L.	= Direct On-Line compressor
INV	= Inverter ODU	WB	= Wet Bulb
DB	= Dry Bulb	IOL	= Inner Overload Relay
Mg-Sw	= Magnetic Contactor	IGBT	= Inverter Gate Bi-Polar Transistor
OCR	= Over Current Relay	IPDU	= Intelligent Power Drive Unit
M/C	= Multi Controller		

Note 9: MPaG $\Rightarrow$ kgf/cm²G conversion multiplier

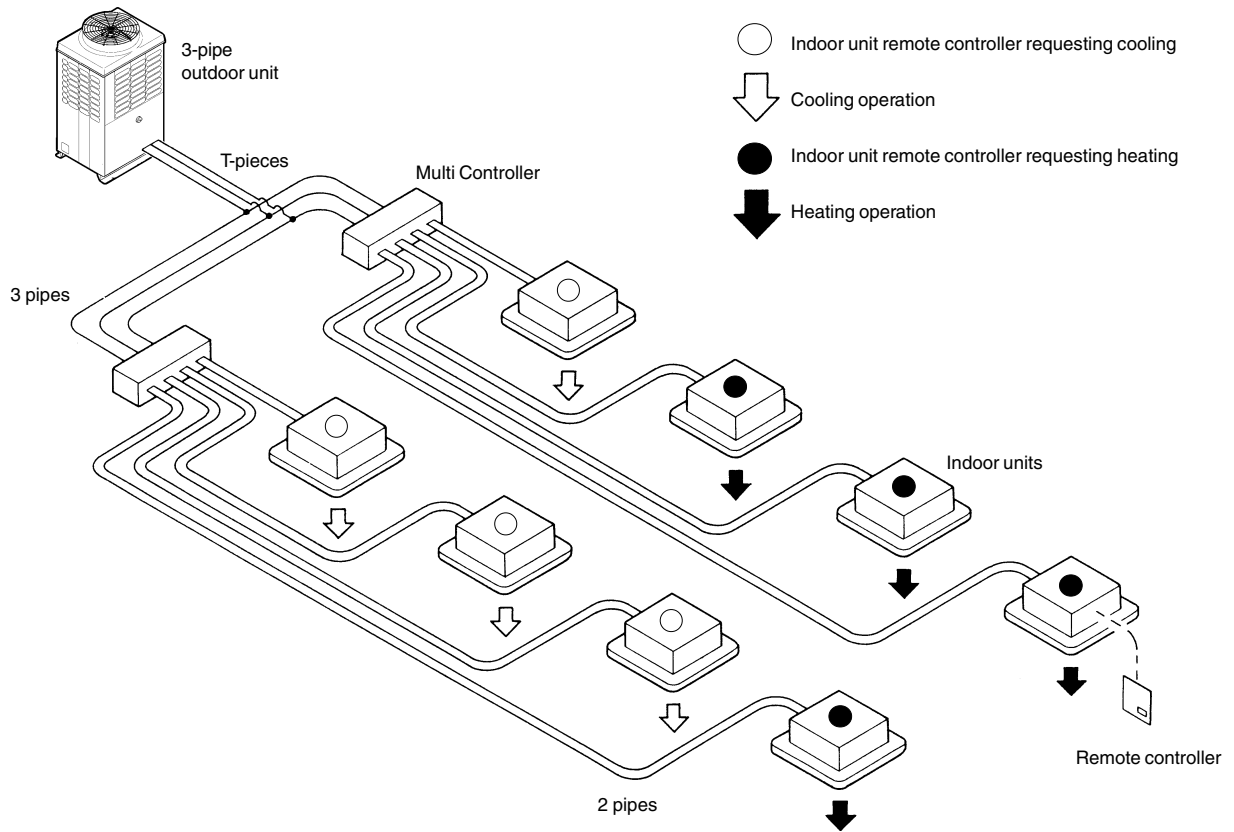
1.0 MPaG = 10.2 kgf/cm²G

Introduction

Super multi system basic components

3-pipe heat pump with simultaneous heating and cooling

This system allows separate operation of each indoor unit in either heating or cooling simultaneously.



Introduction

Outline of control system

The refrigerant and electrical systems of the Super Multi air conditioner are controlled by the Multi Controller and the outdoor unit microprocessors.

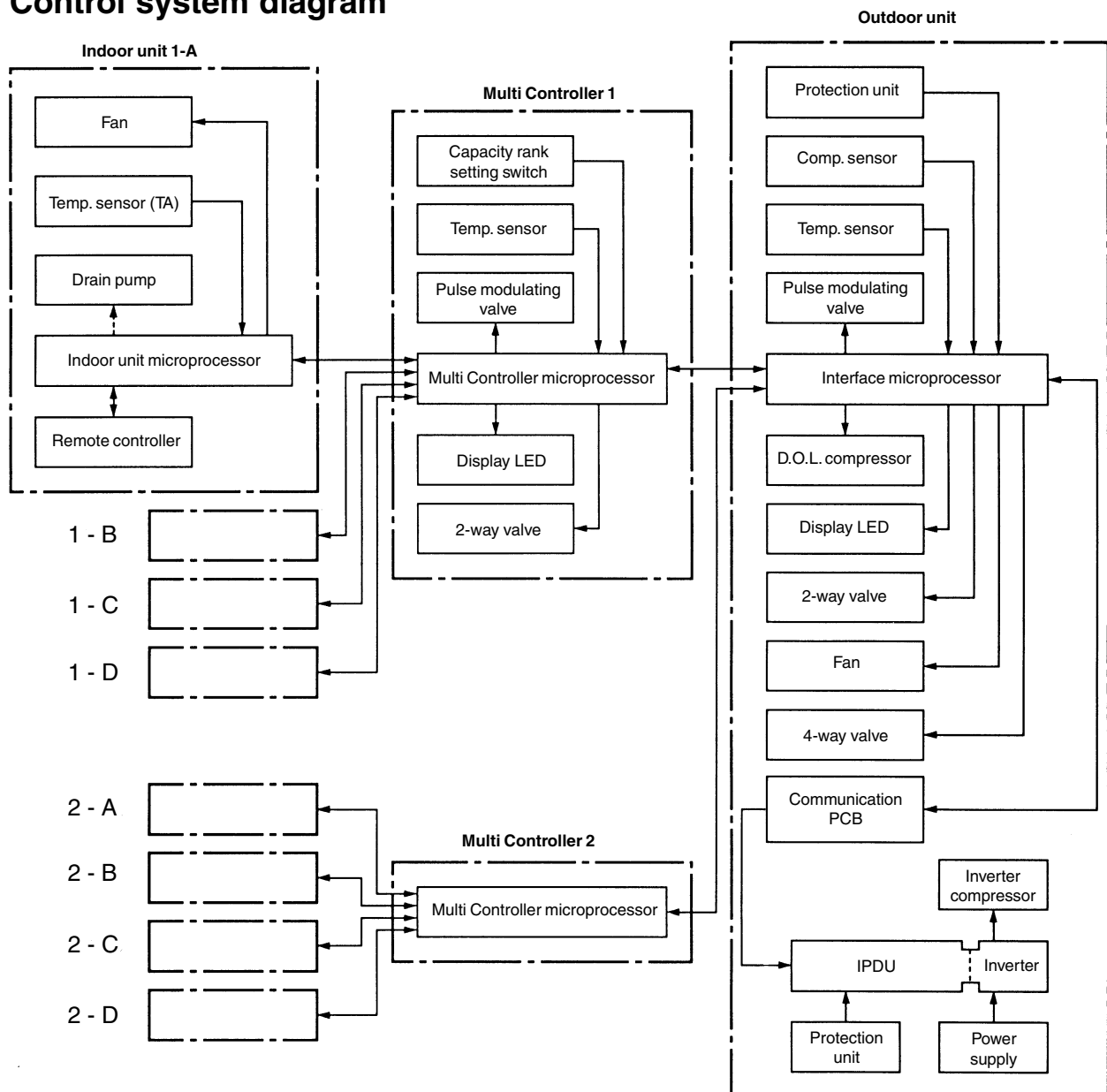
All RAV heat pump, R-407C, 4/5 series indoor units are compatible with the Super Multi system, i.e. 1~5 HP.

For system operation, initially the microprocessor in each indoor unit calculates the difference between the current room temperature (TA) and the requested temperature which has been set on the remote controller. A demand signal is determined and transmitted to the Multi Controller microprocessor in the form of operation commands (i.e. ON/OFF, cooling or heating operation mode, operation demand frequency).

The Multi Controller microprocessor receives operation commands from all indoor units connected, calculates the accumulative operation command and transmits this information to the outdoor unit interface microprocessor.

The interface microprocessor calculates the capacity required for heating or cooling and determines the operation mode of the outdoor unit and the actual frequency of the compressor.

Control system diagram



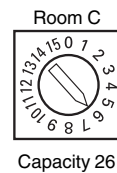
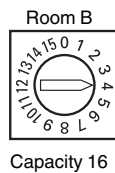
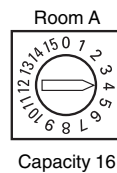
Introduction

Components

Setting of indoor unit capacity codes

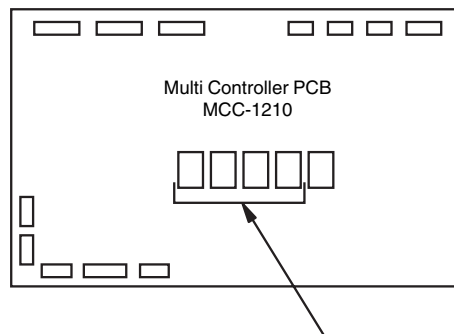
- The setting of the indoor unit capacities is important. Set the correct indoor unit code numbers according to the indoor unit capacity. The capacities are set by the rotary switches on the printed circuit board switch A (unit A), switch B (unit B), switch C (unit C) and switch D (unit D).
- During manufacture, the indoor capacity selection switches are set at '0'.
- Record the indoor capacity codes, indoor unit model names and locations in the installation manual, and on the wiring diagram on the electrical panel cover.

Example:



Indoor unit	Capacity	No connection	10	13	16	20	26	36	46
	Code number	0	2	3	4	5	6	8	10

(Example: Model RAV-364UH-PE, capacity = 36)



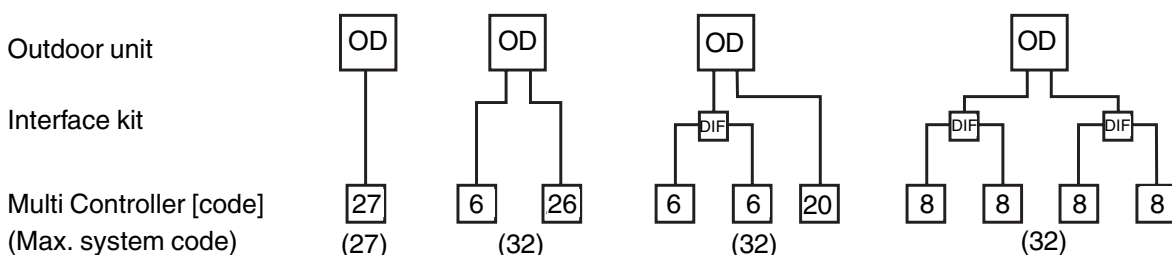
Capacity select switches

- Multiple indoor units may be connected to each outdoor unit, providing the total indoor code does not exceed the limits shown below.

Combination of Multi Controllers and indoor units

Number of Multi Controllers	Maximum No. of indoor units	Indoor unit diversity	Maximum system code	Maximum code per Multi Controller
1	4	135%	27	27
2	8	160%	32	27
3	12			27 (13*)
4	16			13

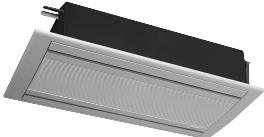
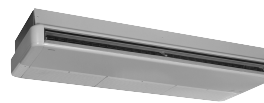
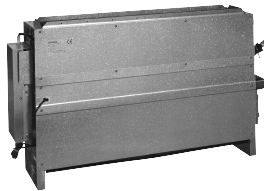
Example of systems with maximum possible code:



Introduction

Components

Indoor units

Type	Appearance	Model name	Capacity code on Multi Controller
Cassette (4-way)		RAV-164UH-PE RAV-264UH-PE RAV-364UH-PE RAV-464UH-PE	4 6 8 10
Cassette (2-way)		RAV-104TUH-1-PE RAV-134TUH-1-PE RAV-164TUH-1-PE	2 3 4
Built-In Horizontal		RAV-104SBH-PE	2
		RAV-164BH-PE RAV-264BH-PE RAV-364BH-PE RAV-464BH-PE	4 6 8 10
Ceiling Suspended		RAV-134CH/CHR-PE RAV-164CH/CHR-PE RAV-264CH/CHR-PE RAV-364CH/CHR-PE RAV-464CH/CHR-PE	3 4 6 8 10
High Wall		RAV-105KH-E RAV-135KH-E RAV-165KH-E RAV-265KH-E	2 3 4 5
High Wall		RAV-264KH-PE	6
Built-In Vertical		RAV-104NH-PE RAV-134NH-PE RAV-164NH-PE RAV-264NH-PE	2 3 4 6
Floor Mounted		RAV-164SH/SHR-PE RAV-264SH/SHR-PE	4 6

Parts specification

Outdoor unit

Specification of 3-pipe, heat recovery outdoor unit

Model name			MAR-F105HTM8-PE
Cooling capacity	kW		25.0 (28.0)
Heating capacity	kW		28.0 (31.5)
Power supply			380-415 V, 3 phase, 50 Hz
Cooling			
Operating current	A		17.4 (19.1)
Power consumption	kW		11.8 (12.8)
EER			2.11 (2.19)
Power factor	%		98 (97)
Heating			
Operating current	A		17.7 (15.5)
Power consumption	kW		11.8 (10.5)
COP			2.37 (3.0)
Power factor	%		96 (98)
Starting current	A		60
Starting method			Direct
Sound level	Sound pressure	dB	58
	Sound power	dB(A)	70
Dimensions (H x W x D)	mm		1700 x 990 x 790
Net weight	kg		285
Colour			Silky Shade (Munsell 1-Y8.5/0.5)/RAL 1013 (DE ~9)
Compressor	Type		Hermetically sealed (twin scroll)
	Oil (volume)	ml	Polyolester NISSEKI RB74AF 74VG (7000)
	Crankcase heater		Internal type
	Motor output	kW	7.5
Fan assembly	Fan		Propeller fan
	Motor output	kW	0.4
	Air flow volume	m³/h	10,000
Refrigeration (charged weight)	kg		R-407C (19)
Piping connection	Liquid	mm	15.9 (flare connection)
	Discharge gas	mm	19.0 (flare connection)
	Suction gas	mm	28.6 (brazing connection)
Max. equivalent piping length	m		120
Max. actual piping length	m		100
Max. piping head	m		50: when the outdoor unit is installed above
	m		20: when the outdoor unit is installed below
Suction accumulator case heater	W		29 (240 V AC)

Notes:

- Specifications are subject to change without notice.
- The specification shown in () denotes operation with an indoor unit load diversity of 135%.
- Operating conditions:

Outdoor temperature °C	Cooling	-5 to 43
	Heating	-15 to 21
Room temperature °C	Cooling	18 to 32
	Heating	15 to 29

- Sound level: 1 m horizontal from centre.
1.5 m vertical from base.

Parts specification

Outdoor unit

Specification of outdoor unit refrigeration cycle parts

Parts description				MAR-F105HTM8-PE	
Compressor model	Model name			MG1300CW-20	
	Motor type			3-phase induction motor	
	Power supply			380-415 V, 3 phase, 50 Hz	
	Output	kW		7.5	
	Pole			2/2 (inverter side/non-inverter side)	
	Coil resistance	Ω		1.49/2.51 (inverter side/non-inverter side)	
	Compressor oil name			Polyolester NISSEKI RB68AF VG 74	
	Amount of oil	ml		7,000	
Fan motor	Model name			STF-200-350A	
	Motor type			1-phase, induction type	
	Power supply			220-240 V, 1 phase, 50 Hz	
	Output	kW		0.4	
	Supply current	A		3.2 ~ 3.5	
	Pole			6	
	Protective device (Relay)	Operation	°C	145 ± 5	
		Reset	°C	86 ± 15	
High pressure switch	Model name			ACB-JB128	
Inverter side	Operating pressure	Operation	MPa	3.2	
		Reset	MPa	2.55	
High pressure switch	Model name			ACB-JA64	
Fixed side	Operating pressure	Operation	MPa	3.2	
		Reset	MPa	2.55	
Low pressure switch	Model name			20PS-1	
	Operating pressure	Operation	MPa	0.025	
		Reset	MPa	0.15	
4-way valve				CHV-0712, coil AC240 V	
Liquid tank			l	(RSH 12) 12	
Accumulator			l	(BT9-1 1/8)	
Check valve	Model name			YCV5-3SPTF-1	
	Maximum pressure			MPa	3.3
Check joint	Model name			R-407C type	
	Maximum pressure			MPa	3.53
Expansion valve				EXV-V-E80HTF-9WS	
Compressor heater				(Internal within the compressor)	
Accumulator heater			W	29 (240 V AC)	
Pressure sensor				Model name: NTP-Q250TF-2	
				Input voltage: DC12 V	
				Output voltage: DC 0.5 - 4.5 V	
Discharge temperature sensor				25°C = 50 kΩ	
				50°C = 18.1 kΩ	
				100°C = 3.35 kΩ	
Suction temperature sensor/ outdoor air temperature sensor				0°C = 34.6 kΩ	
				25°C = 10 kΩ	
				50°C = 3.4 kΩ	
Pulse modulating valve (Cooling bypass PMV1)				EV18RC1, coil DC 12 V	
Pulse modulating valve (PMV2)				EV23RC2, coil DC 12 V	
2-way valve (SV1, SV2, SV4, SV5, SV16, SV17)				NEV202DXF, coil AC 240 V	
2-way valve (SV13, SV14, SV15)				RP100-03, coil AC 240 V	

Note:

Specifications are subject to change without notice.

Parts specification

Outdoor unit

Specification of outdoor unit inverter assembly parts

Model name		MAR-F105HTM8-PE
Power supply		380-415 V, 3 phase, 50 Hz
PC board assembly	PCB (IPDU)	MCC-1342
PC board assembly	PCB (noise filter)	MCC-1366
PC board assembly	PCB (interface)	MCC-1223
PC board assembly	PCB (communication)	MCC-1387
Fan motor capacitor		8 μ F/450 V AC
Electrolytic capacitor		2200 μ F/400 V
Power supply terminal plate	L1, L2, L3	600 V AC, 60 A, 3 pole
Transformer	Interface	TT01
Magnetic contactor	Inverter side	FMCa - 1S
Magnetic contactor	Fixed side	FC - 3
Reactor	CH - 25 - 2FK	
Fuse	6 A	500 V AC
Fuse	20 A	600 V AC
Fuse	3.15 A	250 V AC
Fuse	6.3 A	250 V AC
Fuse	20 A	250 V AC
Fuse	10 A	250 V AC
Thermistor (PTC)		
Relay		15 A, 240 V AC

Note:

Specifications are subject to change without notice.

Parts specification

Multi Controller unit

Specification of 3-pipe Multi Controller parts

Model name		RBM-Y1034F-PE	RBM-Y1044F-PE
Pulse modulating valve		EV23RC7, coil DC 12 V	
Temperature sensor		At 0°C = 32.8 kΩ, 25°C = 10 kΩ, 50°C = 3.6 kΩ	
Float switch		FS-085-0031	
PC board assembly		PCB MCC-1222	
Power supply transformer	Model name	FT69	
	Specification	Primary side: AC 240 V, secondary side: AC 12 V	
Relay (PC board)		G2R-117P, coil DC 12 V	
Heater		10.4 W/m	
		50 W	65 W
Thermal fuse for heater		Cut out at 119°C	
Heater fuse		T1A	
Discharge gas side 2-way valve		RP100-03, coil AC 240 V	
Suction gas side 2-way valve		REV-1506DXFQ6, coil AC 240 V	
2-way valve		NEV202DXF-AC 240 V	
2-way valve		NEV603DXF-AC 240 V	

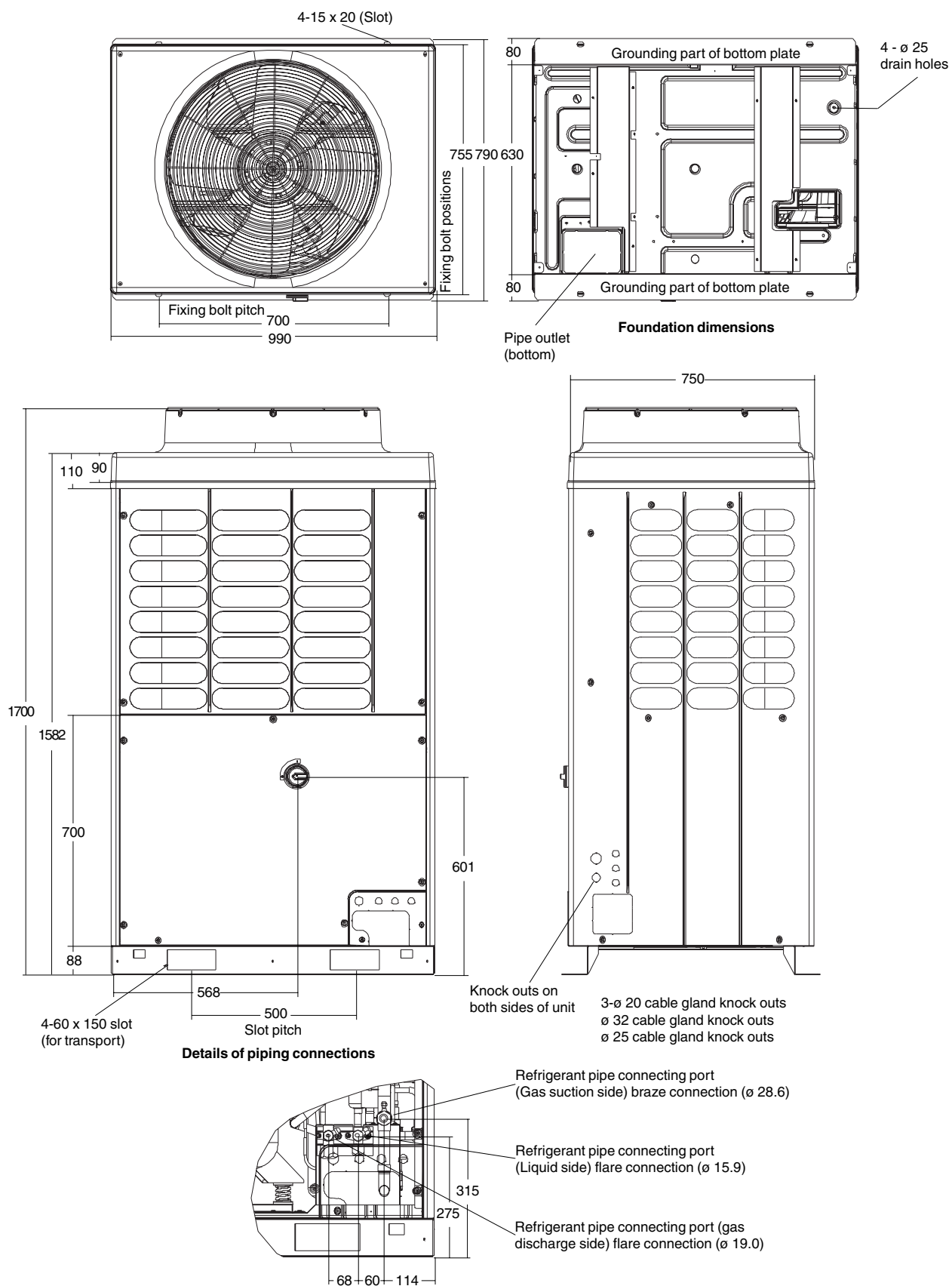
Note:

Specifications are subject to change without notice.

Construction view

Outdoor unit

MAR-F105HTM8-PE Dimensional drawings



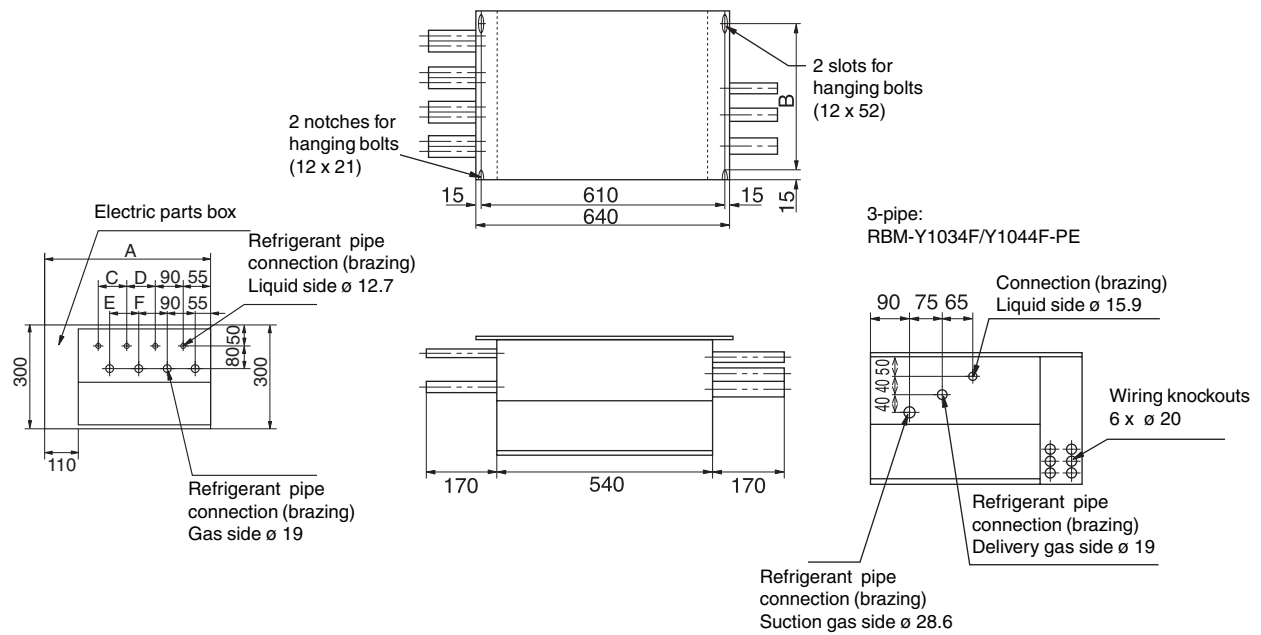
All dimensions in mm

Construction view

Multi Controller

Model	A	B	C	D	E	F
RBM-Y1034F-PE	460	300	-	90	-	90
RBM-Y1044F-PE	530	370	90	90	90	90

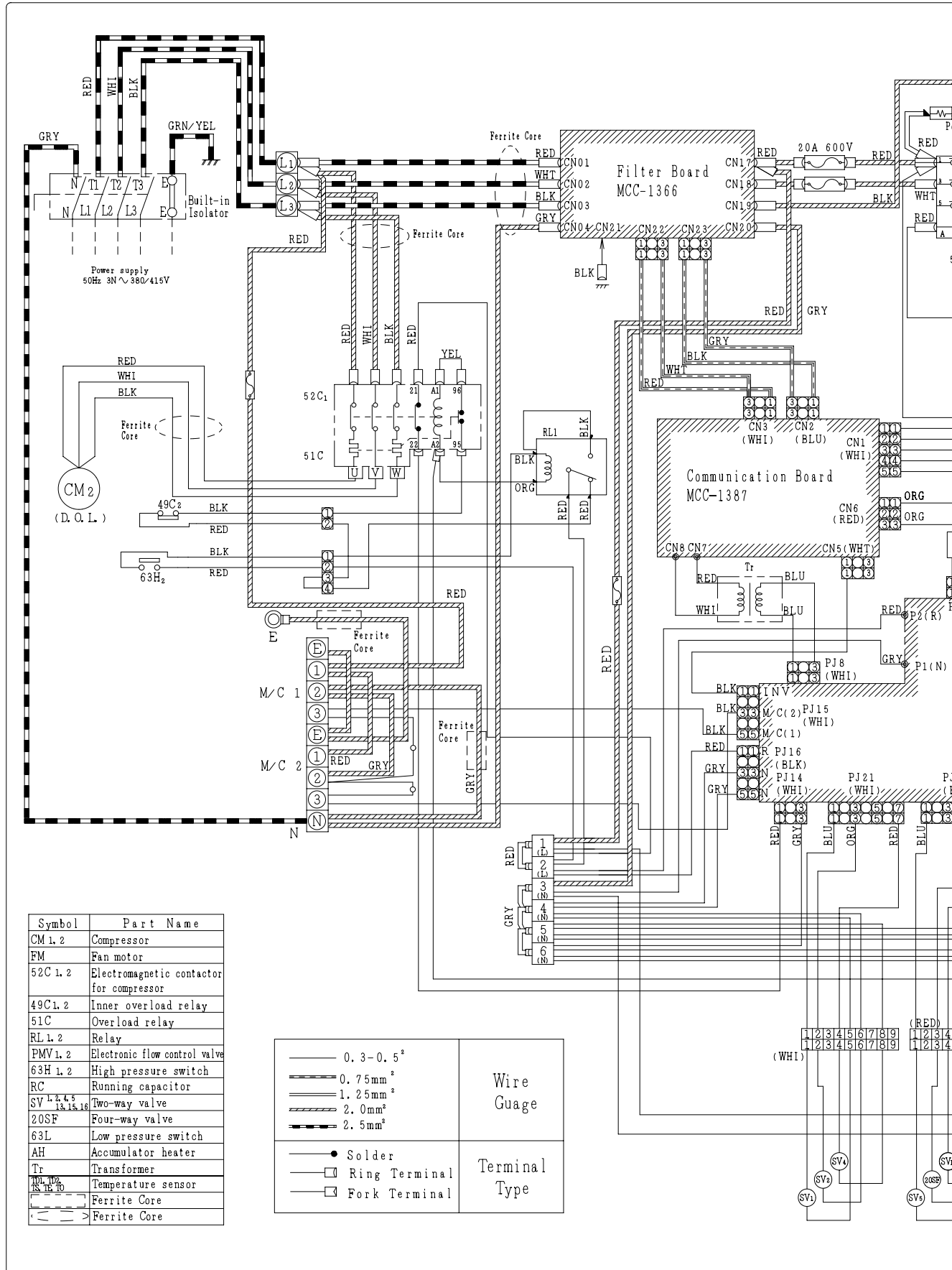
All dimensions are in mm



Wiring diagrams

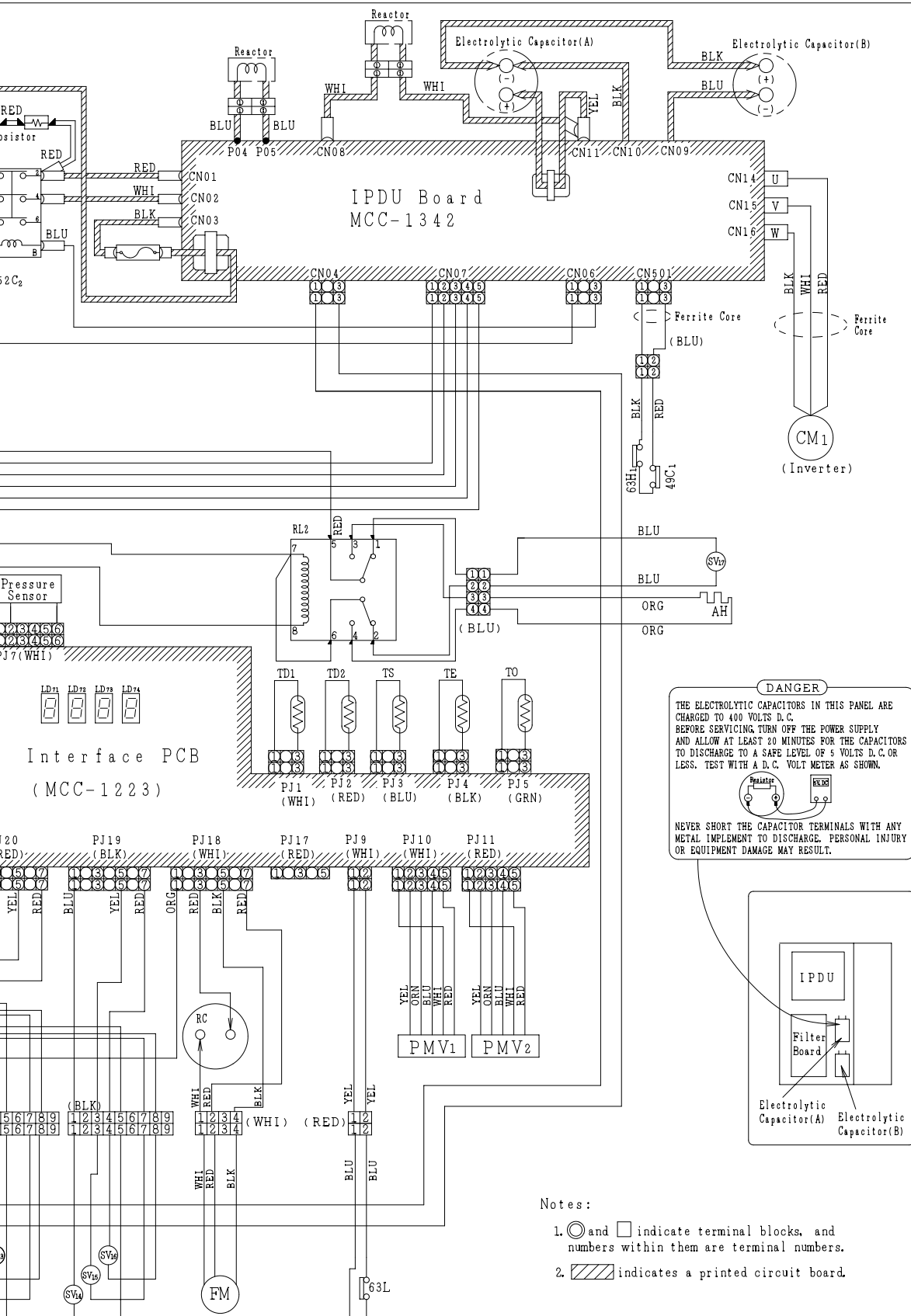
Outdoor unit

MAR-F105HTM8-PE



Wiring diagrams

Outdoor unit

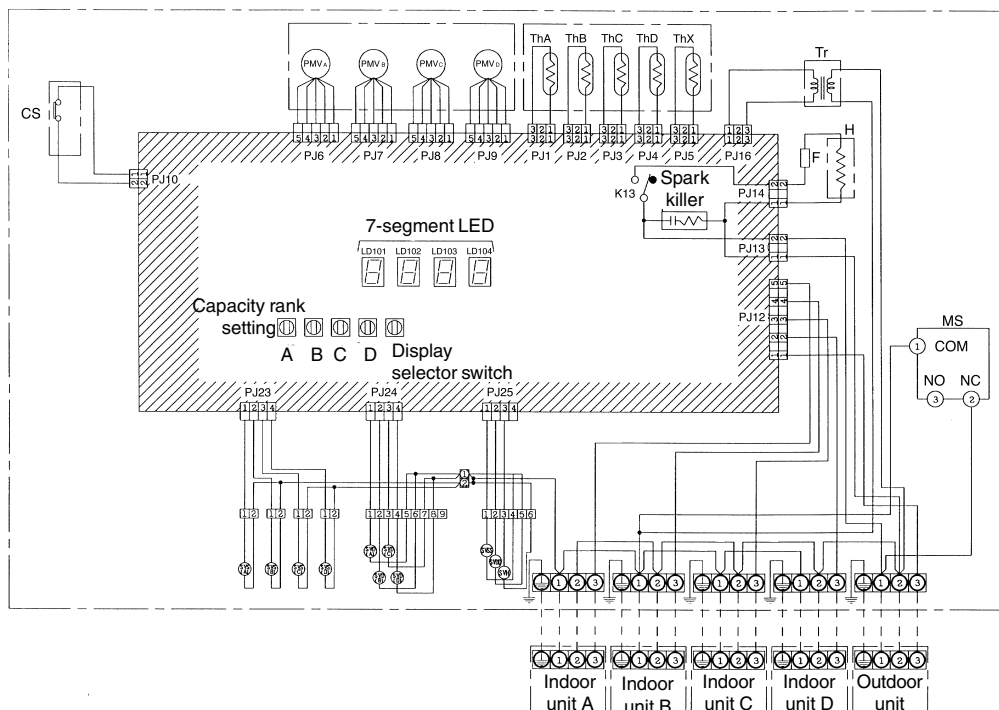


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Wiring diagram

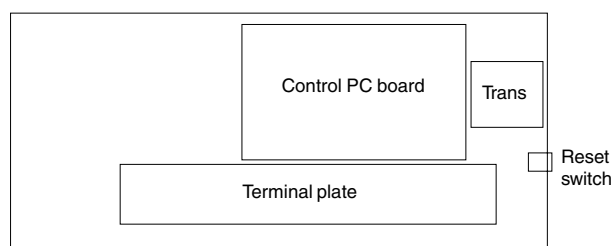
Multi Controllers

3-pipe Multi Controller (RBM-Y1034F-PE, RBM-Y1044F-PE)



Symbol	Part Name
PMVA, B, C, D	Pulse modulating valve
Th A, B, C, D, X	Temperature sensor
Tr	Power transformer
CS	Float switch
H	Heater
MS	Reset switch
F	Fuse (T1A)
SVD (A), (B), (C), (D)	Electrically operated valve for discharge gas side
SVD (A), (B), (C), (D)	Electrically operated valve for suction gas side
SVDD	Electrically operated valve for increasing pressure
SVSS	Electrically operated valve for decreasing pressure
SVH	Electrically operated valve for superheat control
LD 101, 102, 103, 104	Fault indicator LED

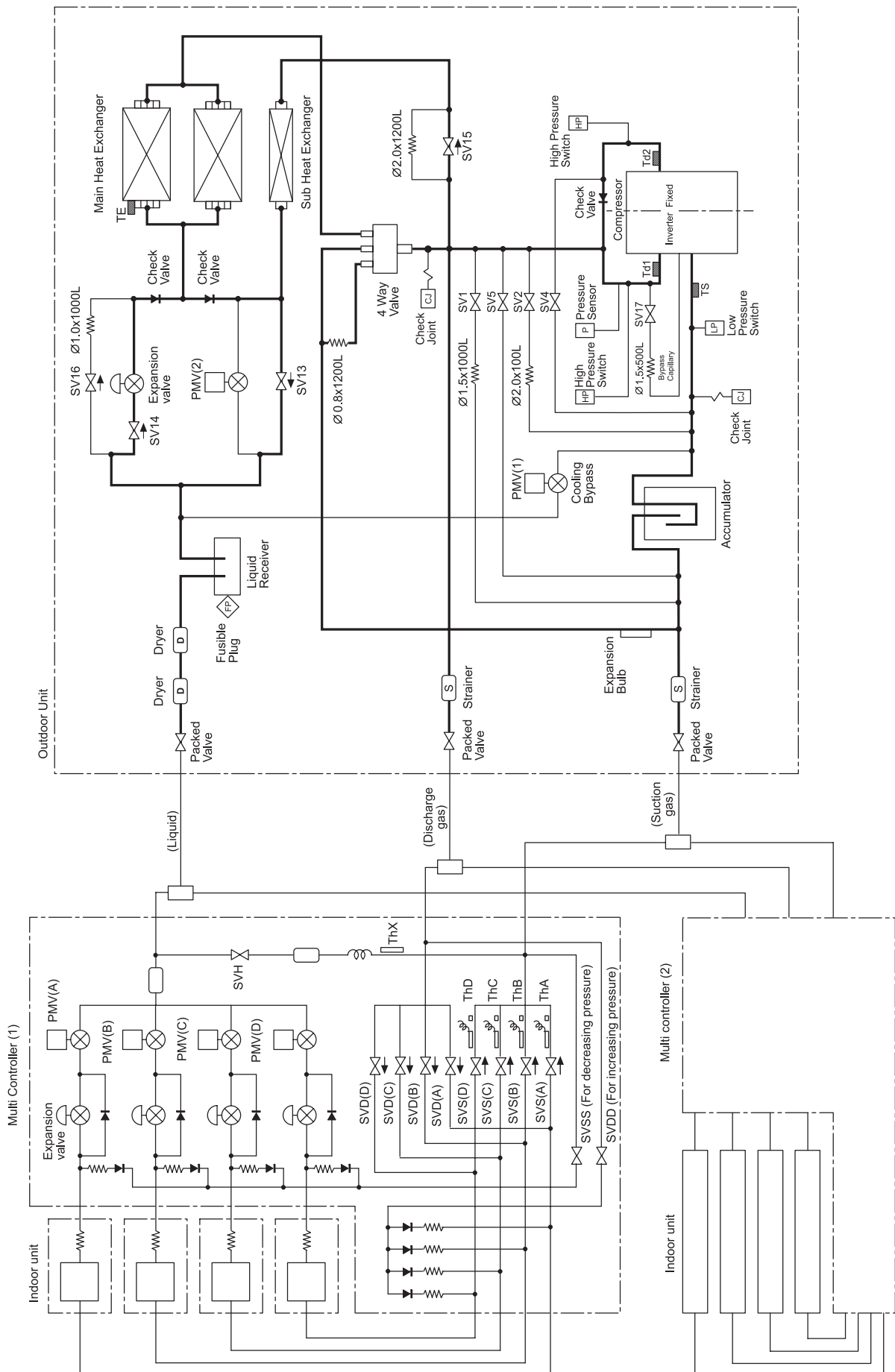
Parts layout



- The dashed lines indicate wiring on site.
- ○ and □ indicate terminal blocks, and numbers within them are terminal numbers.
- ▨ indicates a printed circuit board.
- The frame □ indicates the product body.
- RBM-Y1034F-PE does not have PMVD, SVD(D), SVS(D), ThD or the connection block for indoor unit D. The capacity rank code setting for unit D is to be set to "0".

Refrigeration piping schematic diagram

Outdoor unit

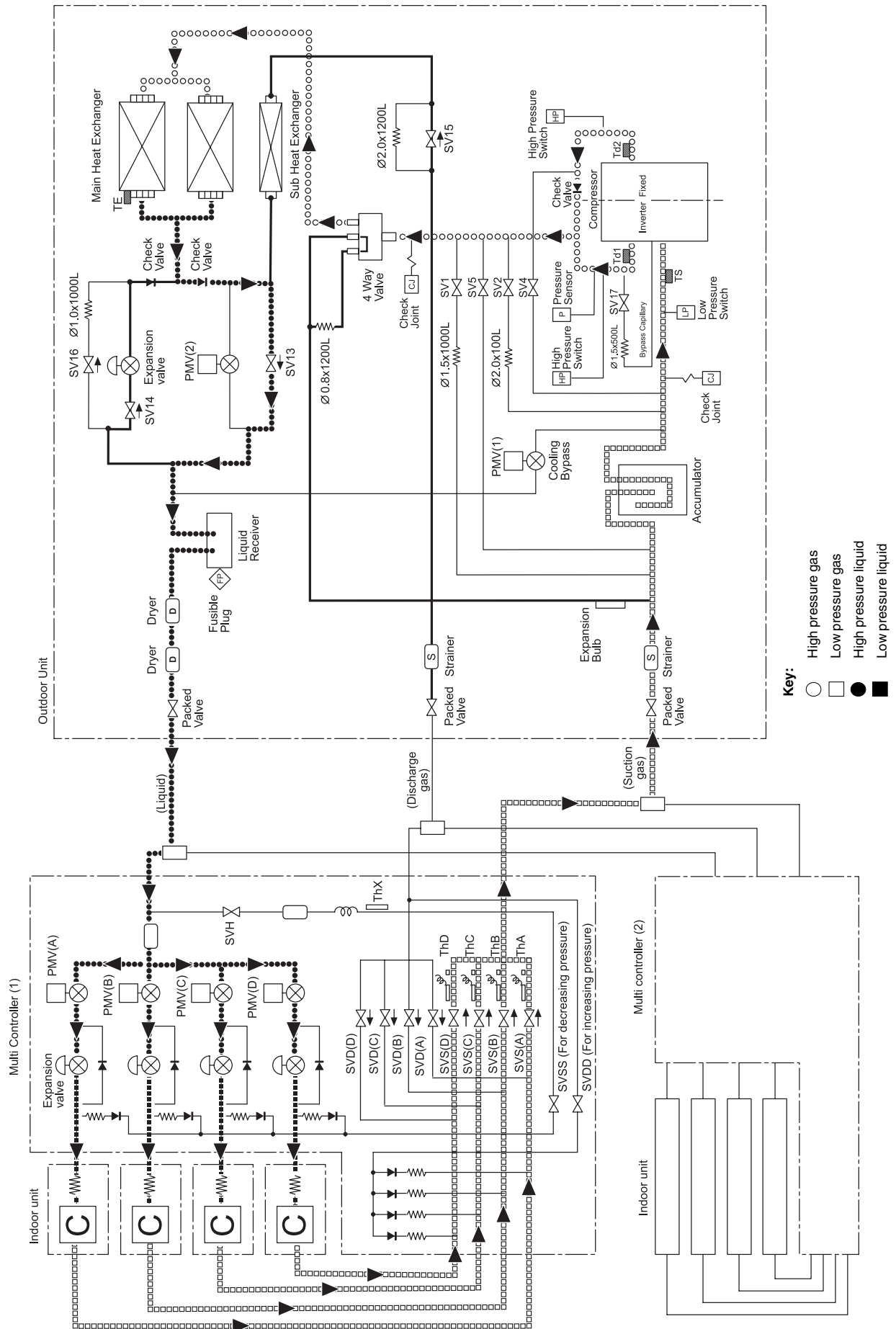


Normal operation - heat mode



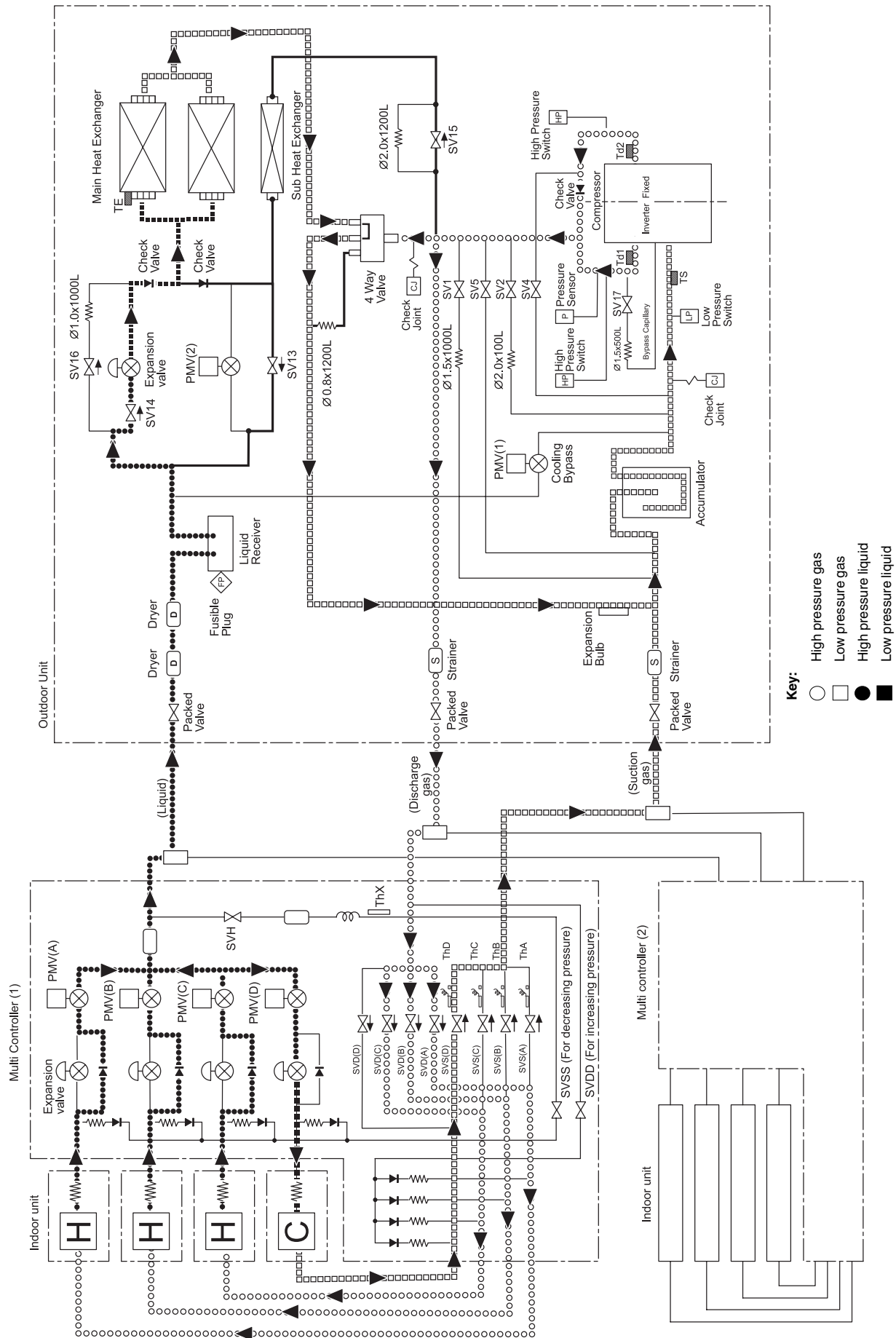
Refrigeration piping schematic diagram

Normal operation - cool mode



Refrigeration piping schematic diagram

Simultaneous operation - mainly heat

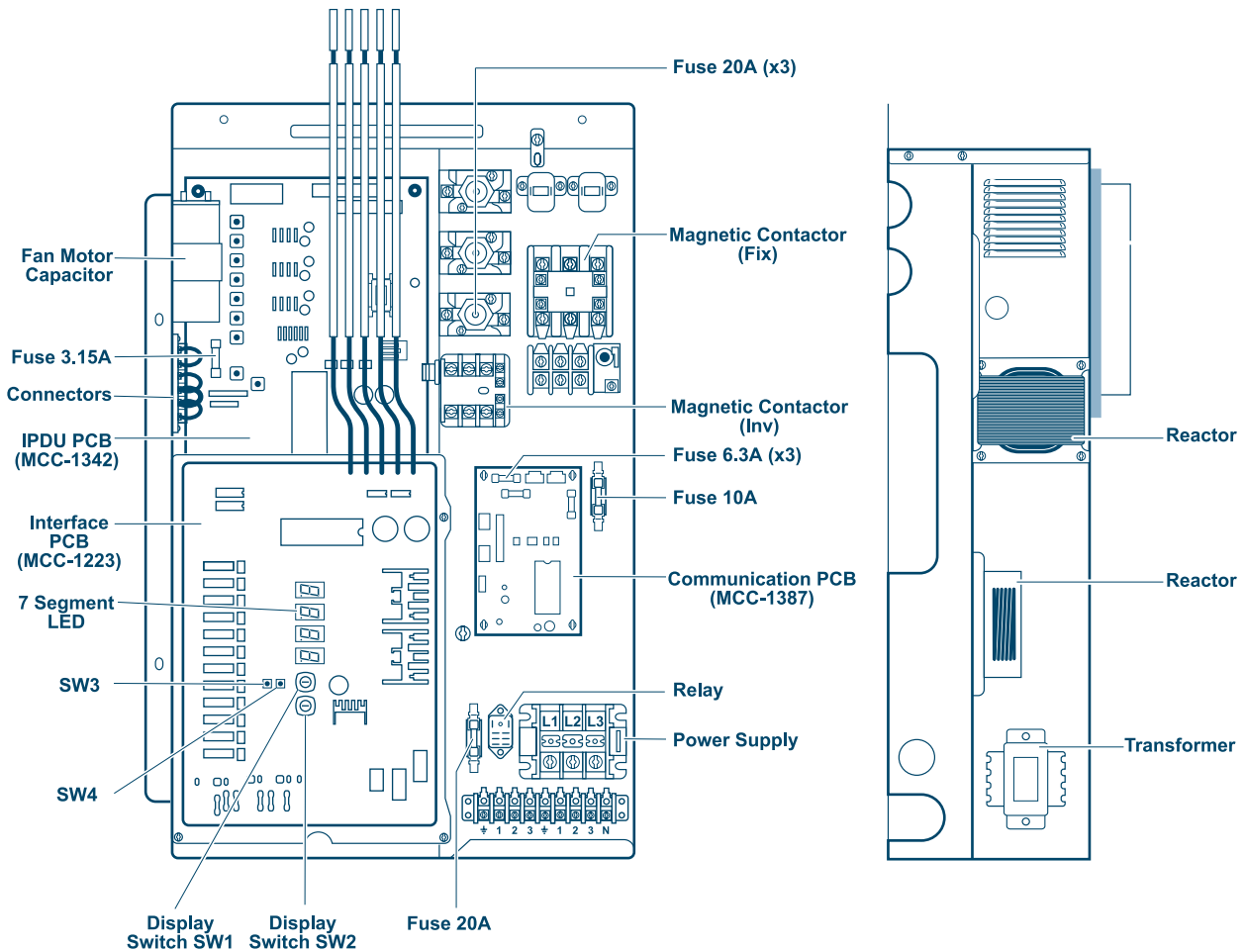


Diagnostic procedure

Outdoor unit

MAR-F104HTM8-PE, 3-pipe outdoor unit

Malfunction judgement is performed using the self-diagnostic function of the outdoor unit. The combination of the display switches (SW1, SW2) and the LEDs (LD 71~LD 74) indicates the diagnostic details.



Located on the electrical inverter cover is a transparent window, which allows the 7-segment LED to be viewed while the unit is in operation.

The transparent window has access holes to allow the adjuster pen, which is attached to the window, access to alter the setting of the display switch SW1 and SW2 while the unit is running.

A slit is positioned at one end of the pen to locate onto the rotary switch.

The adjuster pen can also be used to push switches SW3 and SW4 during the diagnostic procedure check.

A crimp connector is located on the other end of the adjuster pen. This connector can be used to 'short' TP1 or TP2 connectors when performing manual operation of the outdoor valves.



No other component other than that supplied must be used to adjust the rotary switches. Failure to do so may result in damage to the unit, malfunction or risk of electric shock.

Diagnostic procedure

Outdoor unit

- Communication serial signals and the system operating status are displayed by changing display switch SW2.

Display switch SW1 position	Display switch SW2 position	Indication		7-segment LED display			
				LD 71	LD 72	LD73	LD74
0	0	Sending/receiving status of interface control board serial signal		Between M/C (1)	Between M/C (2)	Between inverter	Non-inverter compressor
				Sending only [S]	Not sending or receiving []		Abnormal [E]
				Receiving only [J]			Normal [0]
	1	Operating instruction from M/C (1)	Operating mode	Heating [H], Simultaneous cooling/heating [HC], Cooling [C], Stop []		-----	-----
	2		Instruction frequency	Heating frequency [00-1F] (Refer to conversion table 2)		Cooling frequency [00-1F] (Refer to conversion table 2)	
	3	Outdoor unit operating condition	Operating mode	Heating [H], Cooling [OC], Simultaneous heating [Hc] Defrost [JO], Simultaneous cooling [hC]		-----	-----
	4			M/C operation Off: [0] 1 unit: [1] 2 unit: [2]	Frequency release No: [0] Yes: [1]	-----	-----
	5	Operating instruction from M/C (2)	Operating mode	Heating [H], Simultaneous cooling/heating [HC], Cooling [C], Stop [-]		-----	-----
	6		Instruction frequency	Heating frequency [00-1F] (Refer to conversion table 2)		Cooling frequency [00-1F] (Refer to conversion table 2)	
	9	Inverter operation status		Normal [C]	Operating frequency [0-F], (Refer to table 3)	-----	-----
				Abnormal [E]			
	10	Operating instruction to inverter from interface control board		Normal [C]	Operating frequency [0-F], (Refer to table 3)	-----	-----
				Abnormal [E]		-----	-----
	11	Power supply frequency		50 Hz [5]	[0]	-----	-----
				60 Hz [6]			

Diagnostic procedure

Outdoor unit

2. Display code conversion table - instruction frequency from Multi Controller.

Display code	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Frequency conversion value (Hz)	0	0	0	3.9	6.9	10.0	13.0	16.2	19.3	22.4	25.5	28.6	31.7	34.8	37.9	41.0

Display code	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Frequency conversion value (Hz)	44.1	47.2	50.3	53.4	56.5	59.6	62.7	65.8	68.9	72.0	75.1	78.2	81.3	84.4	87.5	90.0

3. Display code conversion table - operating frequency of inverter compressor

Display code	0	3	4	5	6	7	8	9	A	B	C	D	E	F
Frequency conversion value (Hz)	0	30.5	38.3	41.9	45.5	53.3	61.0	68.8	76.0	83.8	91.6	102.9	110.7	121.5

4. PMV opening information is displayed

Display switch SW1 position	Display switch SW2 position	Indication	7-segment LED display			
			LD 71	LD 72	LD 73	LD 74
1	2	PMV1 opening	[P]	[1]	Displays the degree of PMV opening (0 - 240) as a hexadecimal code: [00] : Closed, [F0] : Fully open	
	3	PMV2 opening	[P]	[2]		

5. Malfunction information is displayed, refer to fault code section for details

Display switch SW1 position	Display switch SW2 position	Indication	7-segment LED display			
			LD 71	LD 72	LD 73	LD 74
2	0	Check code display for the outdoor unit, M/C (1), M/C (2)	[E]	Outdoor unit: [r] M/C (1): [1] M/C (2): [2]	Refer to fault code section for details (page 36)	

Diagnostic procedure

Outdoor unit

6. The indoor unit capacity rank setting is displayed.

Display switch SW1 position	Display switch SW2 position	Indication		7-segment LED display			
				LD 71	LD 72	LD 73	LD 74
3	0	Outdoor unit HP		[9]	8 HP [8], 10 HP [A]	[H]	[P]
	1	Capacity rank of indoor unit connected to M/C (1)	Unit A	[A]	Displays the code number of the indoor unit registered to each M/C. Refer to the table below (7) for indoor unit model capacity rank.		
	2		Unit B	[b]			
	3		Unit C	[C]			
	4		Unit D	[d]			
	5	Capacity rank of indoor unit connected to M/C (2)	Unit A	[A]			
	6		Unit B	[b]			
	7		Unit C	[C]			
	8		Unit D	[d]			

7. Indoor unit code No. conversion table

Code No.	0	2	3	4	5	6	8	10
Capacity rank	No correction	10	13	16	20	26	36	46

(Indoor unit model example: Model RAV-364UH-PE, capacity rank = 36)

8. Sensor temperature display

Display switch SW1 position	Display switch SW2 position	Indication	7-segment LED display			
			LD 71	LD 72	LD 73	LD 74
4	0	Pressure sensor data	[P]	[d]	Refer to table 9 conversion chart	
	1	ThD1 sensor data	[d]	[1]	Refer to table 10 conversion chart	
	2	ThD2 sensor data	[d]	[2]		
	3	ThS sensor data	[S]	[1]	Refer to table 11 conversion chart	
	4	TE sensor data	[h]	[E]		
	5	TO sensor data	[h]	[0]		

Diagnostic procedure

Outdoor unit

9. Outdoor unit - pressure sensor data

Display	Pressure (kgf/cm ² G)	Display	Pressure (kgf/cm ² G)	Display	Pressure (kgf/cm ² G)	Display	Pressure (kgf/cm ² G)
17	0.0	67	11.0	96	17.5	C6	24.0
1E	1.0	6B	11.5	9A	18.0	C9	24.5
25	2.0	6E	12.0	9E	18.5	CD	25.0
2D	3.0	72	12.5	A1	19.0	D1	25.5
34	4.0	76	13.0	A5	19.5	D4	26.0
3B	5.0	79	13.5	A9	20.0	D8	26.5
43	6.0	7D	14.0	AC	20.5	DC	27.0
4A	7.0	81	14.5	BD	21.0	DF	27.5
51	8.0	84	15.0	B4	21.5	E3	28.0
58	9.0	88	15.5	B7	22.0	E7	28.5
5C	9.5	8B	16.0	BB	22.5	EA	29.0
60	10.0	8F	16.5	BE	23.0	EE	29.5
63	10.5	93	17.0	C2	23.5	F2	30.0

These are sample displays. intermediate displays to those above are possible.

10. Outdoor unit - ThD1 and ThD2 sensor data

Display	Temp. °C	Display	Temp. °C	Display	Temp. °C	Display	Temp. °C
05	-5.0	28	40.0	7C	85.0	C6	130.0
07	0.0	30	45.0	87	90.0	CB	135.0
09	5.0	38	50.0	90	95.0	D0	140.0
0C	10.0	41	55.0	9A	100.0		
0F	15.0	4A	60.0	A2	105.0		
12	20.0	54	65.0	AB	110.0		
17	25.0	5E	70.0	B2	115.0		
1C	30.0	68	75.0	B9	120.0		
22	35.0	72	80.0	C0	125.0		

These are sample displays. Intermediate displays to those above are possible.
Display [00] signifies sensor open circuit.

11. Outdoor unit - ThS, TE, TO sensor data

Display	Temp. °C	Display	Temp. °C	Display	Temp. °C	Display	Temp. °C
2A	-20.0	7C	9.0	A2	22.0	C0	35.0
36	-15.0	7F	10.0	A5	23.0	C2	36.0
42	-10.0	82	11.0	A7	24.0	C4	37.0
51	-5.0	85	12.0	AA	25.0	C6	38.0
60	0.0	88	13.0	AC	26.0	C8	39.0
63	1.0	8B	14.0	AF	27.0	C9	40.0
66	2.0	8E	15.0	B1	28.0	D1	45.0
69	3.0	91	16.0	B3	29.0	D8	50.0
6C	4.0	94	17.0	B6	30.0	DE	55.0
6F	5.0	97	18.0	B8	31.0	E3	60.0
73	6.0	9A	19.0	BA	32.0	E7	65.0
76	7.0	9C	20.0	BC	33.0	EA	70.0
79	8.0	9F	21.0	BE	34.0		

These are sample displays. Intermediate displays to those above are possible.
Display [00] signifies sensor open circuit.

Diagnostic procedure

Outdoor unit circuit test procedure

12. Outdoor unit circuit test procedure

- These systems have a feature which enables them to check that the wiring and piping connections are aligned with each other. This is carried out by allowing refrigerant to flow to one indoor unit at a time and monitoring that indoor unit's coil sensor for a corresponding drop in temperature. Each indoor unit is tested in turn and where two Multi Controllers are installed each Multi Controller is tested in turn.
- This test would normally be used at the commissioning stage.
- Procedure for initialising the circuit test.
 1. Turn the power off.
 2. Ensure the capacity codes are set correctly, capacity switches set to '0' are not tested.
 3. Put the outdoor display switches SW1 and SW2 to 9 and Multi Controller(s) display switch to 6.
 4. Turn the power back on.
 5. Set all the remote controllers to cool mode and 29°C.
 6. Press the on/off button to start all the indoor units (the outdoor LEDs show '1020').
 7. Press the outdoor unit switch SW3, and hold for 3 seconds.
 8. The system is now in self-testing (all 8 LEDs will be flashing rapidly).
 9. The system will stop at the end of the test.
- In the event of cross wiring/piping the system will indicate which units are faulty, see table below:

Outdoor display switch SW1 and SW2 set to position 9.

One and two Multi Controllers

Display	Multi Controller	Fault	Units that are indicated failed the test
1020	All	None	
1A20	1	Unit A	
1B20		Unit B	
1C20		Unit C	
1D20		Unit D	
102A	2	Unit A	
102B		Unit B	
102C		Unit C	
102D		Unit D	

Three or more Multi Controllers

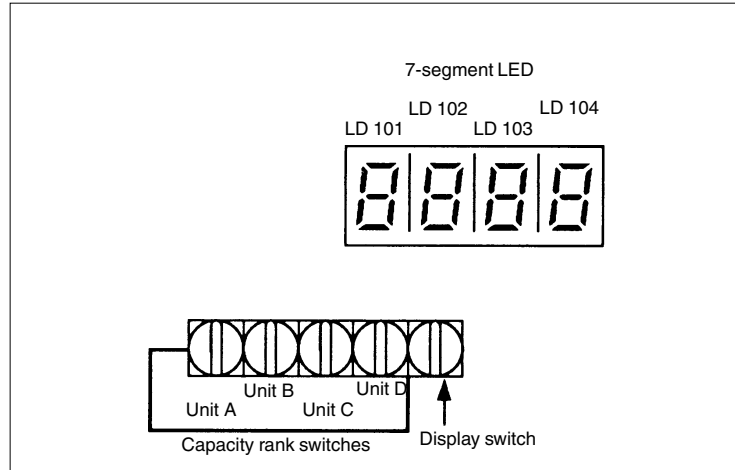
Display	Multi Controller	Fault	Units that are indicated failed the test
1020	All	None	
1A20	1	Unit A or B	
1B20		Unit C or D	
1C20	2	Unit A or B	
1D20		Unit C or D	
102A	3	Unit A or B	
102B		Unit C or D	
102C	4	Unit A or B	
102D		Unit C or D	

Diagnostic procedure

Multi Controller

The combination of the display switch and the four 7-segment LEDs (LD 101-LD 104) indicates the diagnostic details.

MCC - 1223



Multi Controller control board

Switch position	Indication	7 segment LED display															
		LD 101	LD 102	LD 103								LD 104					
0	Serial signal	-	-	-	Receiving from unit B	Sending to unit B	-	Sending to unit A	Receiving from unit A	Receiving from outdoor unit	-	Receiving from unit D	Sending to unit D	-	Sending to unit C	Receiving from unit C	Sending to outdoor unit
1	Fault codes	[E]	-	Fault code display (normal [00]). Refer to fault code section for details.													
2	Frequency instructions to the outdoor unit	Instructed demand frequency of cooling or heating [00 - 1F]. Refer to the Multi Controller display conversion table (1) for display switch position "2"															
3	Oil retrieval, superheat control	-	-	-	Oil retrieval unit B	Superheat control unit B	-	Superheat control unit A	Oil retrieval unit A	-	-	Oil retrieval unit D	Superheat control unit D	-	Superheat control unit C	Oil retrieval unit C	-
	Defrost			Displays [dF] during defrost operation													
4	Operation mode	M/C operating mode: Heating [H-], Cooling [C-]*		Outdoor unit operating mode: Heating [H-], Cooling [-C], Simultaneous Heating [Hc], Simultaneous Cooling [hC], Stop [--], Defrost [J-]													
5	Restart timer	-	-									Normal display [0], Restart timer counting [1]					
6	Circuit test	[c]	[k]	Displays unit being tested [A b C d]								Indicates faulty unit connection [A b C d]					
7	PMV A	[P]	[A]	Displays the degree of PMV opening (0-240) as a hexadecimal code: [00]: Closed, [FO]: Fully open													
8	PMV B	[P]	[b]														
9	PMV C	[P]	[c]														
10	PMV D	[P]	[d]														
11	ThA	[h]	[A]	Displays sensor temperature Refer to the Multi Controller sensor temperature conversion table (1).													
12	ThB	[h]	[b]														
13	ThC	[h]	[c]														
14	ThD	[h]	[d]														
15	ThX	[h]	[H]														

* and Simultaneous Cooling/Heating [HC], Defrost [J-]

Diagnostic procedure

Multi Controller

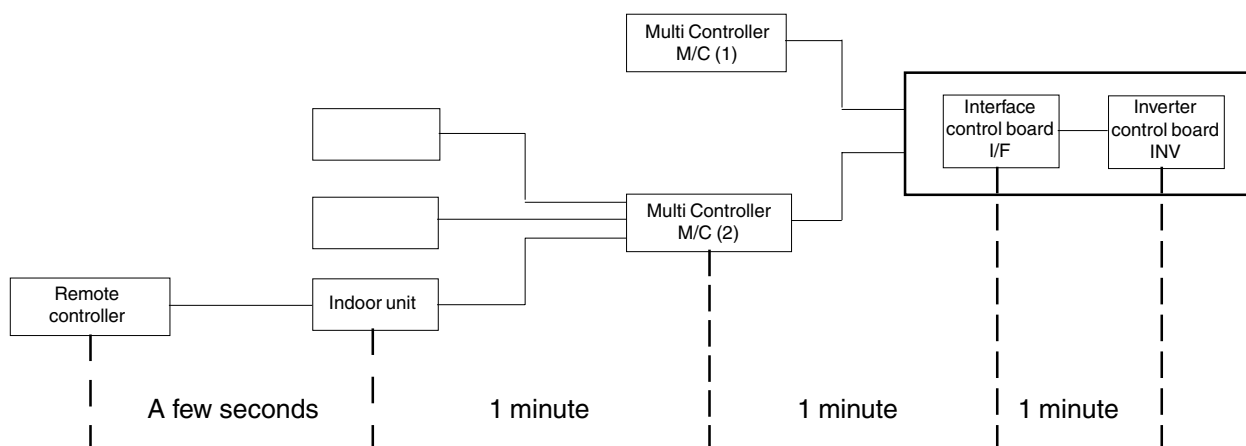
1. Display switch set to position “11”, “12”, “13” and “14”

Multi Controller - ThA, B, C, D and X sensor conversion table

Display	Temp. °C	Display	Temp. °C	Display	Temp. °C	Display	Temp. °C
18	-10.0	44	1.0	70	12.0	9C	23.0
1C	-9.0	48	2.0	74	13.0	A0	24.0
20	-8.0	4C	3.0	78	14.0	A4	25.0
24	-7.0	50	4.0	7C	15.0	A8	26.0
28	-6.0	54	5.0	80	16.0	AC	27.0
2C	-5.0	58	6.0	84	17.0	B0	28.0
30	-4.0	5C	7.0	88	18.0	B4	29.0
34	-3.0	60	8.0	8C	19.0	B8	30.0
38	-2.0	64	9.0	90	20.0		
3C	-1.0	68	10.0	94	21.0		
40	0.0	6C	11.0	98	22.0		

Display [00] signifies sensor open circuit.

Control board communication diagram



Remote controller fault code display time lapse for system malfunctions

Serial signal abnormalities:	Between the I/F and the INV	More than 3 minutes
	Between the M/C and the I/F	More than 2 minutes
	Between the indoor unit and the M/C	More than 1 minute
	Between the remote controller and indoor unit	A few seconds
INV abnormality:		More than 3 minutes
I/F abnormality:		More than 2 minutes
M/C and indoor unit abnormality:		More than 1 minute

Fault code display

Self-diagnostic function

The remote controller, Multi Controller and outdoor unit have the facility to display fault codes which are used to determine any malfunction of the system.

- The remote controller is provided with a “check” button and a check display.
- The Multi Controllers and outdoor units are provided with display switches and LED displays.

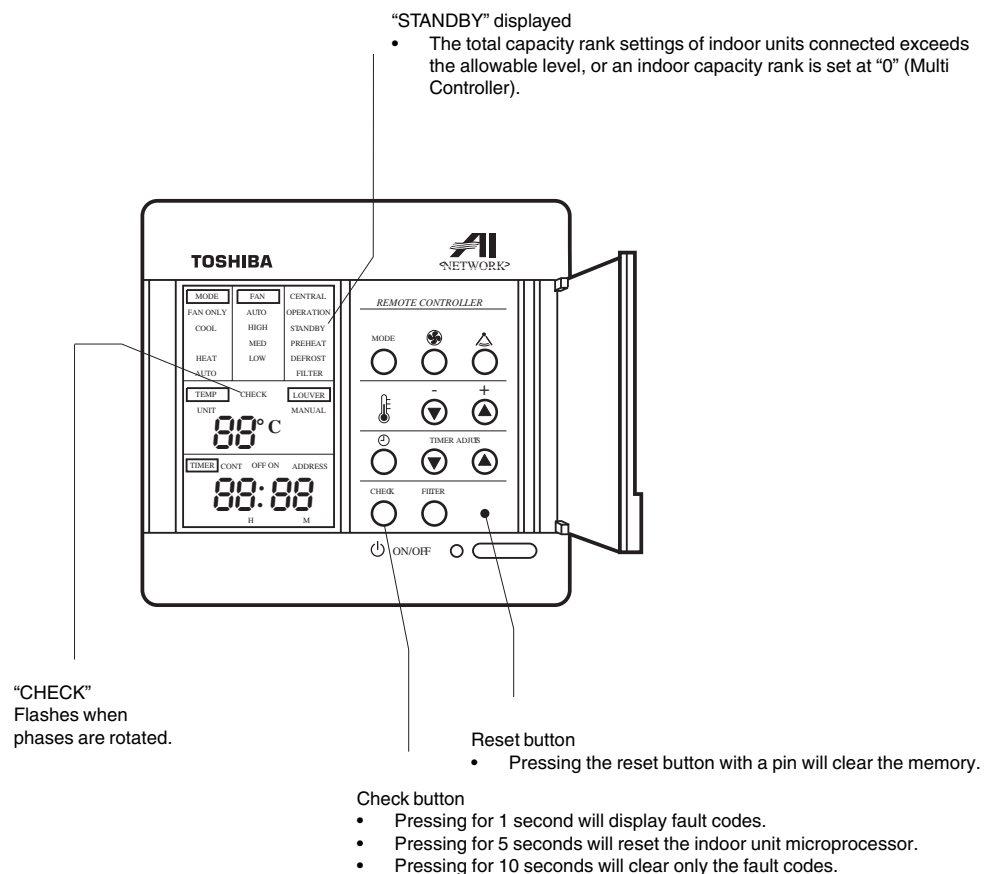
Initially malfunctions can be identified from the remote controller check display.

Details of the Multi Controller and outdoor unit malfunctions can be determined from their control boards. (Multi Controller malfunctions are also displayed on the outdoor unit interface control board).

Fault code identification

When a malfunction has been identified, do not reset the system. Press the “CHECK” button on the remote controller and observe the display.

The location and cause of the malfunction can be determined from the fault code.



Fault code display

Self-diagnostic function

Details of the malfunction check display

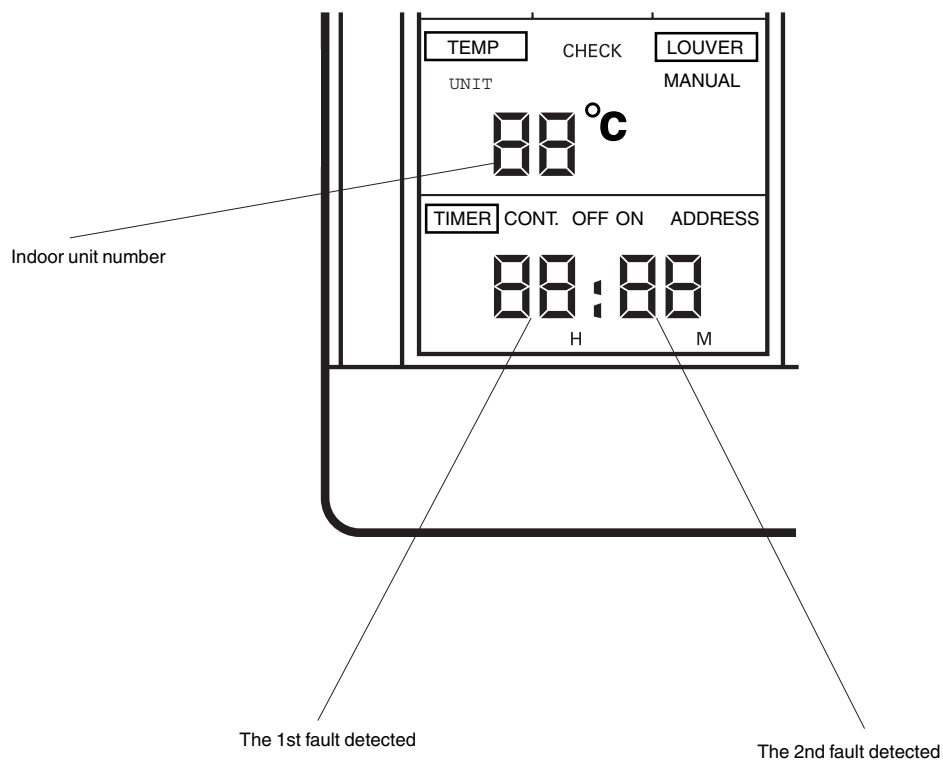
The following fault code display occurs when the check button is pressed for 1 second.

(Note: Up to 16 indoor units can be connected to one remote controller using group control.)

2 fault codes for each of the 16 indoor units can be displayed at 2 second intervals.

Example:

Display for fault codes:



Fault code display

Self-diagnostic function

Remote controller fault code

Multi Controller fault code

Outdoor fault code

04	No communication signal between Interface PCB and IPDU	→	04	No communication signal between Interface PCB and IPDU	→	04	No communication signal between Interface PCB and IPDU
	No communication signal between M/C and O/D			No communication signal between M/C and O/D.			
	No communication signal between I/D and M/C						
0b	Drain pump fault - I/D unit						
0C	TA sensor fault						
0d	TC sensor fault						
08	Reverse TC temperature change						
09	No TC temperature change						
11	Motor short circuit						
12	Indoor PC board short circuit						
b5	External input display fault		8A	Multi Controller PCB error			
	(Low level refrigerant leak if RBC-RD1-PE fitted)		88	Communication error between outdoor unit and Multi Controller			
b6	External interlock display fault		80	ThA sensor fault	←	80	ThA sensor fault
	(High level refrigerant leak if RBC-RD1-PE fitted)	→	81	ThB sensor fault	←	81	ThB sensor fault
97	Central management communication short circuit		82	ThC sensor fault	←	82	ThC sensor fault
98	Central management address set-up fault		83	ThD sensor fault	←	83	ThD sensor fault
99	No communication I/D to R/C		84	ThX sensor fault	←	84	ThX sensor fault
			0b	Drain pump fault - M/C unit	←	0b	Drain pump fault - M/C unit
15	Refer to M/C		89	Indoor units capacity codes too high or set to 0	←	89	Over capacity
						Er	[E][r] fault code refers to Outdoor unit
1C	Refer to O/D	→	1C	Refer to O/D	→	08	Four-way valve alarm.
						A0	Discharge temp. sensor (TD1) short circuit
						A1	Discharge temp. sensor (TD2) short circuit
						A2	Suction temp. sensor (TS) short circuit
						A4	External air sensor (THo) short circuit
						A5	Outdoor heat exchanger sensor (TE) short circuit
						A6	Discharge temp. (TD1 and TD2) protective operation
						A7	Suction temp. (TS) protective operation
						AA	High pressure sensor (Pd) short circuit
						Ad	DOL compressor fault
						AE	Low pressure fault (Ps)
						AF	Outdoor Unit power source phase order miswiring
						1C	Extension IC, EEPROM short circuit
14	Refer to O/D	→	14	Refer to O/D	→	14	G-Tr short-circuit protective operation
17	Refer to O/D	→	17	Refer to O/D	→	17	Current detection circuit
21	Refer to O/D	→	21	Refer to O/D	→	21	High pressure SW circuit
1d	Refer to O/D	→	1d	Refer to O/D	→	1d	Compressor error
1F	Refer to O/D	→	1F	Refer to O/D	→	1F	Inverter malfunction
d3	Refer to O/D	→	d3	Refer to O/D	→	d3	TH sensor circuit - Inverter microprocessor (IPDU)
dA	Refer to O/D	→	dA	Refer to O/D	→	dA	Heat sink overheat protective operation (IPDU)

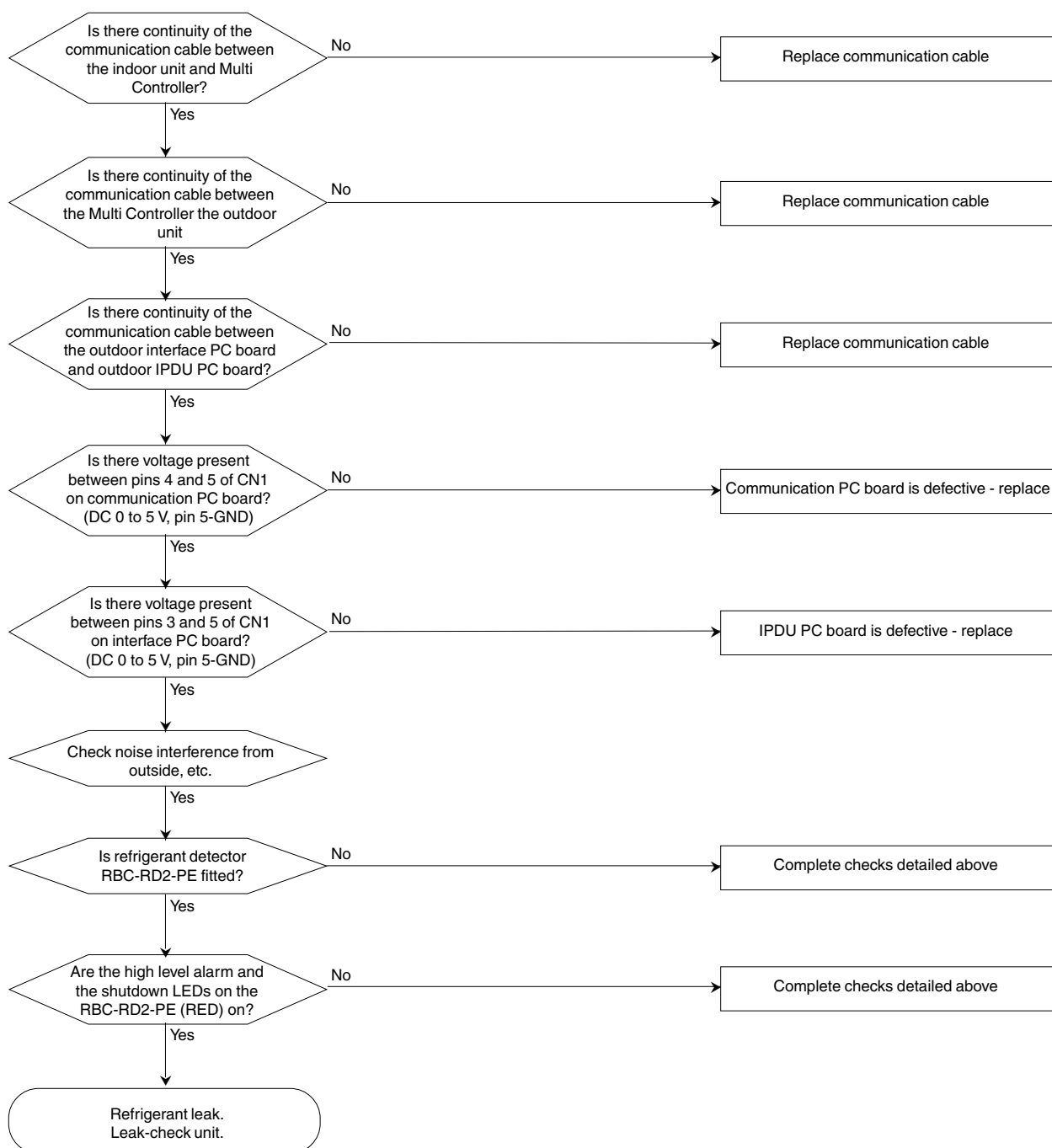
NOTE:

- To retrieve fault codes from the outdoor unit ensure rotary switch SW1 is set to position '2' and SW2 is set to position '0'.
- To retrieve fault codes from the Multi Controller ensure the display switch is set to position '1'.

Troubleshooting

Diagnostic procedure for check code

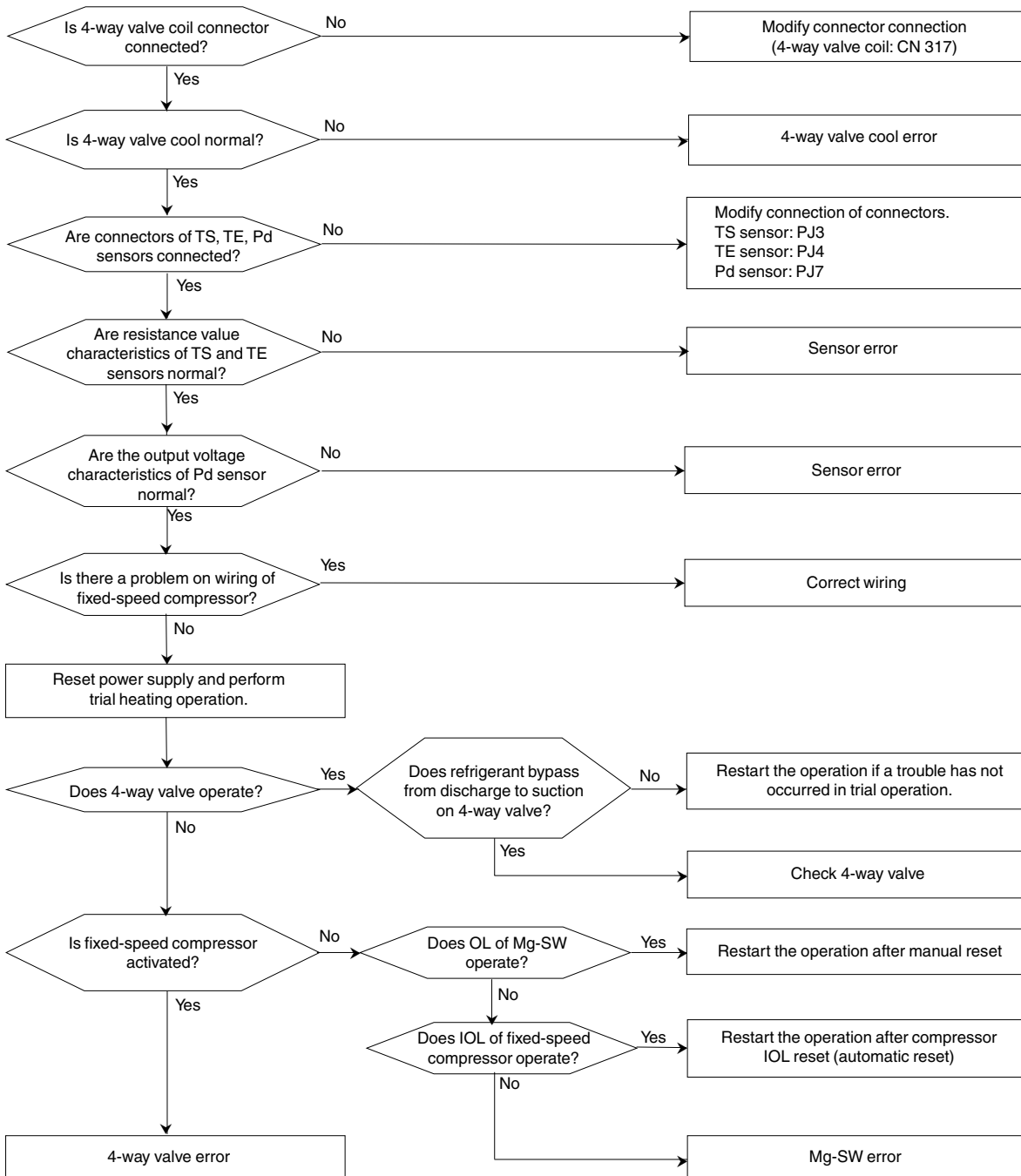
Check code	Status of air conditioner	Operation cause
[04] Inverter serial signal short circuit Inter-unit communication	Outdoor unit stops Indoor unit fan continues	1. Wiring fault between indoor and Multi Controller or Multi Controller and outdoor interface PC board. 2. Defective communication PC board. 3. Defective IPDU PC board. 4. Noise from outside. 5. High level refrigeration leak (RBC-RD2-PE).



Troubleshooting

Diagnostic procedure for check code

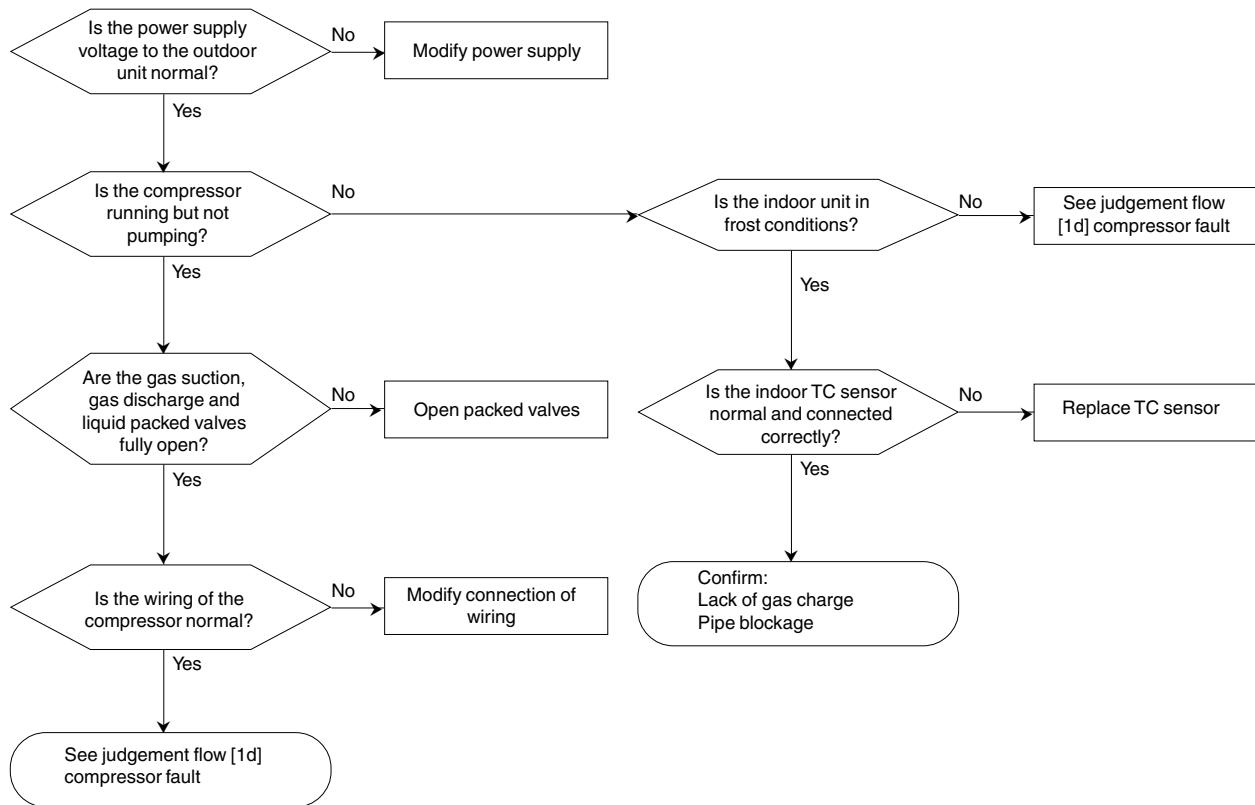
Check code	Operation cause
[08] 4-way valve reversal alarm	1. 4-way valve operation error. 2. TS sensor/TE sensor error. 3. Pd sensor error. 4. Fixed-speed compressor Mg-SW operation error.



Troubleshooting

Diagnostic procedure for check code

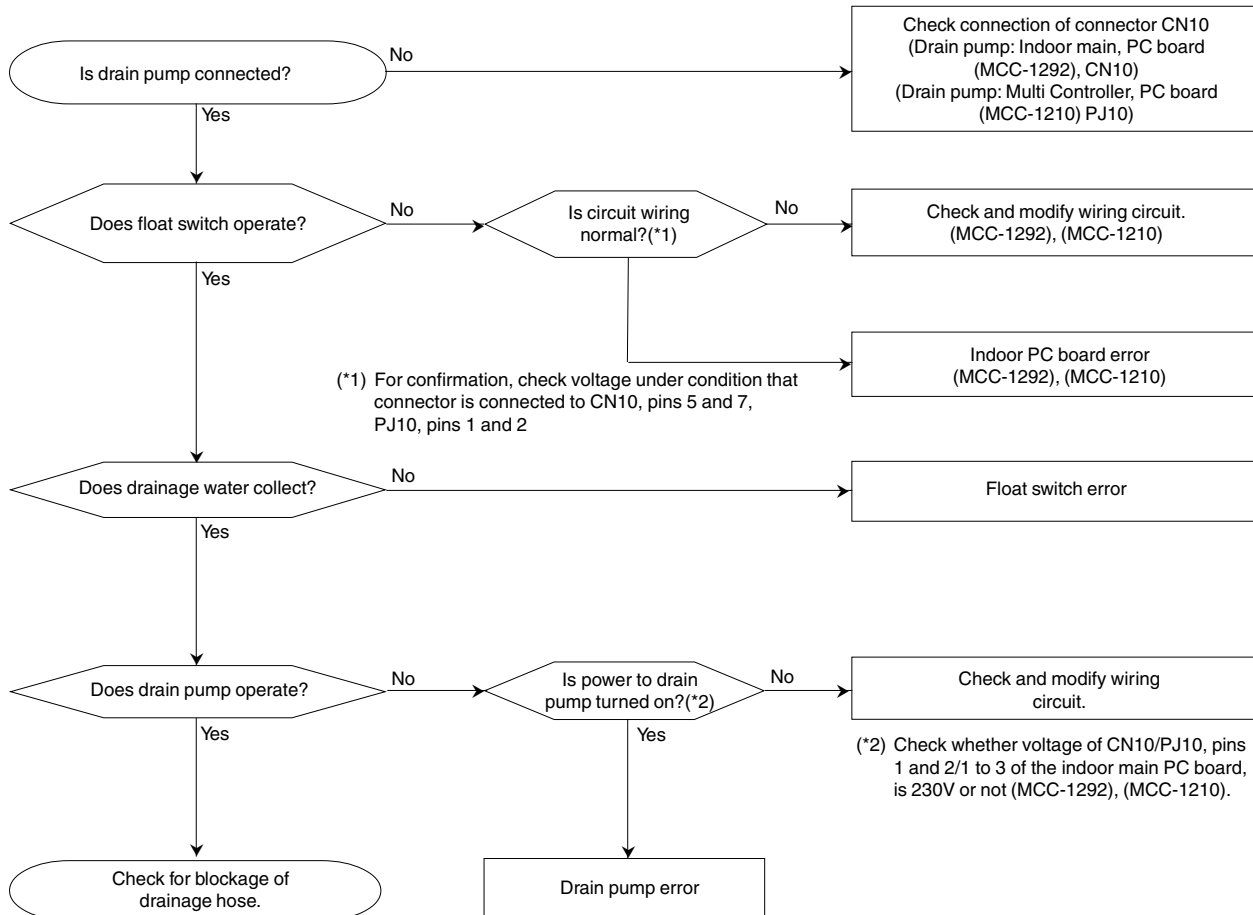
Check code	Operation cause
[09] No temperature change in TC temperature	1. Compressor running but not pumping. 2. Bi-Metal thermostat protection device. 3. Compressor wiring. 4. Faulty TC temperature sensor.



Troubleshooting

Diagnostic procedure for check code

Check code	Status of air conditioner	Operation cause
[0b] Drain pump fault Indoor unit or Multi Controller	Operation stops for Multi Controller fault. Indoor unit fan continues for unit fault.	1. Float switch disconnection. 2. Drain pump operation error. 3. Drain hose blockage.



Check code	Operation cause
[0C] Indoor sensor (TA) short or open circuit alarm	TA sensor open/short

TA sensor open/short has been detected. Check disconnection of connector connection (TA sensor: CN04) circuit and resistance value characteristics of sensor.

When the sensors are normal, replace indoor PC board (MCC-1292).

Check code	Status of air conditioner	Operation cause
[0D] Indoor sensor (TC) short or open circuit alarm	Indoor unit fan stops. 'Preheat defrost' is displayed on R/C.	TC sensor open/short

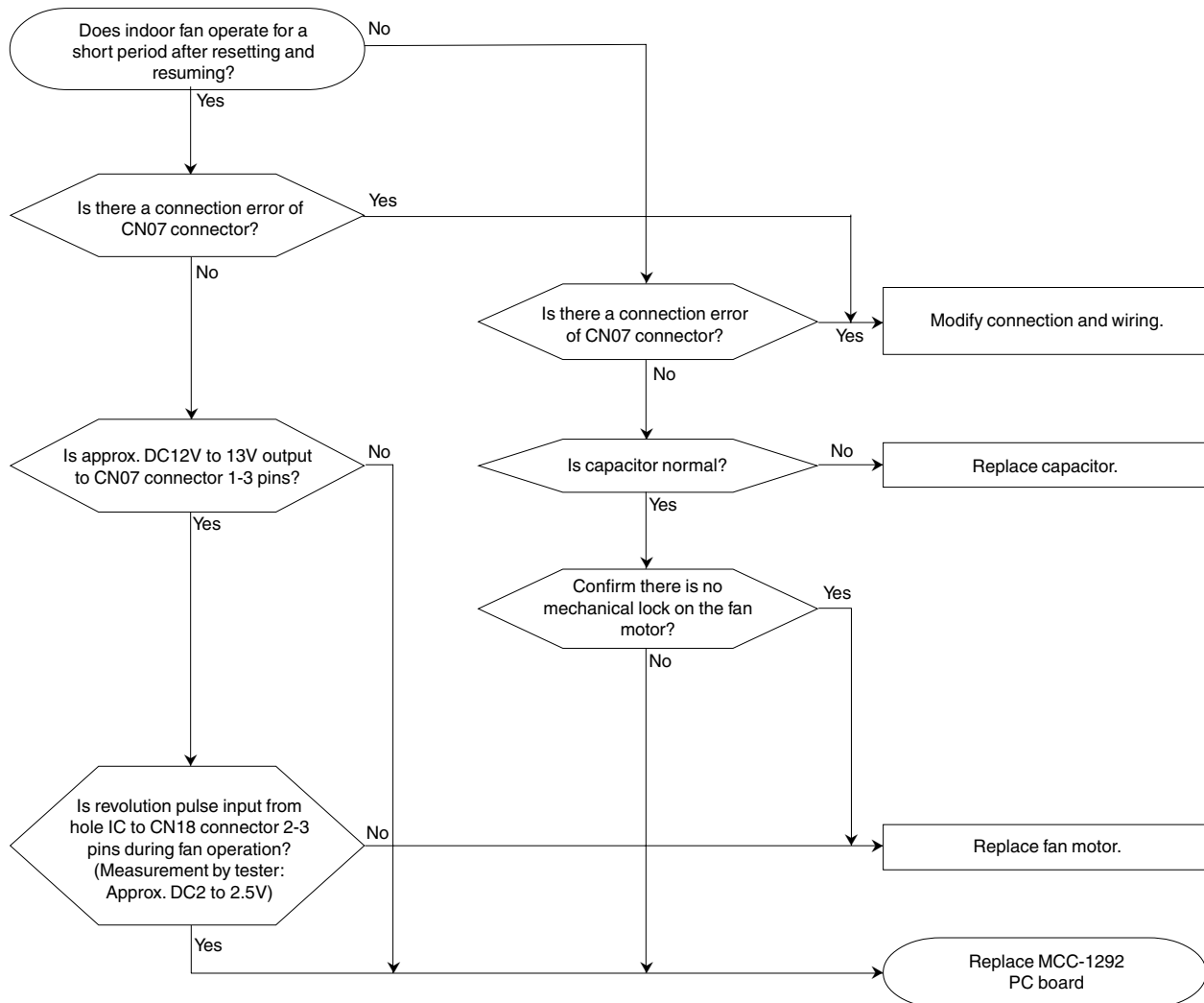
TC sensor open/short has been detected. Check disconnection of connection (TC sensor: CN05) circuit and resistance value characteristics of sensor.

When the sensors are normal, replace indoor PC board (MCC-1292).

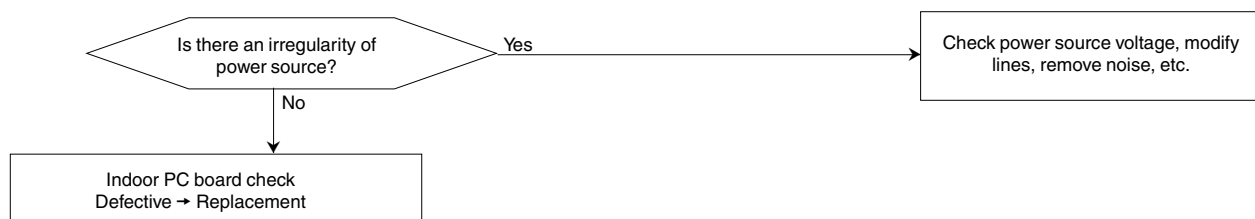
Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[11] Indoor fan motor short circuit alarm	1. Fan motor circuit connection error. 2. Capacitor error. 3. Fan motor error. 4. Defective indoor PC board.



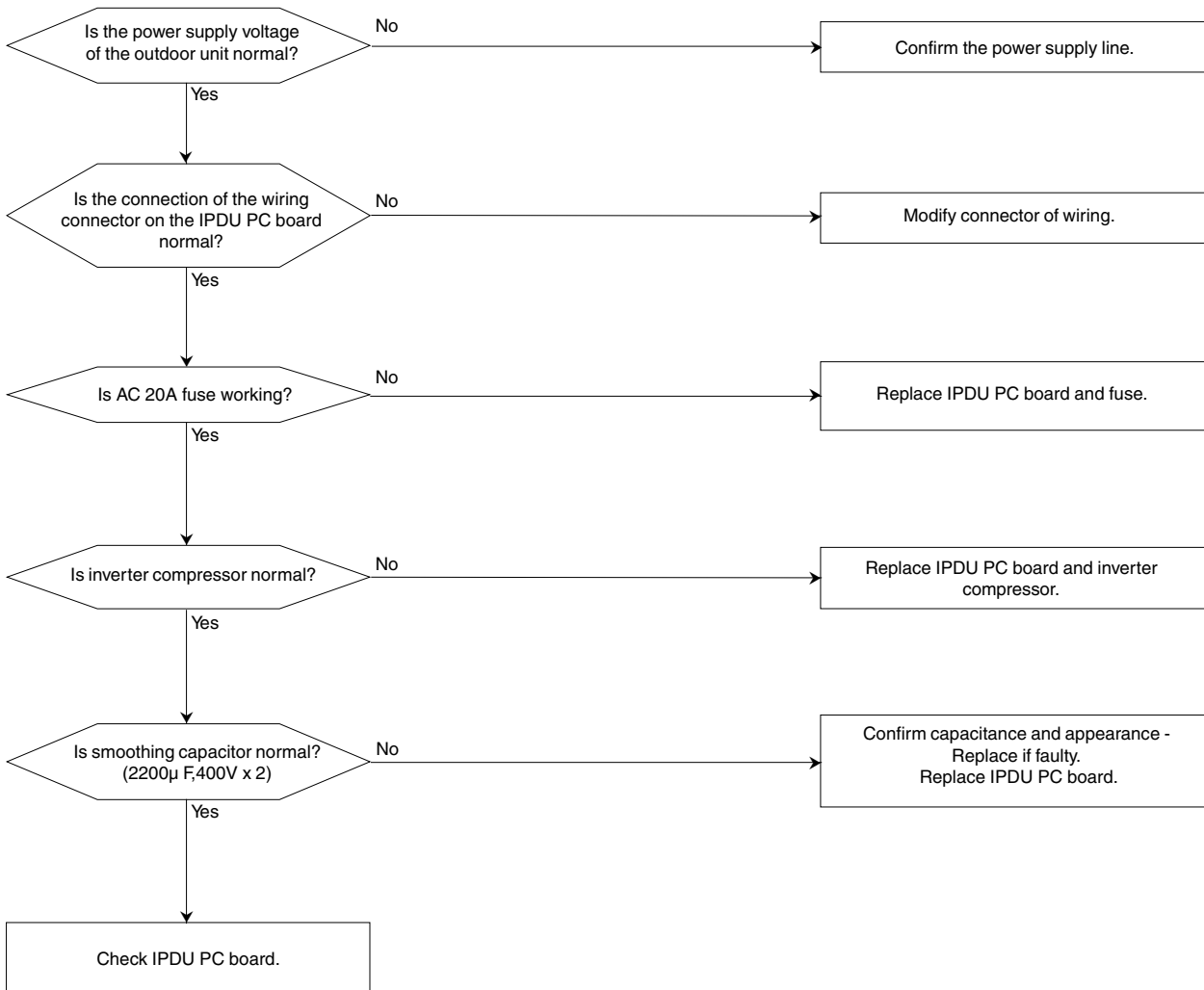
Check code	Operation cause
[12] Indoor PC board short circuit alarm	1. Irregularity of power source. 2. Noise of peripheral equipment. 3. Defective indoor PC board.



Troubleshooting

Diagnostic procedure for check code

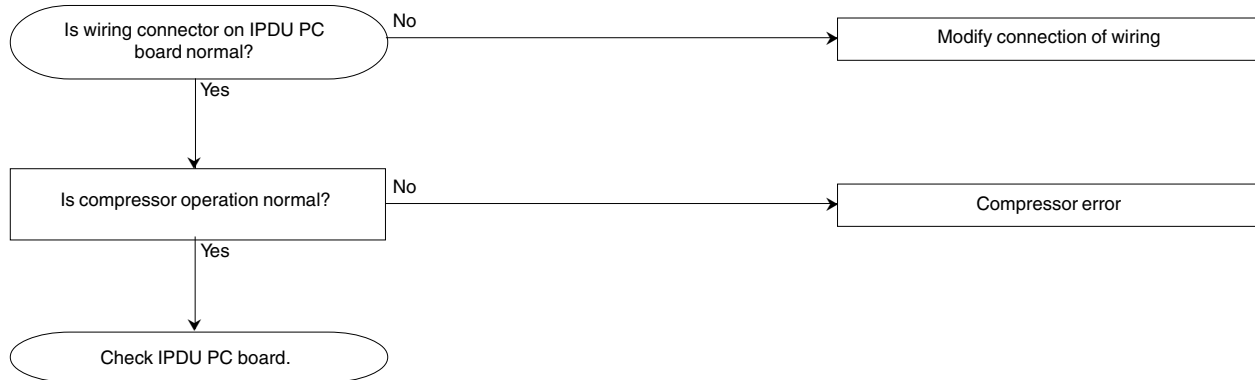
Check code	Status of air conditioner	Operation cause
[14] IPDU short circuit protective operation alarm (Power Transistor)	Operation stops	1. Outdoor unit power supply error. 2. Wiring error on IPDU PC board. 3. AC fuse disconnection. 4. Inverter compressor error. 5. Defective IPDU PC board.



Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[17] Current detection circuit alarm Refer to outdoor unit	1. Defective wiring of IPDU PC board. 2. Defective IPDU PC board.



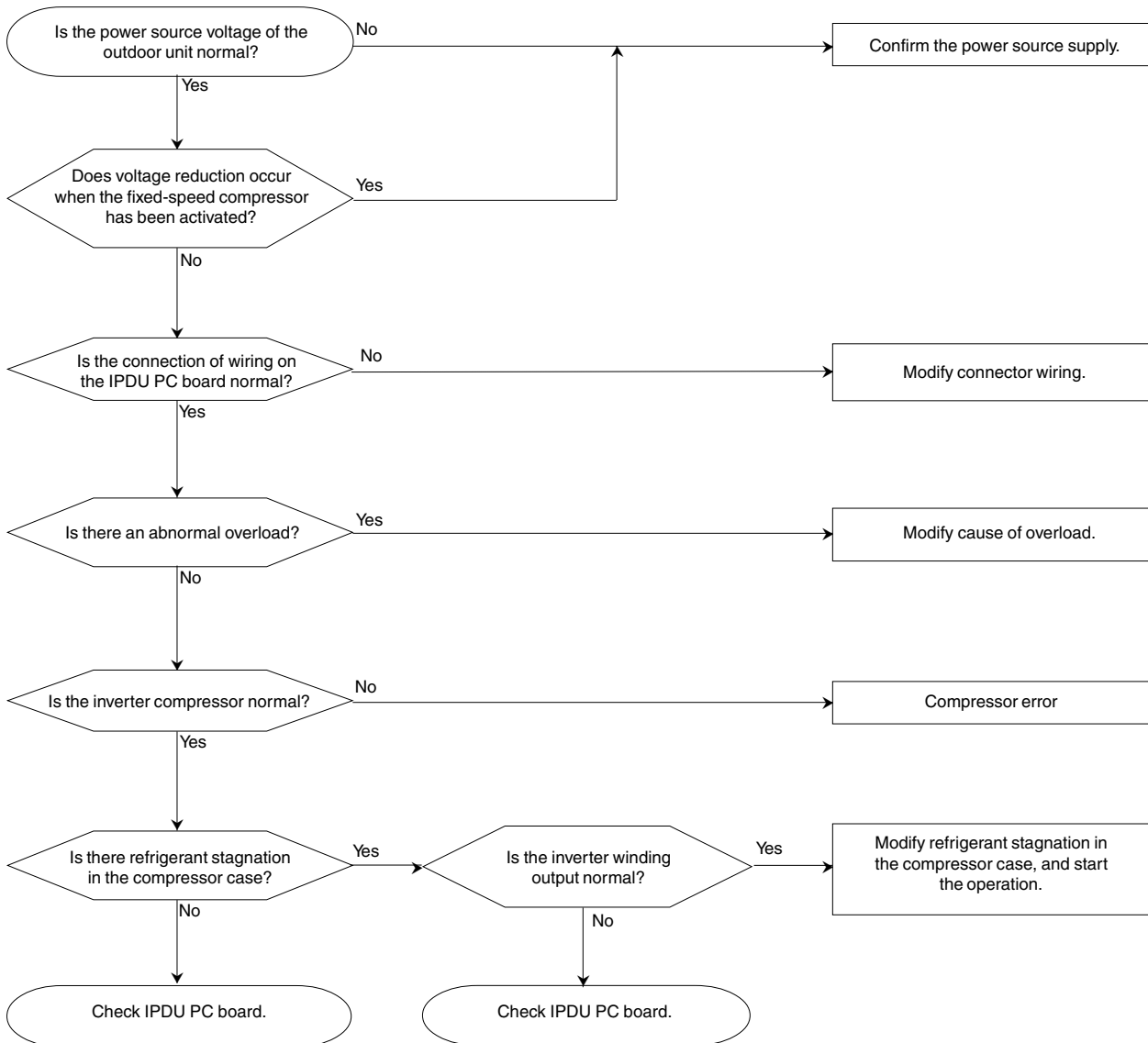
Check code	Operation cause
[1C] Extension IC, EEPROM short circuit alarm (Outdoor unit error)	1. Outdoor unit power source error. 2. Interface PC board error.



Troubleshooting

Diagnostic procedure for check code

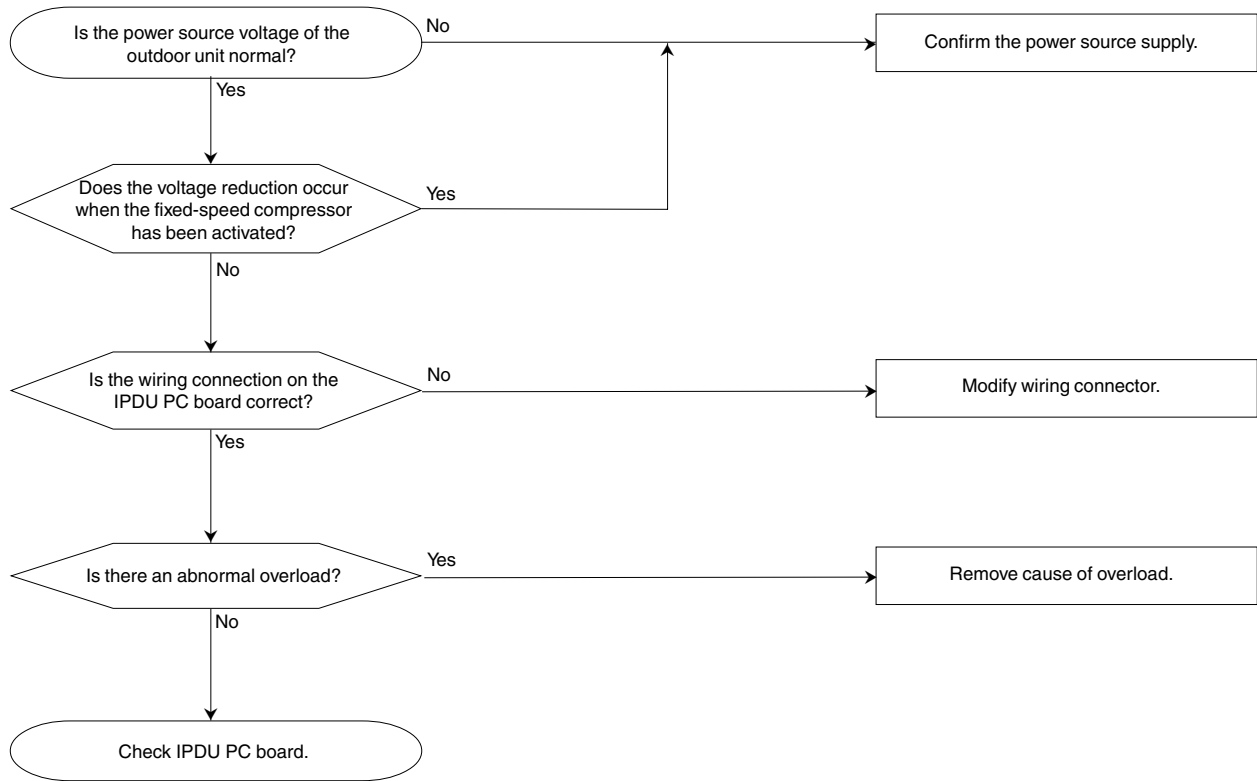
Check code	Status of air conditioner	Operation cause
[1d] Compressor error alarm Refer to outdoor unit	Operation stops	1. Outdoor unit power source error. 2. Inverter compressor circuit system error. 3. Inverter compressor error. 4. Inverter compressor refrigerant stagnation. 5. Defective IPDU PC board.



Troubleshooting

Diagnostic procedure for check code

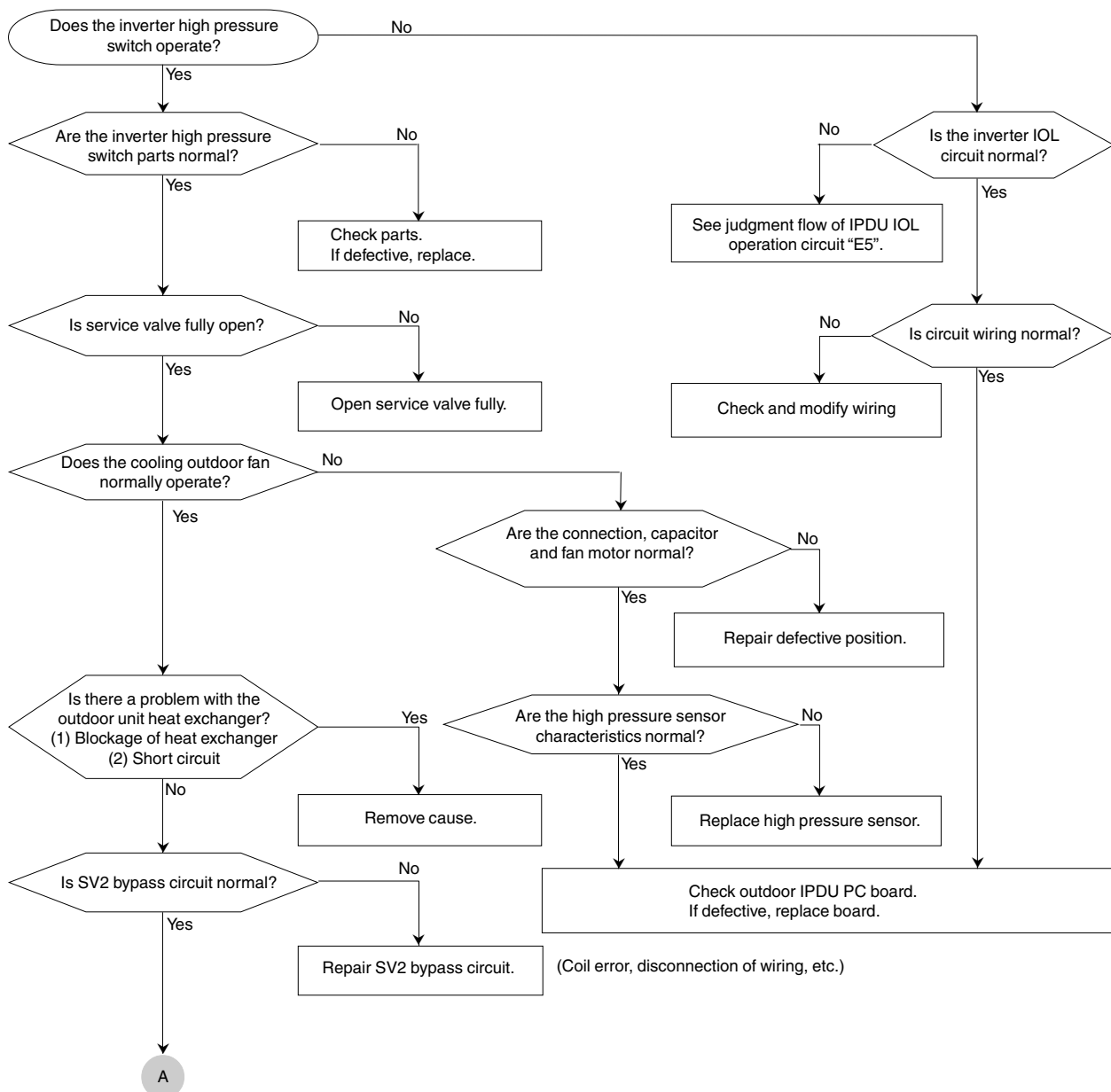
Check code	Status of air conditioner	Operation cause
[1F] Compressor break down Refer to outdoor unit	Operation stops	1. Outdoor unit power source error. 2. Inverter compressor circuit system error. 3. IPDU PC board error.



Troubleshooting

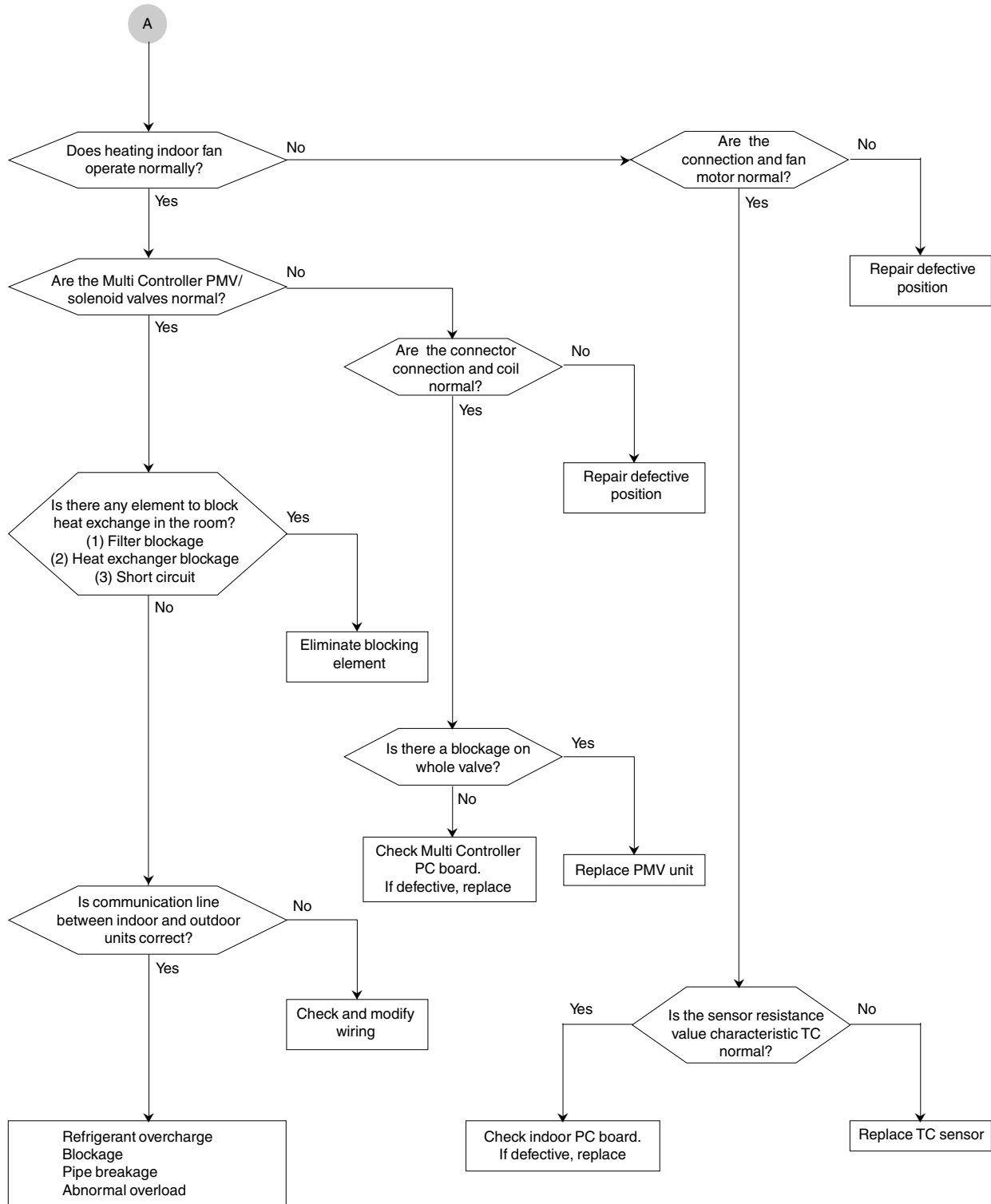
Diagnostic procedure for check code

Check code	Operation cause
[21] Inverter high pressure SW system alarm	1. Inverter high pressure SW error. 2. Inverter IOL operation. 3. Service valve closed. 4. Outdoor fan, capacitor error. 5. Indoor/Outdoor PMV blockage. 6. Outdoor heat exchanger blockage. 7. SV2 circuit blockage. 8. Indoor – Outdoor communication error. 9. Pd sensor error. 10. Refrigerant overcharge.



Troubleshooting

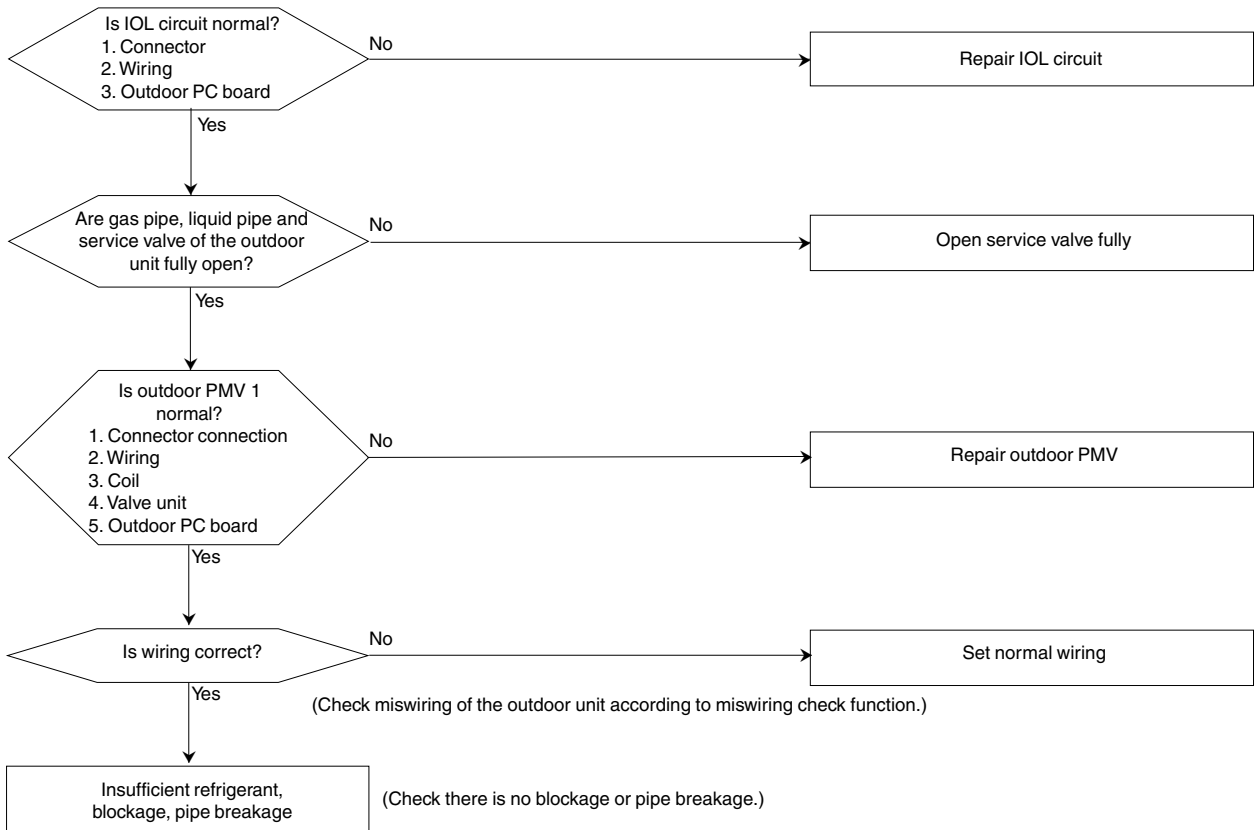
Diagnostic procedure for check code



Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[E5] Inverter IOL operation	1. Inverter IOL operation. 2. Service valve closed. 3. Outdoor PMV 1 error. 4. Miswiring of communication between indoor and outdoor unit.



Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[80] Gas temp. sensor (Tha) short circuit	Tha sensor open/short

Open/short of Tha sensor was detected. Check connector (Tha sensor: PJ1) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the Multi Controller.

Check code	Operation cause
[81] Gas temp. sensor (Thb) short circuit	Thb sensor open/short

Open/short of Thb sensor was detected. Check connector (Thb sensor: PJ2) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the Multi Controller.

Check code	Operation cause
[82] Gas temp. sensor (Thc) short circuit	Thc sensor open/short

Open/short of Thc sensor was detected. Check connector (Thc sensor: PJ3) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the Multi Controller.

Check code	Operation cause
[83] Gas temp. sensor (Thd) short circuit	Thd sensor open/short

Open/short of Thd sensor was detected. Check connector (Thd sensor: PJ4) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the Multi Controller.

Check code	Operation cause
[84] Gas temp. sensor (Thx) short circuit	Thx sensor open/short

Open/short of Thx sensor was detected. Check connector (Thx sensor: PJ5) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the Multi Controller.

Troubleshooting

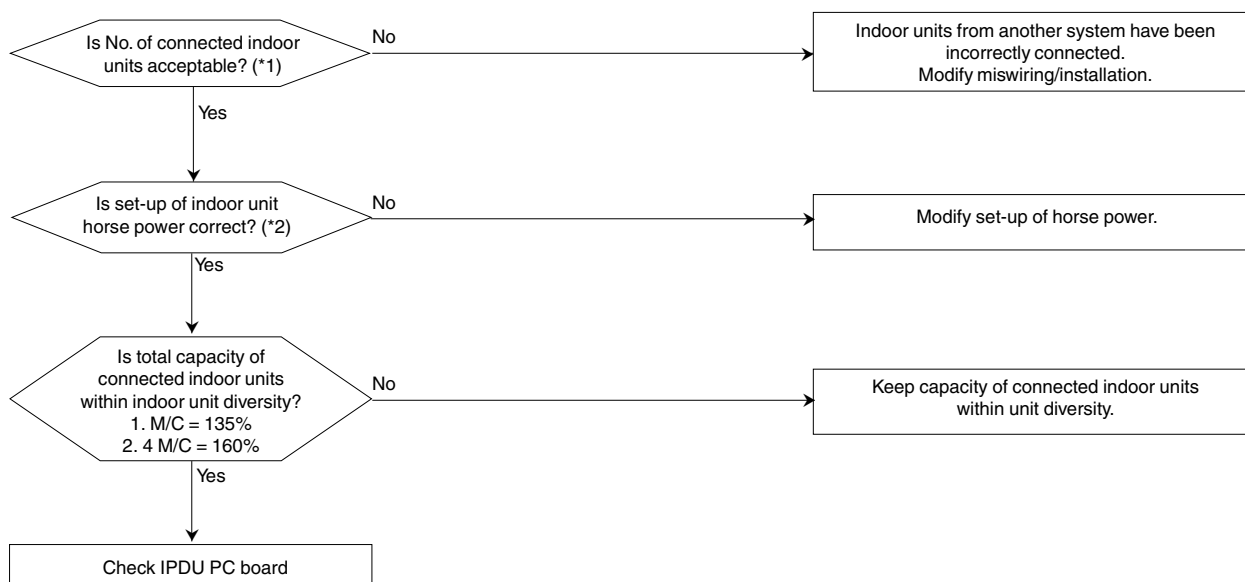
Diagnostic procedure for check code

Check code	Operation cause
[88] Communication alarm between Multi Controller and outdoor unit	1. Error connection between Multi Controller and outdoor unit. 2. Outdoor PC board error (MCC-1223). 3. Multi Controller PC board error (MCC-1210). 4. Multi Controller does not recognise outdoor capacity.

Check lead connection between indoor unit and Multi Controller.

If no error is detected, replace Multi Controller board (MCC-1210) or outdoor PC board (MCC-1223).

Check code	Operation cause
[89] Indoor over capacity	1. No. of connected indoor units/connected over capacity. 2. Incorrect set-up of indoor unit horse power or set to 0.



(*1)

Number of Multi Controllers	Maximum No. of indoor units	Indoor unit diversity	Maximum system code	Maximum code per Multi Controller
1	4	135%	27	27
2	8	160%	32	27
3	12			27 (13*)
4	16			13

(*2)

Indoor unit	Capacity	No connection	10	13	16	20	26	36	46
	Code number	0	2	3	4	5	6	8	10

(Example: Model RAV-364UH-PE, capacity = 36)

Troubleshooting

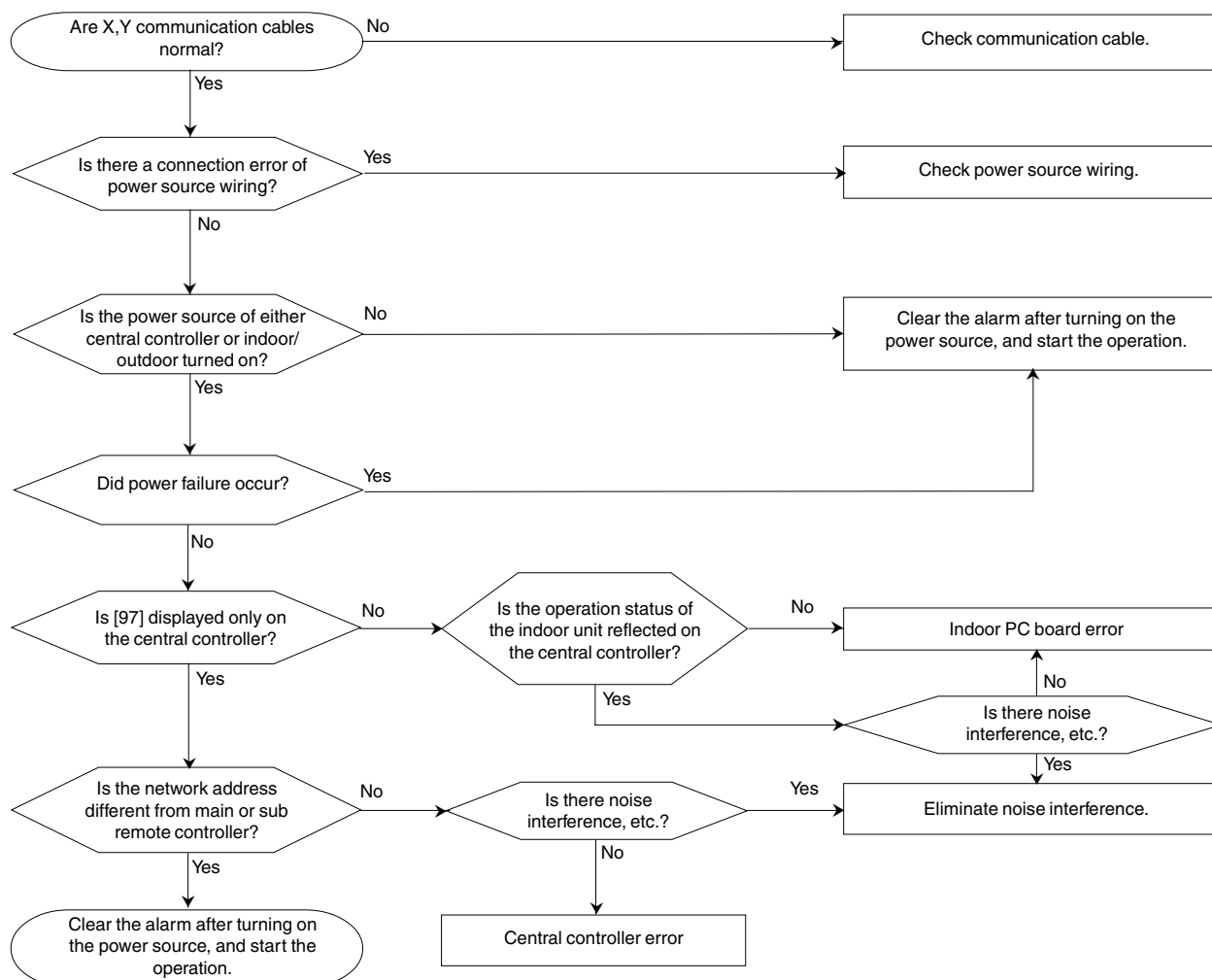
Diagnostic procedure for check code

Check code	Operation cause
[8A] Multi Controller PC board malfunction	1. Multi Controller PC board error.

Check lead connection between indoor unit and Multi Controller and outdoor unit.

If no error is detected, replace Multi Controller board (MCC-1210).

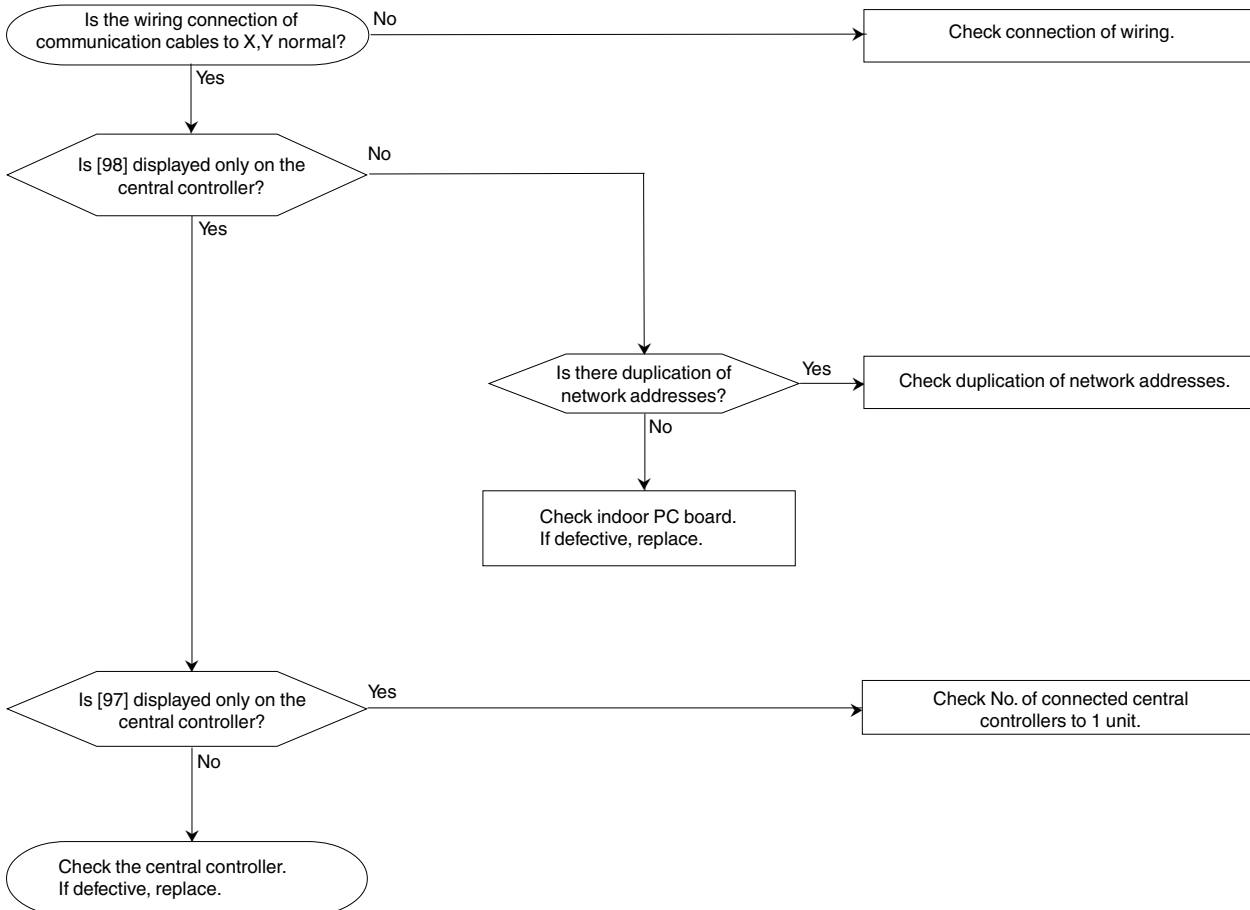
Check code	Operation cause
[97] Central management communication short circuit	1. Connection error of (XY) for outdoor units. 2. Power source system error of the central controller and indoor unit. 3. Noise of peripheral devices. 4. Power failure. 5. Indoor PC board error, central controller error.



Troubleshooting

Diagnostic procedure for check code

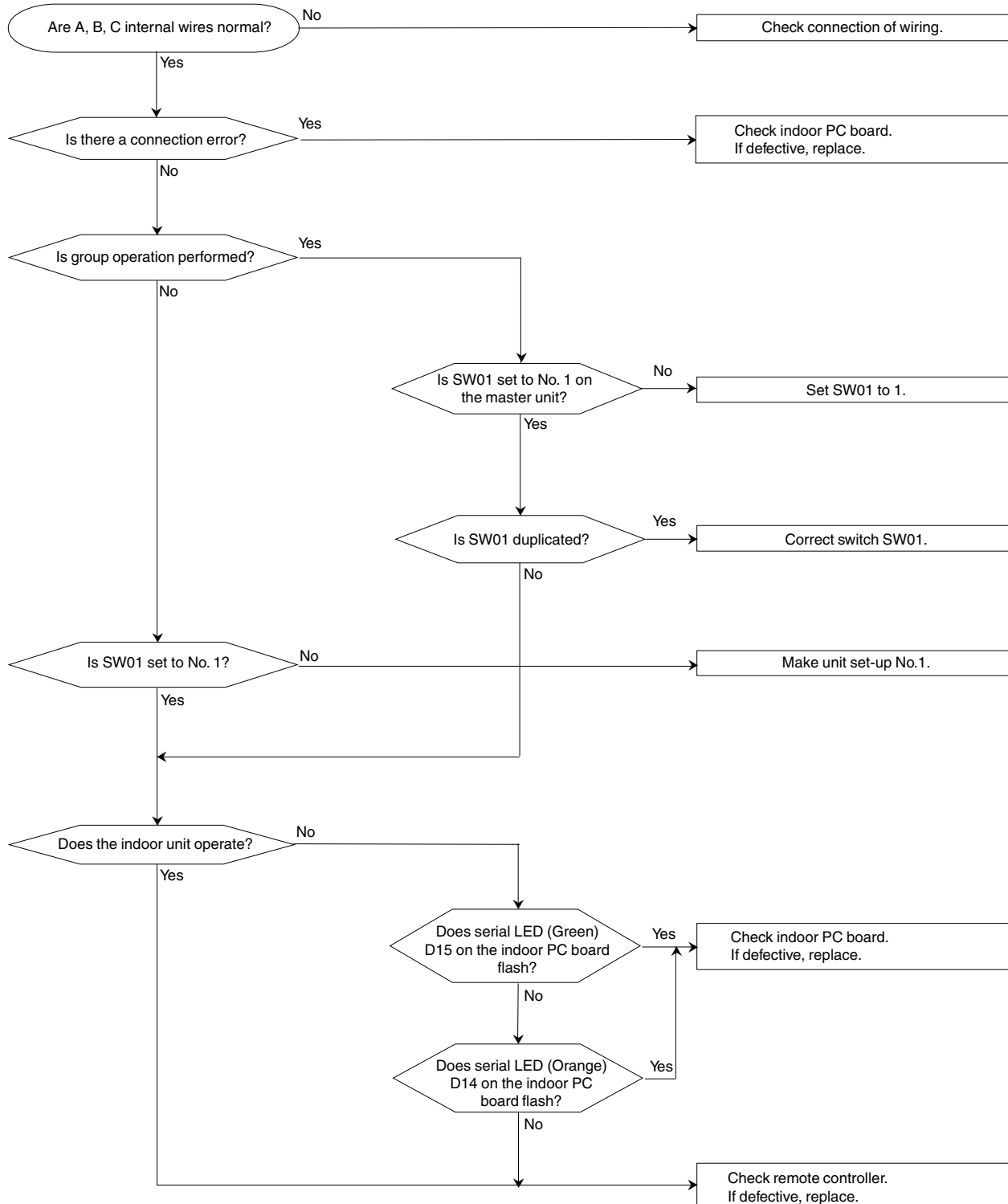
Check code	Operation cause
[98] Central management address set-up fault	1. Miswiring of XY communication cables. 2. Duplicated network addresses. 3. Indoor PC board error, central controller error.



Troubleshooting

Diagnostic procedure for check code

Check code	Status of air conditioner	Operation cause
[99] Remote controller serial signal circuit alarm	Indoor unit stops	1. Remote controller circuit connection error. 2. Duplicated indoor No.1 units. 3. Remote controller error. 4. Indoor PC board error.



Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[A0] Discharge temp. sensor (TD1) short circuit	TD1 sensor open/short

Open/short of TD1 sensor was detected. Check connector (TD1 sensor: PJ1) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the outdoor unit.

Check code	Operation cause
[A1] Discharge temp. sensor (TD2) short circuit	TD2 sensor open/short

Open/short of TD2 sensor was detected. Check connector (TD2 sensor: PJ2) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the outdoor unit.

Check code	Operation cause
[A2] Suction temp. sensor (TS) short circuit	TS1 sensor open/short

Open/short of TS1 sensor was detected. Check connector (TS1 sensor: PJ3) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the outdoor unit.

Check code	Operation cause
[A4] External air sensor (Tho) short circuit	Tho sensor open/short

Open/short of Tho sensor was detected. Check connector (Tho sensor: PJ5) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the outdoor unit.

Check code	Operation cause
[A5] Outdoor heat exchange sensor (TE) short circuit	TE sensor open/short

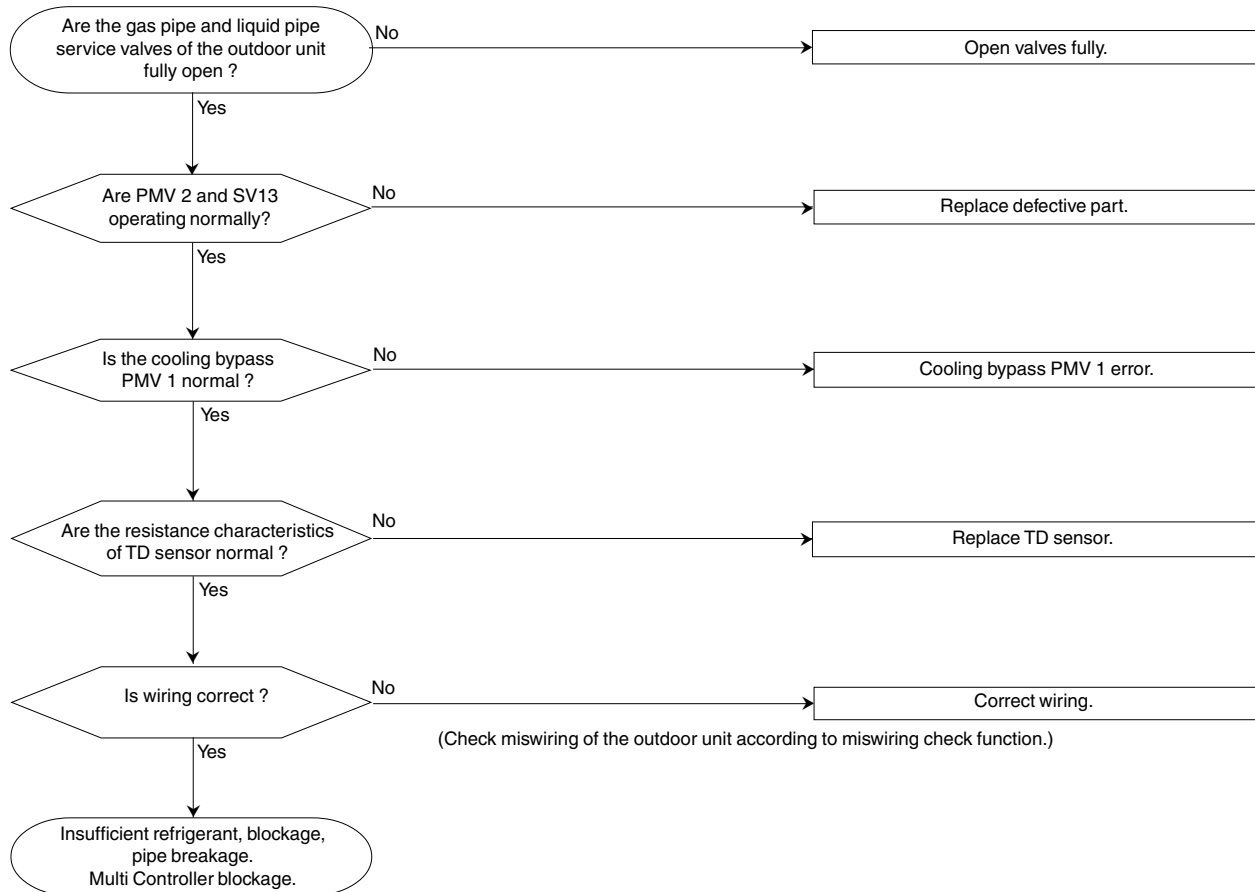
Open/short of TE sensor was detected. Check connector (TE sensor: PJ4) and characteristics of sensor resistance value.

When sensor is normal, replace interface PC board of the outdoor unit.

Troubleshooting

Diagnostic procedure for check code

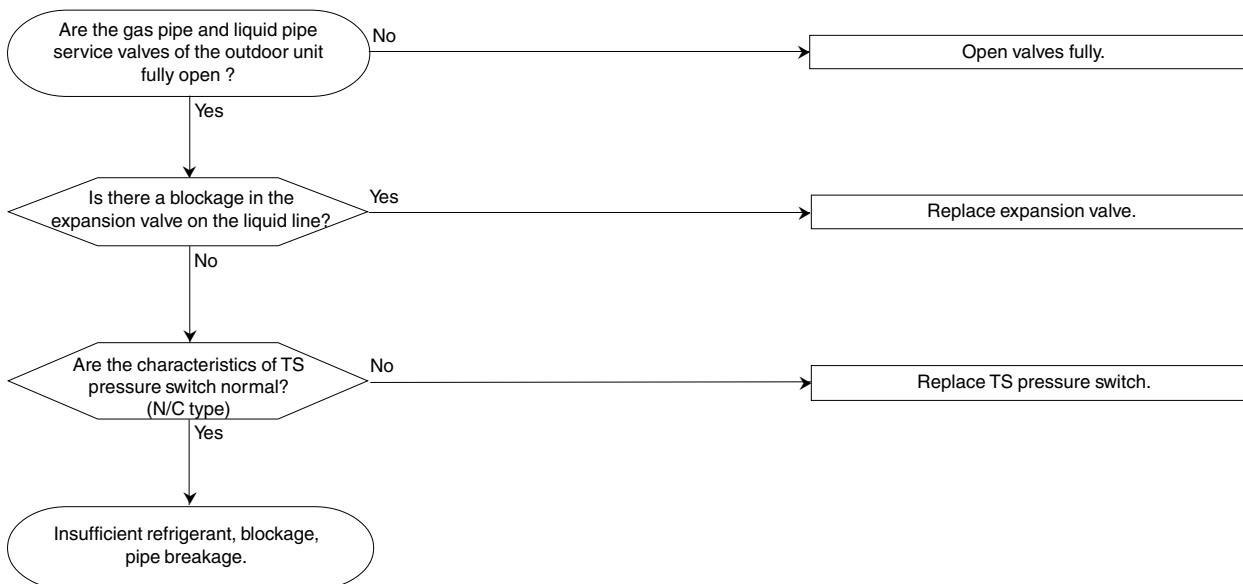
Check code	Operation cause
[A6] Discharge temperature TD1 and TD2 alarm (>130°C)	1. Outdoor unit service valve closed. 2. Cooling bypass PMV error. 3. TD sensor error. 4. Insufficient refrigerant, blockage in pipe. 5. Blockage of PMV assembly on the liquid line.



Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[A7] Suction temperature (TS) alarm (>40°C)	1. Outdoor unit service valve closed. 2. Blockage PMV assembly on the liquid line. 3. TS pressure switch error. 4. Insufficient refrigerant, blockage in pipe.



Check code	Operation cause
[AA] High pressure sensor (Pd) short circuit	Pd sensor output voltage alarm

If there is an abnormal output voltage of Pd sensor: Check connector circuit (Pd sensor: PJ7) and output voltage of the sensor.

If the sensor is normal, replace the outdoor interface PC board.

Sensor characteristics:

Input 12 V DC ($\pm 10\%$) across red and black leads of the pressure sensor.

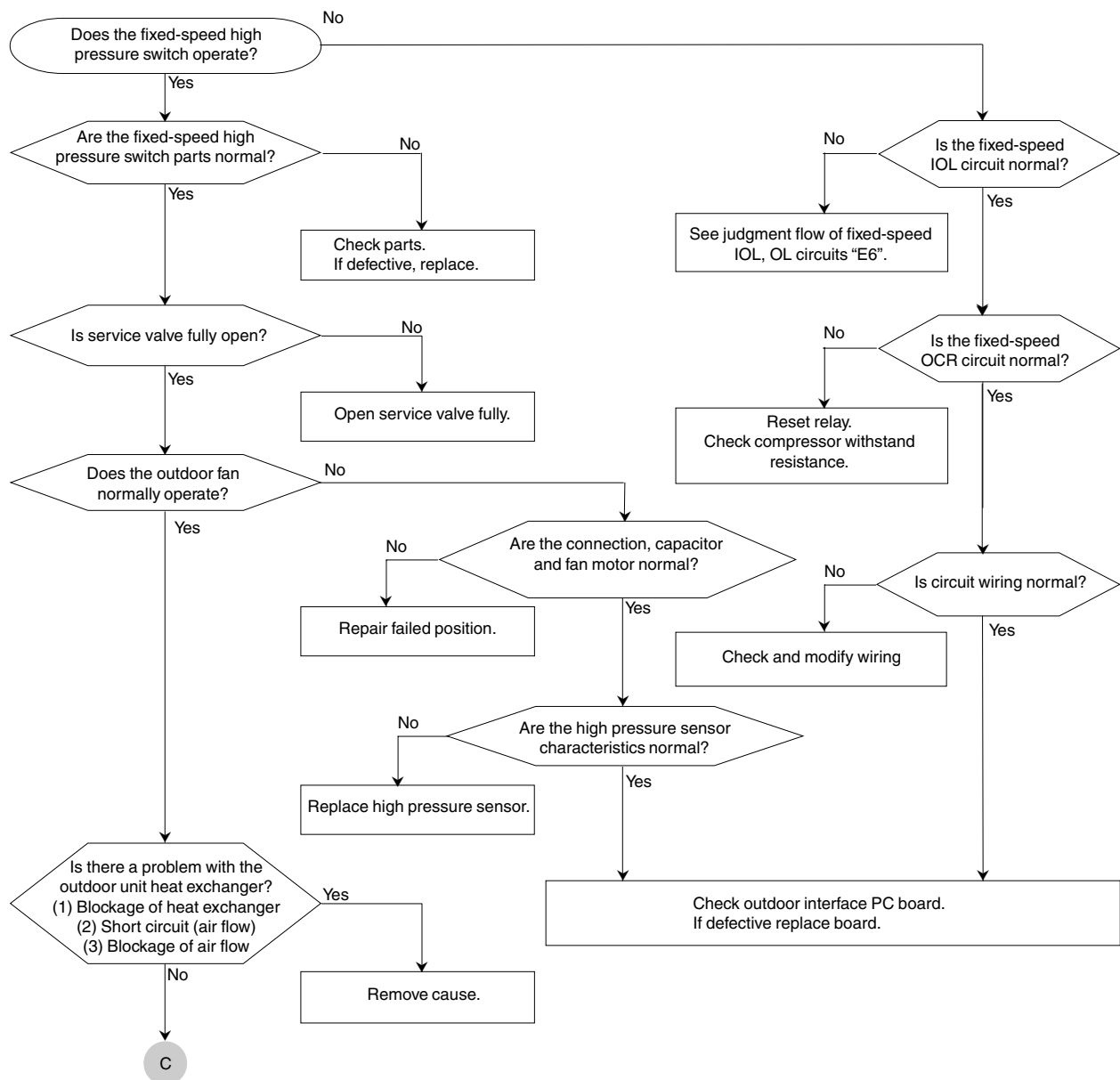
Output 0.5 to 4.5 V at 0 to 2.45 MPa across white and black leads of the pressure sensor.

Troubleshooting

Diagnostic procedure for check code

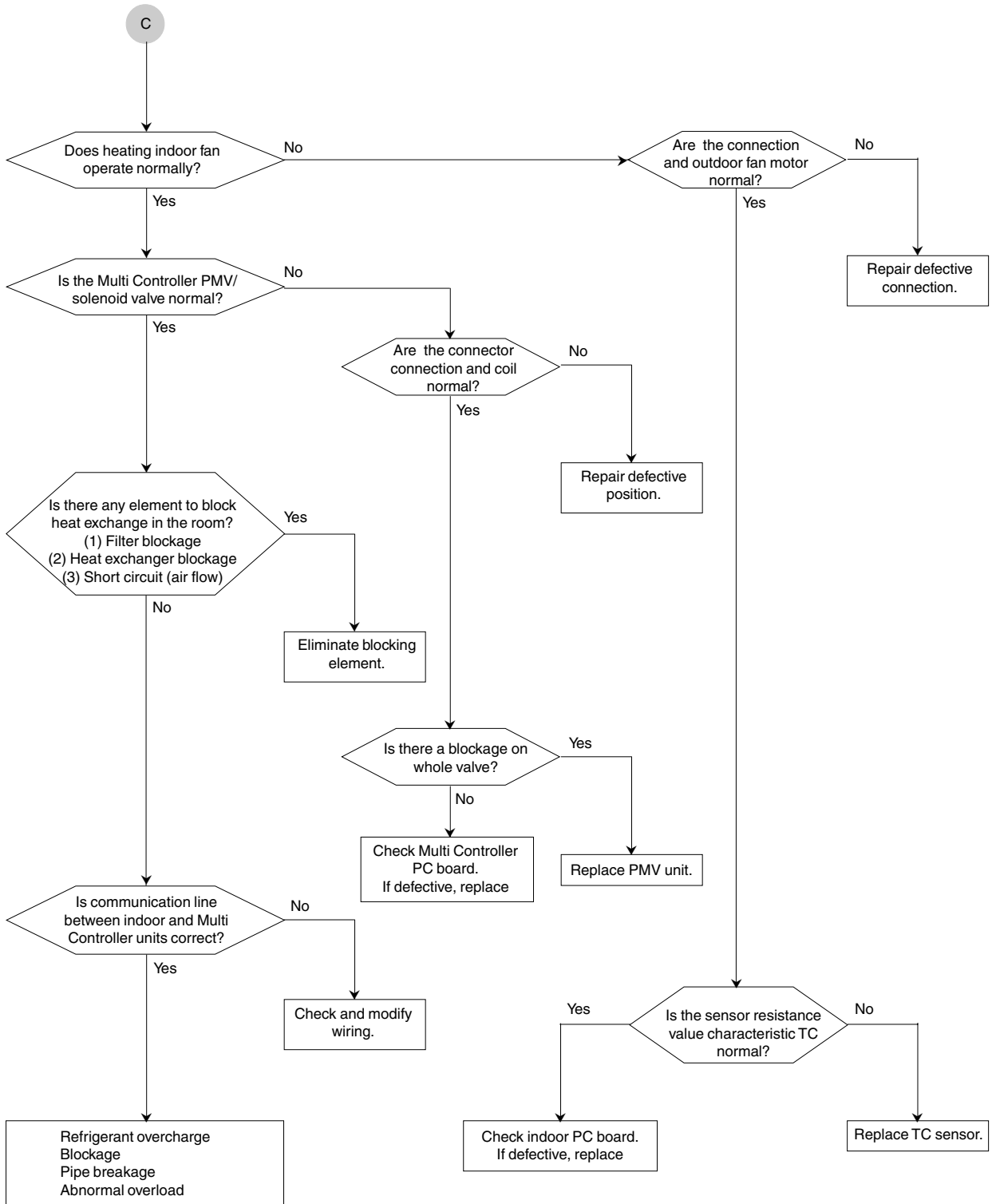
Check code	Operation cause
[Ad] Fixed-speed compressor fault	1. Fixed-speed compressor high pressure SW error. 2. Fixed-speed compressor IOL operation. 3. Service valve closed. 4. Outdoor fan, capacitor error. 5. Multi Controller/outdoor PMV blockage. 6. Outdoor heat exchanger blockage. 7. SV2 circuit blockage. 8. Miswiring of communication between indoor and outdoor unit. 9. Pd sensor error. 10. Refrigerant overcharge.

OCR: Overload Current Relay



Troubleshooting

Diagnostic procedure for check code

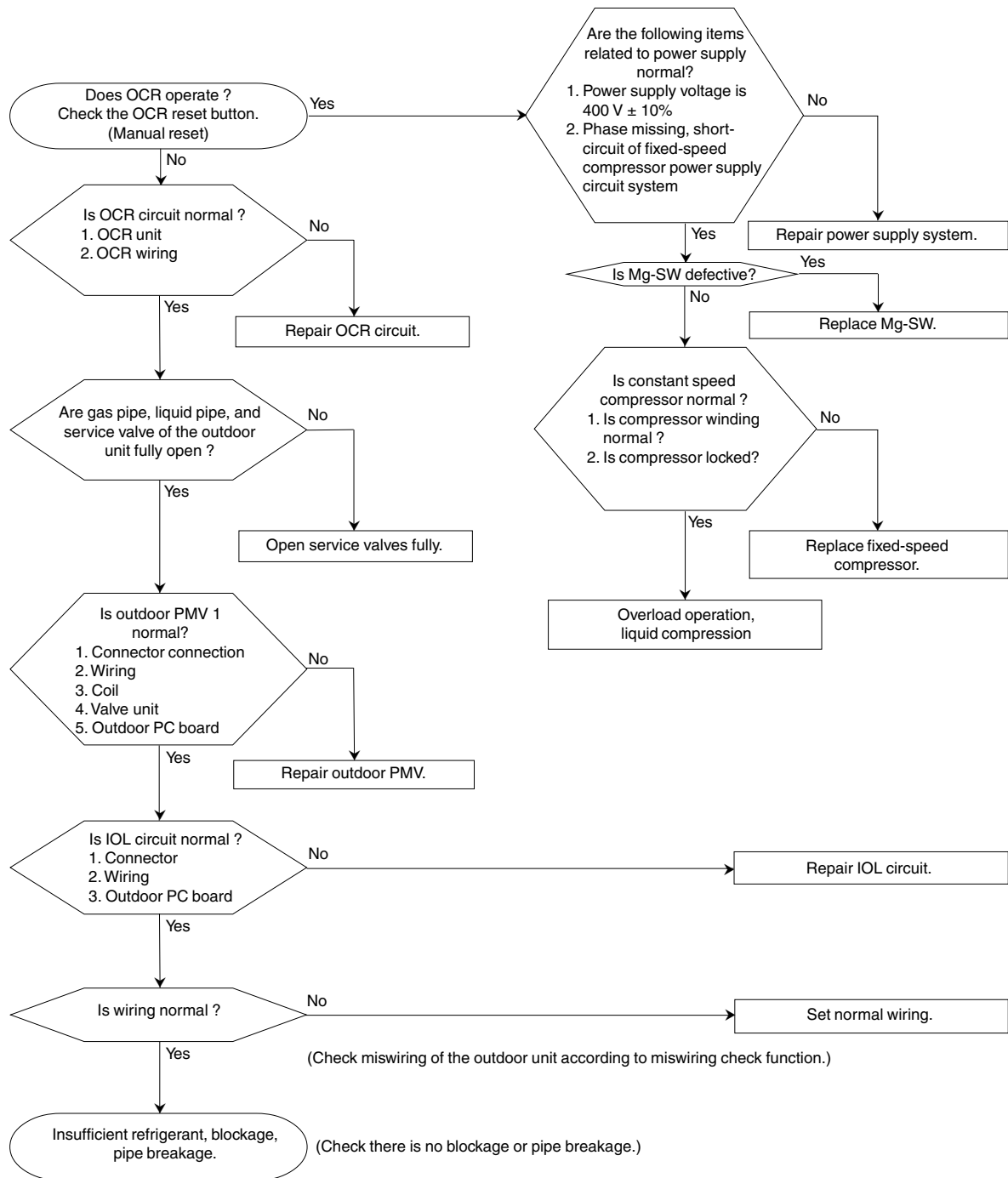


Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[E6] Fixed-speed IOL, OL alarm	1. Power supply error. 2. Fixed-speed compressor IOL operation. 3. Service valve closed. 4. Blockage in Multi Controller PMV/outdoor PMV 1. 5. Miswiring of communication between indoor and outdoor units.

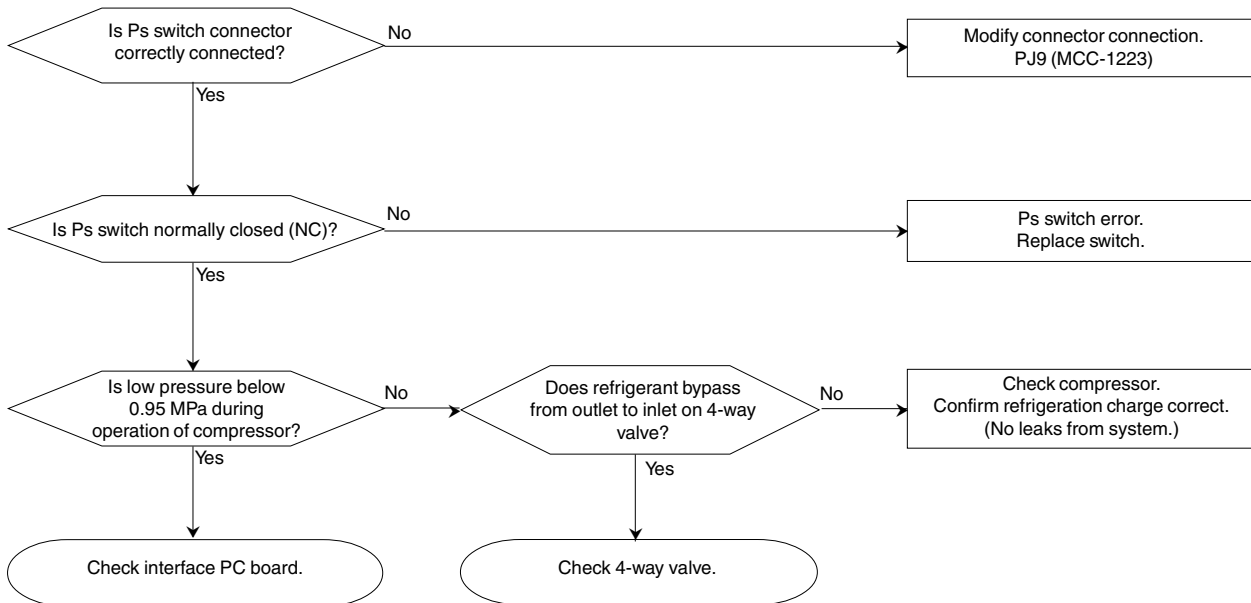
OCR: Overload Current Relay



Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[AE] Low pressure fault (Ps) Detected by low pressure switch <0.02 MPa	1. Ps pressure switch error. 2. Ps switch connection error. 3. Compressor compressing error. 4. 4-way valve error. 5. Refrigeration undercharge. 6. Gas leak.



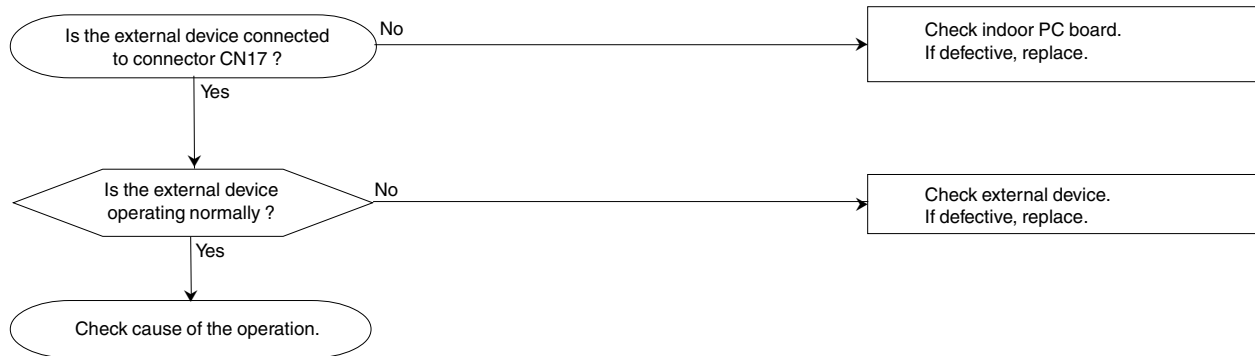
Check code	Operation cause
[AF] Outdoor unit power source phase order miswiring	Abnormal power phase order or missing phase of power supply to the outdoor unit.

There is an abnormal power phase order or missing phase of power supply to the outdoor unit. Check power supply wiring.

Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[b5] External input display fault [b6] External interlock display fault Fault on CN17 - fault on external device (accessory)	1. External device error. 2. Indoor PC board error.



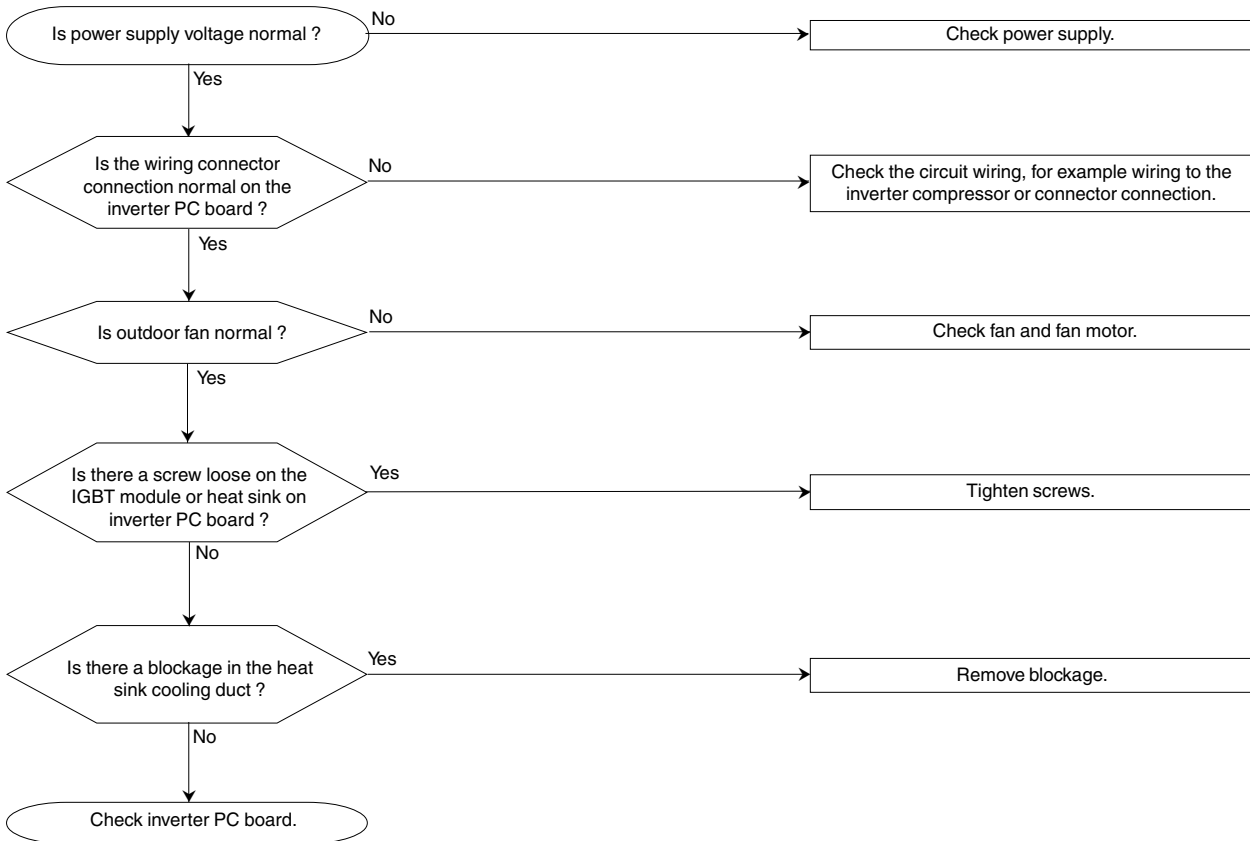
Check code	Operation cause
[d3] TH sensor alarm Refer to outdoor unit	Error of temp. sensor incorporated in IGBT (Inverter Gate Bi-Polar Transistor)

There is an error with the temp. sensor incorporated in IGBT. Check connectors CN07 on IPDU PC board and CN600 on interface PC board. If there is no problem, replace IPDU PC board.

Troubleshooting

Diagnostic procedure for check code

Check code	Operation cause
[dA] Abnormal overheat of heat sink Refer to outdoor unit	1. Power supply failure. 2. Outdoor fan error. 3. Heat sink installation failure. 4. Blockage of heat sink cooling duct. 5. Defective inverter PC board.



Control features

Outdoor unit

No.	Control	Description	Diagram
1	Defrost (Heating mode)	- During heating operation, defrost is controlled by the outdoor unit heat exchanger temperature sensor (TE). - When the cumulative operating time with TE in zone [A] has amounted to 25 minutes, defrost operation starts. Subsequent defrosts occur after a cumulative time of 55 minutes. When heating operation resumes after defrost, the timer is reset. - The maximum defrost period after TE returns to zone [B] is 1 minute. - Heating operation resumes immediately after TE returns to zone [C].	The diagram shows a temperature curve (Temp. in °C) over time. The y-axis has markers at -3, 8, and 12. The curve starts at a high temperature, drops into zone A (below -3°C), then rises through zone B (between -3°C and 8°C) to zone C (above 12°C). Arrows indicate the direction of temperature change.
2	Oil retrieval (Cooling mode)	- During cooling operation, oil retrieval is performed if one or more Indoor units are OFF. - If this condition is satisfied, when cooling operation commences, oil retrieval takes place for 1 minute. - If cooling operation continues, oil retrieval is repeated for 1 minute every 60 minutes. - During oil retrieval the following occurs: - All Multi Controller PMVs are opened. The degree of opening is dependant on the capacity of the Indoor unit . - If the D.O.L. compressor is OFF the inverter compressor speed increases to SD. - When cooling operation ends the timer is reset.	The diagram shows a timeline for 'Cooling mode'. It indicates a 60 min. interval followed by a period of '< 60 min.' before the compressor ('Compr. off') stops. Within the 60 min. interval, there is a 1 min. 'Oil retrieval' period. After the compressor stops, there is another 1 min. 'Oil retrieval' period followed by a 'Timer reset'.
3	Superheat control (Cooling mode)	- Superheat is controlled by the Multi Controller for cooling operation. The level of superheat is determined by the difference between the evaporator outlet temperature for each indoor unit (ThA, B, C, D) and the saturation temperature (ThX). - If the superheat of an indoor unit enters zone [Y], the relevant M/C PMV opens further to compensate for this increase. The degree of additional opening is dependant on the capacity of the indoor unit. - When zone [Z] is reached the PMV opening is held at the level set in zone [Y]. - When the level of superheat returns to zone [X] the control is cancelled.	The diagram shows a superheat curve (Superheat in °C) over time. The y-axis has markers at 9 and 13. The curve starts at a low level (zone X), rises into zone Y (above 13°C), then into zone Z (between 9°C and 13°C), and finally returns to zone X (below 9°C). Arrows indicate the direction of superheat change.

Control features

Outdoor unit/Multi Controller

No.	Control	Description
4	Simultaneous cooling/heating operation	a. Changeover between cooling and heating operations for each indoor unit. The solenoid valves inside the Multi Controller are energised when operation commands are sent from the indoor units. - SVS valve (suction side) opens for a cooling command. - SVD valve (discharge side) opens for a heating command. b. Determination of the outdoor unit operation mode and frequency of the compressor. These are determined by the difference between the total requested cooling command frequencies and the total requested heating command frequencies of all Indoor units. c. Outdoor unit refrigerant control during cooling dominant mode. Refrigerant flow is controlled by PMV2. The degree of opening is controlled by the difference of all requested cooling and heating command frequencies and the compressor operation frequency. This flow control is required to divide cooling exhaust heat appropriately between indoor units in heating mode and the outdoor unit heat exchanger. High pressure control is performed to ensure that the capacity of the indoor units in heating is maintained. This is controlled by the outdoor unit fan control and the changeover from the main heat exchanger to the sub heat exchanger. d. Outdoor unit refrigerant control during heating dominant mode. Superheat control is performed by the outdoor unit expansion valve. Evaporating temperature control is performed by the outdoor unit fan control to ensure that the capacity of the Indoor units in cooling mode is maintained. Changeover from expansion valve to capillary control is performed when there is a minimal difference between the cooling and heating demands and the heat absorption of the outdoor unit heat exchanger is low.

Valve and sensor function and operation

Multi Controller

Multi Controller - valve functions

Symbol	Name of valve	Description of function
PMV (A, B, C, D)	Flow rate control valve	This opens to the extent corresponding to the capacity and performance required of each indoor unit.
SVS (A, B, C, D)	Suction gas valve	Opens when commands from the indoor units require cooling operation.
SVD (A, B, C, D)	Discharge gas valve	Opens when commands from the indoor units require heating operation.
SVSS	Pressure reducing solenoid valve	Opens when the number of units in heating operation is reduced (heating to shutdown or cooling).
		Opens when defrost operation starts.
		Opens during oil retrieval control in cooling operation.
SVDD	Pressure increasing solenoid valve	Opens when the number of units in heating operation is increased (shutdown or cooling to heating).
		Opens while control is performed to prevent liquid in discharge pipe from being blocked.
SVH	Superheat solenoid valve	Opens when unit in cooling operation is present.

Multi Controller - valve operation

Outdoor operation mode	Indoor operation mode	SVD	SVS	PMV
Shutdown	Shutdown, fan	OFF	OFF	Fully closed
During operation	Shutdown, fan	OFF	ON	Fully closed
	Cooling	OFF	ON	Rated opening
	Cooling thermo-control (@ setpoint) OFF	OFF	ON	Fully closed
	Heating	ON	OFF	Rated opening
	Heating thermo-control (@ setpoint) OFF	ON	OFF	Fully closed

Valve and sensor function and operation

Outdoor unit

Outdoor unit - valve functions

Symbol	Name of valve	Description of function
20SF (4-way valve)	4-way reversing valve Cooling operation	OFF: Normal operation. ON: Low load condition when sub heat exchanger is required. Used in combination with SV15.
	4-way reversing valve Heating operation	ON: Normal operation. OFF: Defrost operation.
SV1	High-pressure release solenoid valve	Heating operation. Energised when TE > 11°C
SV2	Hot gas bypass solenoid valve	Opens when Pd < 13 kgf/cm ² G (cooling) (For cooling-dominant mode and sub heat exchanger operation Pd < 17 kgf/cm ² G - 3-pipe only). Opens when outdoor ThO < 0°C
SV4	Solenoid valve for D.O.L. compressor start-up	Opens when the D.O.L. compressor starts for unloading.
SV5	Gas balance	Opens when compressor stops.
SV13	Cooling solenoid valve	Opens for cooling-only operations.
SV14	Solenoid valve for expansion valve.	SV14 opens for heating operation.
SV15	Sub heat exchanger solenoid valve	Opens for sub heat exchanger requirement.
SV16	Solenoid valve for capillary	SV16 opens for low load conditions during heating-dominant operation.
SV17	Solenoid valve for compressor	Open when compressor ON. Closed when compressor OFF.
PMV1	Cooling bypass flow control valve	Opens when ThD1 or ThD2 > 110°C. Opens when ThS > 20°C.
PMV2	Outdoor flow control valve	Opens during cooling-dominant operation. Degree of opening depending on cooling/heating demands.

Valve and sensor function and operation

Valve operation

Outdoor unit - valve operation

Outdoor operation mode		20SF	SV15	SV13	SV14	SV16	SV17	PMV2
Cooling operation	Outdoor main heat exchanger	OFF	OFF	ON	OFF	OFF	ON	-
	Sub heat exchanger	ON	ON	ON	OFF	OFF	ON	-
Heating operation	-	ON	OFF	OFF	ON	OFF	ON	-
Simultaneous cooling-dominant operation	Outdoor main heat exchanger	OFF	OFF	OFF	OFF	OFF	ON	Opening degree control
	Sub heat exchanger	ON	ON	OFF	OFF	OFF	ON	Opening degree control
Simultaneous heating-dominant operation	Expansion valve	ON	OFF	OFF	ON	OFF	ON	-
	Capillary	ON	OFF	OFF	OFF	ON	ON	-

Manual operation of valves

- Note:**
1. Ensure that the system is OFF before valves are manually operated.
 2. Valves will reassume their required position unless electrically isolated.

a. Multi Controller

Display switch position	Valve	Short point	
0	PMV A	TP 1	TP 2
1	PMV B	Selected valve is fully opened for 2 minutes.	Selected valve is fully closed for 2 minutes
2	PMV C		
3	PMV D		

b. Outdoor unit

SW1	SW2	Short point	Operation	Function
0	N/A	TP1	Valves will automatically assume their required position after 2 minutes.	PMV 1 fully open for 2 minutes.
1				PMV 2 fully open for 2 minutes.
0		TP2		PMV 1 fully closed for 2 minutes.
1				PMV 2 fully closed for 2 minutes.
2	0	TP3	Push SW04	Solenoid valves manually energised in sequence when pressed.
			Press SW03	Solenoid valves automatically energised in sequence (1 second intervals)

Note:

During Manual/Automatic operation of outdoor valves, SV17 will not be energised. To confirm operation of SV17 operation is as follows: Energised valve when compressor ON.

Valve and sensor function and operation

Sensor and switch operation

Multi Controller sensor operation

Temperature sensor	According to a difference between the evaporator outlet temperature and (ThA, B, C, D) saturation temperature (ThX): a. Performs superheat control. (cooling operation) b. Determines the degree of opening of PMV 2. (cooling-dominant operation)
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Outdoor unit sensor and switch operation

Discharge pressure sensor	Determines the frequency of the inverter compressor and the ON/OFF status of D.O.L. compressor. Controls the outdoor unit fan operation. Controls the hot gas bypass SV2. Controls the changeover from outdoor unit main heat exchanger to sub heat exchanger.
Suction temperature sensor (ThS)	Controls the degree of opening of PMV 1. Stops operation when ThS > 40°C for 10 minutes (Fault code [A7]).
Discharge temperature sensor (ThD1, ThD2)	Controls the degree of opening of PMV 1. Stops operation when the discharge temperature > 130°C (Fault code [A6]).
Outdoor unit heat exchanger sensor (TE) Heating operation	Controls the defrost operation. Controls the outdoor unit fan operation. Controls the high pressure release valve SV1.
Outdoor temperature sensor (ThO)	Controls the hot gas bypass SV2. Controls the outdoor unit fan operation. Prevents heating operation when ThO > 25°C.
High pressure switch	Stops operation when Pd > 30 kgf/cm ² G.
Low pressure switch	Stops operation when suction pressure < 0.25 kgf/cm ² G: • After 30 seconds for cooling operation. • After 10 minutes for heating operation. The low pressure switch is ignored during defrost operation.

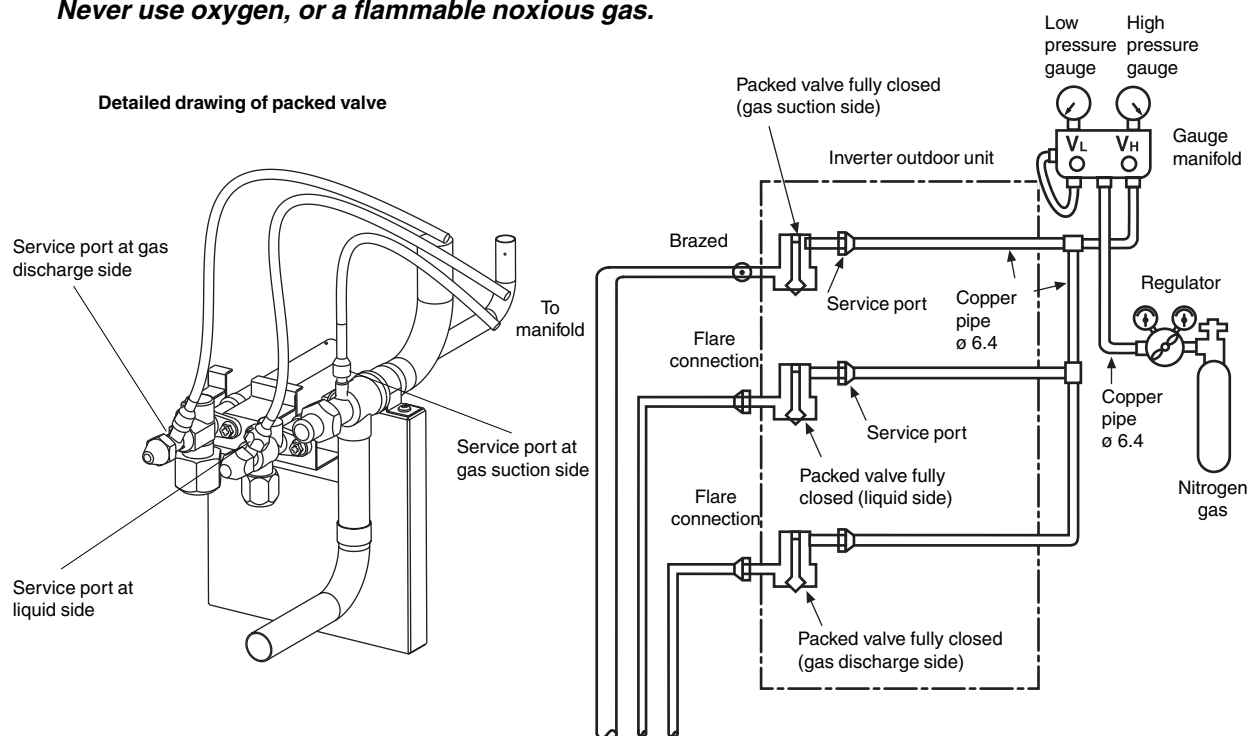
Refrigerant pipe installation

Leak test

Airtight test

- ⚠ Carry out an airtight test after the refrigerant piping is complete. For an airtight test, connect a nitrogen gas bottle as shown, and apply pressure (use oxygen-free nitrogen, OFN).
- ⚠ The pressure test must be completed before supplying power to ensure that the Multi Controller PMVs (pulse modulating valves) are open.
- ⚠ The test must be completed with the indoor units, Multi Controllers and outdoor unit connected.
 - Be sure to carry out the test from the service ports of the packed valves at the discharge gas, liquid and suction gas side.
 - Keep all of the valves at discharge gas, liquid and suction gas sides fully closed. Nitrogen may enter the cycle of the outdoor unit. Therefore, retighten the valve rod before applying pressure. (For all valves.)
 - For each refrigerant line, apply pressure gradually at the discharge gas, liquid and suction gas sides.

Never use oxygen, or a flammable noxious gas.



To detect a large leakage

Step 1: 0.3 MPa (3 kg/cm² G) Apply pressure for 3 minutes or more

Step 2: 1.5 MPa (15 kg/cm² G) Apply pressure for 3 minutes or more

To detect a fine leakage

Step 3: 3.0 MPa (30 kg/cm² G) Apply pressure for 24 hours

- Check for a reduction in pressure.

If there is no reduction in pressure this is acceptable.

If there is a reduction in pressure check for a leakage.

(Note: If there is a difference of ambient temp. between when the pressure was applied and 24 hours later, then pressure could change by approx. 0.01 MPa (0.1 kg/cm² G) - so correct the pressure change.

Refrigerant pipe installation

Vacuumping

Leak position check

If a pressure drop is detected, check for leakage at connecting points. Locate the leakage by listening, feeling, using foaming agent, etc. - then rebraze or retighten.

Air purge

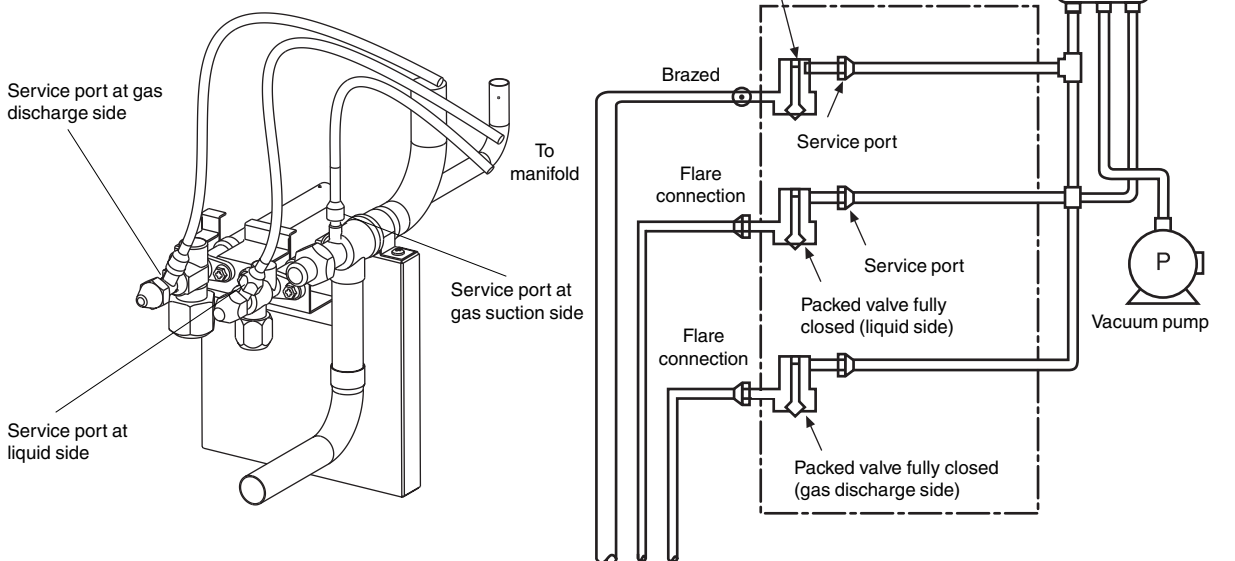


The air purge must be completed before supplying power to ensure the Multi Controller's PMVs are open.

Using a vacuum pump, complete an air purge. Never use refrigerant gas.

- After the airtight test, discharge the nitrogen gas.
- Connect a gauge manifold to the service port at discharge gas, liquid and suction gas sides, and connect a vacuum pump as shown.
- Be sure to vacuum at discharge gas, liquid and suction gas sides.

Detailed drawing of packed valve



- Use a vacuum pump with high vacuum carry-over degree (-0.013×10^5 Pa; 0.750 mm Hg or less) and large displacement (40 l/min. or more).
- Ensure to create a vacuum at -0.013×10^5 Pa (0.75 mm Hg) at the discharge gas, liquid and suction gas.
- After the procedure has been completed, replace the vacuum pump with a refrigerant bottle and add the refrigerant if required.

Refrigerant pipe installation

Charging the system with additional refrigerant

Adding the refrigerant

After the airtight test, replace the vacuum pump with a refrigerant bottle to charge the system.

Calculating the additional refrigerant required

The Super Multi contains sufficient refrigerant to operate an installation having 5 metres of pipework (when delivered). The refrigerant amount at shipment does not include the refrigerant needed for the piping - so first calculate this amount, and then add it.

The total refrigerant charge must be calculated by the weight and within a tolerance of ± 50 g.

Refrigerant charge amount shipped from the factory:

Outdoor unit model name	MAR-F105HTM8-PE
Charging amount (kg)	19.0
Maximum gas charge (kg)	36.3

The amount of additional refrigerant is calculated from the actual length of the liquid pipe.

To calculate the additional refrigerant volume, refer to the diagram and follow the steps below:

- (i) The main pipe length is taken as the addition of pipes X, Y and Z.
- (ii) The sub pipe length is taken as the addition of the two longest of the four (if 4 Multi Controllers).
- (iii) The branch pipe lengths must be individually calculated using the 8 longest pipes.
- (iv) Do not attempt to add gas above the maximum shown in the table above.
- (v) For systems with one Multi Controller ignore the sub-pipe section within the additional gas charge calculation.
- (vi) When using three Multi Controllers, it is important that a reducer is used on the pipework for the third Multi Controller. Pipework before the reducer is classed as main piping and after as sub piping.

Charging the system

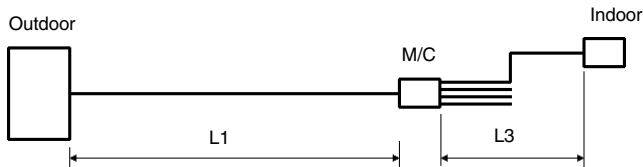
- Keeping the outdoor unit valve closed, charge the refrigerant from the service port on the liquid side.
- If the specified amount of refrigerant cannot be charged - fully open all the outdoor unit's valves, then perform the cooling operation with the valve at the gas side slightly closed.
- If leaks cause a shortage of refrigerant - recover the refrigerant from the system, and recharge with new refrigerant to the total refrigerant charge.
- Prolonged operation with either an overcharge or deficiency in refrigerant will lead to a loss of performance, increased running costs and will cause damage to the machine. Warranty will be void.

Additional refrigerant

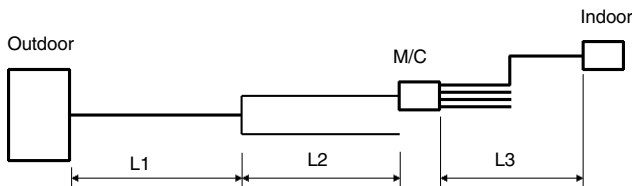
Piping

Additional refrigerant

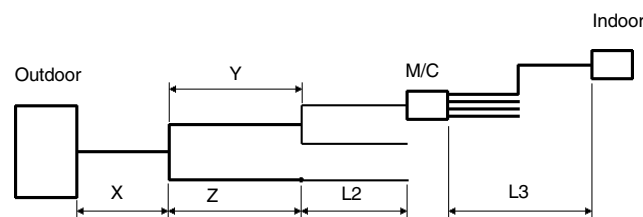
One Multi Controller



Two Multi Controllers



Three Multi Controllers



Four Multi Controllers

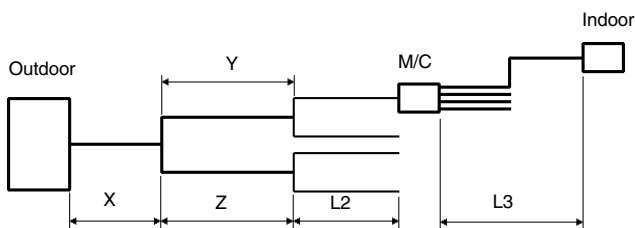


Table 1 - Branch pipes

RAV-10* : 0.030 kg/m

RAV-13* : 0.030 kg/m

RAV-16* : 0.030 kg/m

RAV-26* : 0.045 kg/m

RAV-36* : 0.045 kg/m

RAV-46* : 0.045 kg/m

Example:

RAV-464CH-PE ⑦ RAV-46*

L1 = Main pipe (X + Y + Z)

L2 = Sub pipe

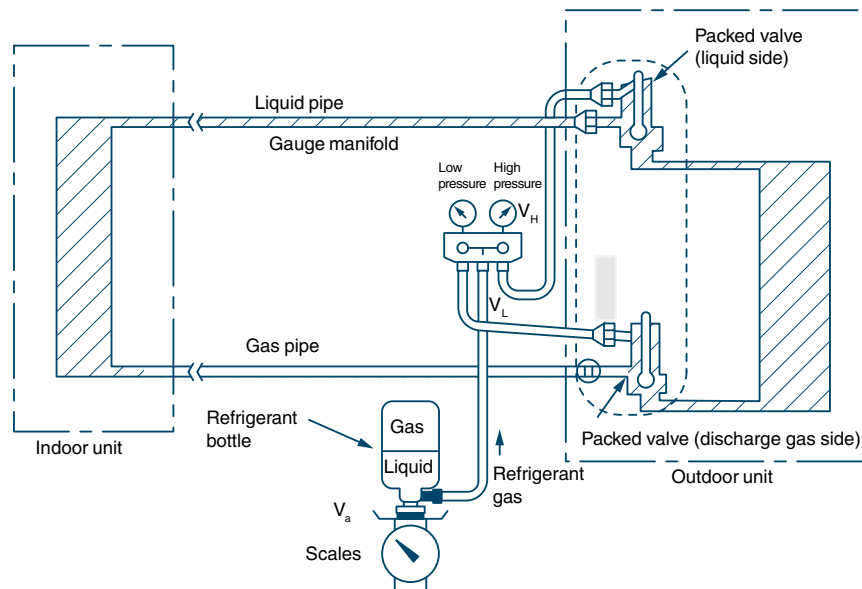
L3 = Branch pipe

Pipe		Additional gas/metre	Additional gas
MAR-F105 - main pipe L1 (X+Y+Z) (minus 2 m)	x	0.19 kg/m	=
1st longest sub pipe L2 (minus 1 m)	x	0.125 kg/m	=
2nd longest sub pipe L2 (minus 1 m)	x	0.125 kg/m	=
1st longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
2nd longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
3rd longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
4th longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
5th longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
6th longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
7th longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
8th longest branch pipe L3 (minus 2 m)	x	Refer to Table 1	=
Total additional gas charge		=	kg

Refrigerant pipe installation

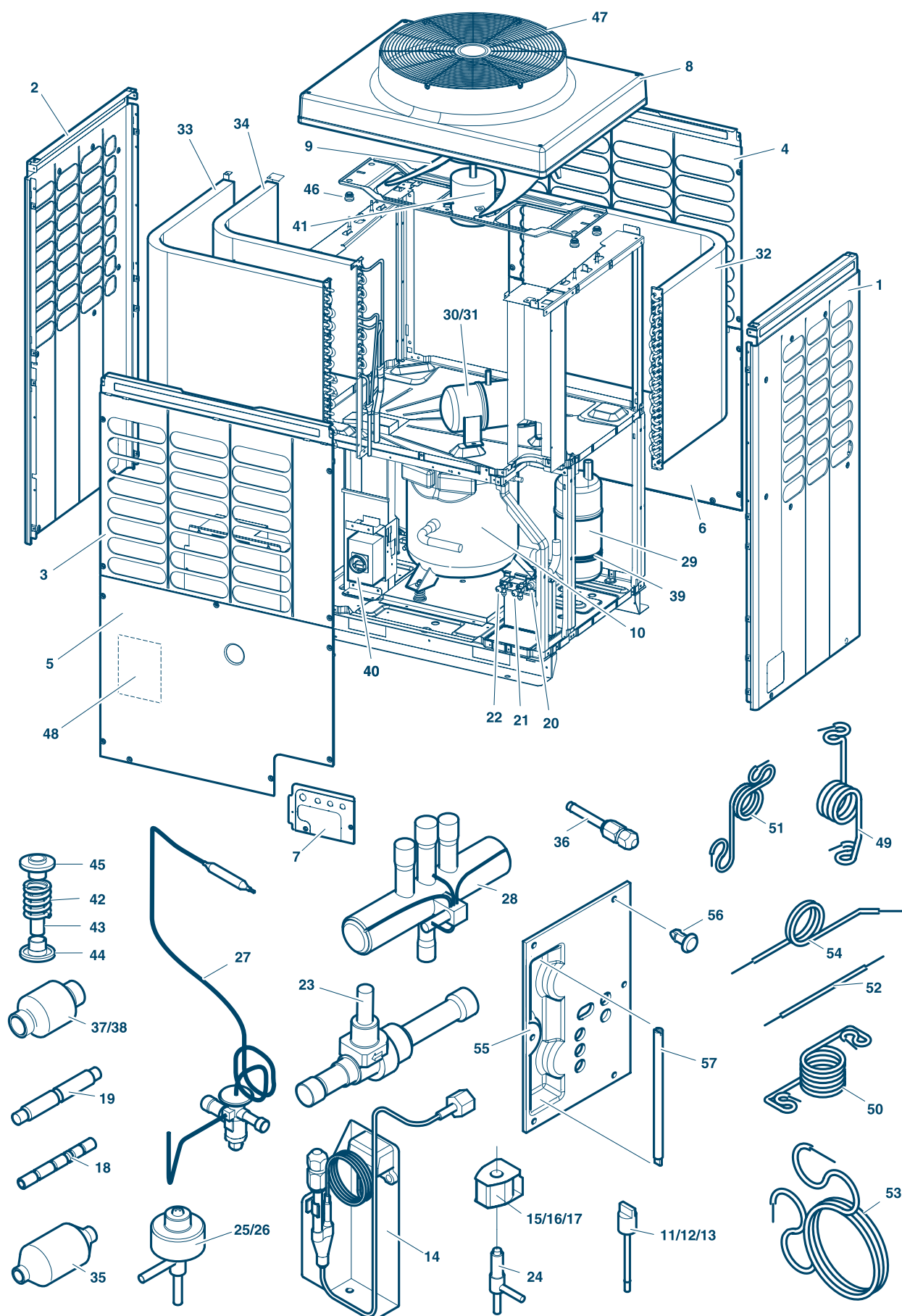
Additional refrigerant charging method

- (1) Loosely connect the refrigerant cylinder hose to the gauge manifold, then open the source valve V_H on the cylinder, purge the air in the hose, and then tighten the hose.
- (2) As shown in the diagram below, turn the refrigerant cylinder upside down, open the valve V_H on the gauge manifold, and then charge the liquid side pipe with refrigerant in the liquid state. (Note that with some types of refrigerant cylinders, the liquid refrigerant will be output through siphoning action with the cylinder in the normal upright position.) If the proper charging amount cannot be reached, close the valve V_H , turn the refrigerant cylinder upright, open the liquid-side and balance packed valves completely, and open the gas-side packed valve only half way. Begin the cooling operation, open valve V_L , and then charge the gas-side pipe with refrigerant in the liquid state.
- (3) While watching the scales display, quickly close the valve V_L completely when the system has been charged with the proper amount of additional refrigerant. Then close the source valve V_a on the cylinder, and open the gas-side packed valve completely.
- (4) Record the amount of additional refrigerant that was added to the system on the nameplate inside the front panel (lower) of the outdoor unit.



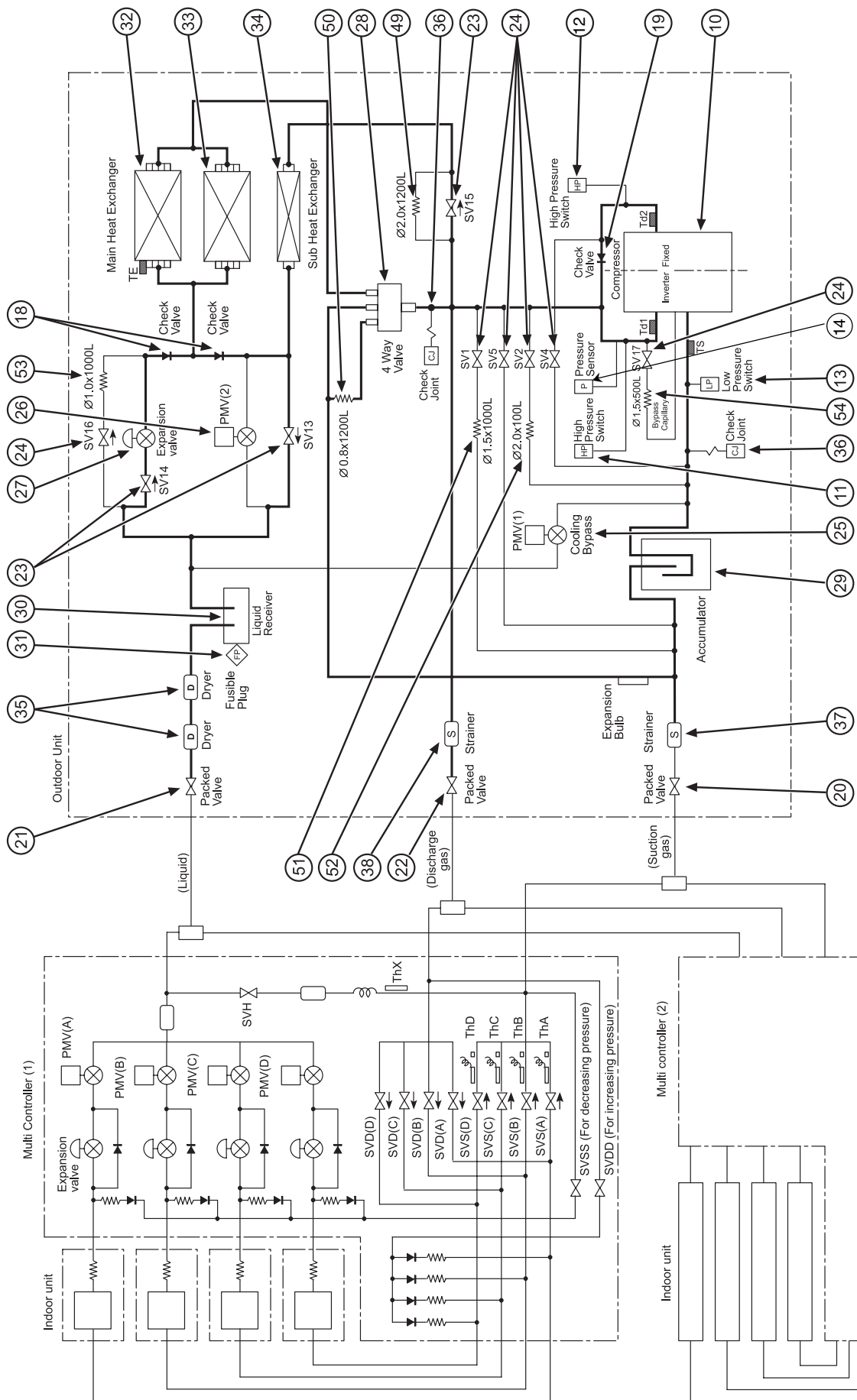
Exploded view and service parts

Outdoor unit



Exploded view and service parts

Outdoor unit



Exploded view and service parts

Outdoor unit

Ref.	Service part No.	Description
01	43A00020	Plate side (Right)
02	43A00021	Plate side (Left)
03	43A00022	Air-in cabinet (Front)
04	43A00023	Air-in cabinet (Back)
05	43A00024	Cabinet-down (Front)
06	43A00025	Cabinet-down (Back)
07	43A00026	Service panel
08	43A00027	Air-out cabinet
09	43A20004	Propeller fan - PY631
10	43A41504	Compressor - 10 HP (INV)
11	43A49011	Switch, high pressure
12	43A49006	Switch, high pressure
13	43151230	Switch, low pressure
14	43A49012	Pressure sensor service kit
15	43A46038	Solenoid coil assembly
16	43A46039	Solenoid coil assembly
17	43A6040	Solenoid coil assembly
18	43146252	Check valve
19	43A46032	Check valve - Dia. 15 mm
20	43146351	Service valve (1-1/8")
21	43146503	Packed valve (5/8")
22	43146504	Packed valve (3/4")
23	43146398	Valve body
24	43046151	2-way valve
25	43A46041	Pulse modulating valve (PMV 1)
26	43A46042	Pulse modulating valve (PMV 2)
27	43A46010	Expansion valve
28	43A46007	4-way valve

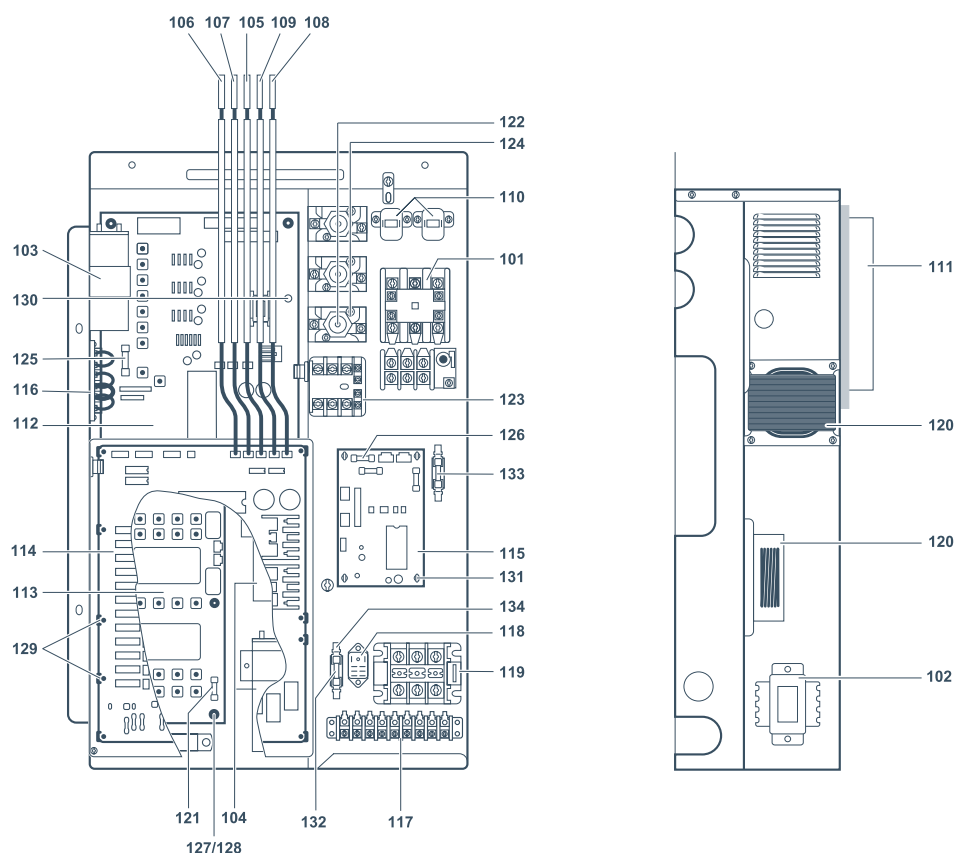
Ref.	Service part No.	Description
29	43A48005	Accumulator
30	43A48006	Liquid tank
31	43A60013	Fusible plug
32	43A43018	Condenser assembly - right
33	43A43019	Condenser assembly - left
34	43A44089	Sub condenser
35	43A45003	Dryer-XH-10C
36	43A46036	Check joint
37	43147529	Strainer
38	43147537	Strainer
39	43A57002	Accumulator heater
40	43A51005	Isolator
41	43A21022	Fan motor
42	43107200	Spring
43	43195198	Spacer
44	43195186	Rubber cushion - under
45	43195185	Rubber cushion - upper
46	43A95001	Rubber support cushion
47	43A19011	Fan guard
48	43A16007	Wiring diagram
49	43A55008	Capi-form-SV15
50	43A55009	Capi-form-4WV
51	43A55010	Capi-form-SV1
52	43A55011	Capi-SV2 (straight)
53	43A55012	Capi-form-SV16
54	43A47034	Capi-form-compressor
55	43A86001	EP window
56	43A89002	Push fastener
57	43A89003	Adjuster pen

Note:

Part 35 - Replace both dryers at the same time.

Exploded view and service parts

Outdoor unit - inverter assembly



Ref.	Service part No.	Description
101	43A52005	Magnetic contactor - FC - 3
102	43A58009	Transformer - TT01
103	43A55004	MF - capacitor - 8 μ F/450 V AC
104	43A55005	Capacitor - electrolytic - 2200 μ F/400 V
105	43A50030	Temperature sensor TS - blue
106	43A50031	Temperature sensor THO - green
107	43A50032	Temperature sensor TE - black
108	43A50015	Temperature sensor TD1 - white
109	43A50016	Temperature sensor TD2 - red
110	43A50033	Thermistor (PTC)
111	43A69016	Heatsink
112	43A69035	PC board (IPDU) MCC-1342
113	43A69018	PC board (noise filter) MCC-1366
114	43A69036	PC board (interface) MCC-1223
115	43A69037	PC board (communication) MCC-1387
116	43A60034	Terminal-6P
117	43A60035	Terminal-9P

Ref.	Service part No.	Description
118	43A34001	Relay
119	43A60016	Terminal block - 3P/60 A
120	43A55006	Reactor - CH - 25 - 2FK
121	43A60020	Fuse - 6 A
122	43A60021	Fuse - 20 A
123	43A52006	Magnetic contactor - FMCa - 1S - 02
124	43A60022	Fuse holder - 30 A
125	43A60023	Fuse - 3.15 A
126	43A60024	Fuse - 6.3 A
127	43A63002	Bushing
128	43A63003	Collar
129	43A63004	Supporter (PCB)
130	43A63005	Supporter assembly
131	43A63006	Supporter assembly
132	43A60036	Fuse - 20 A
133	43A60037	Fuse - 10 A
134	43A60038	Fuse-block

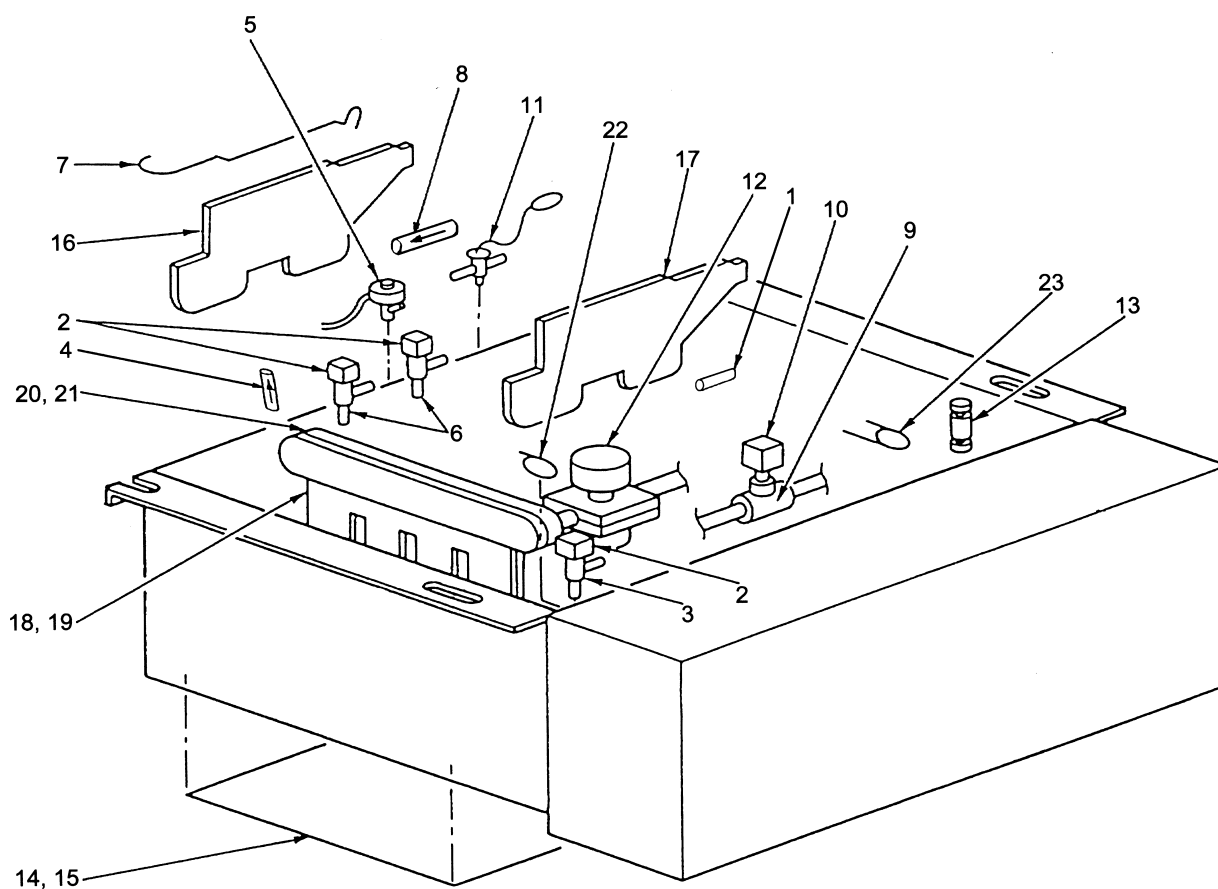
Note:

Part 104 - Replace both capacitors at the same time.

Exploded view and service parts

3-pipe Multi Controller

3-pipe Multi Controller (RBM-Y1034F-PE, RBM-Y1044F-PE)



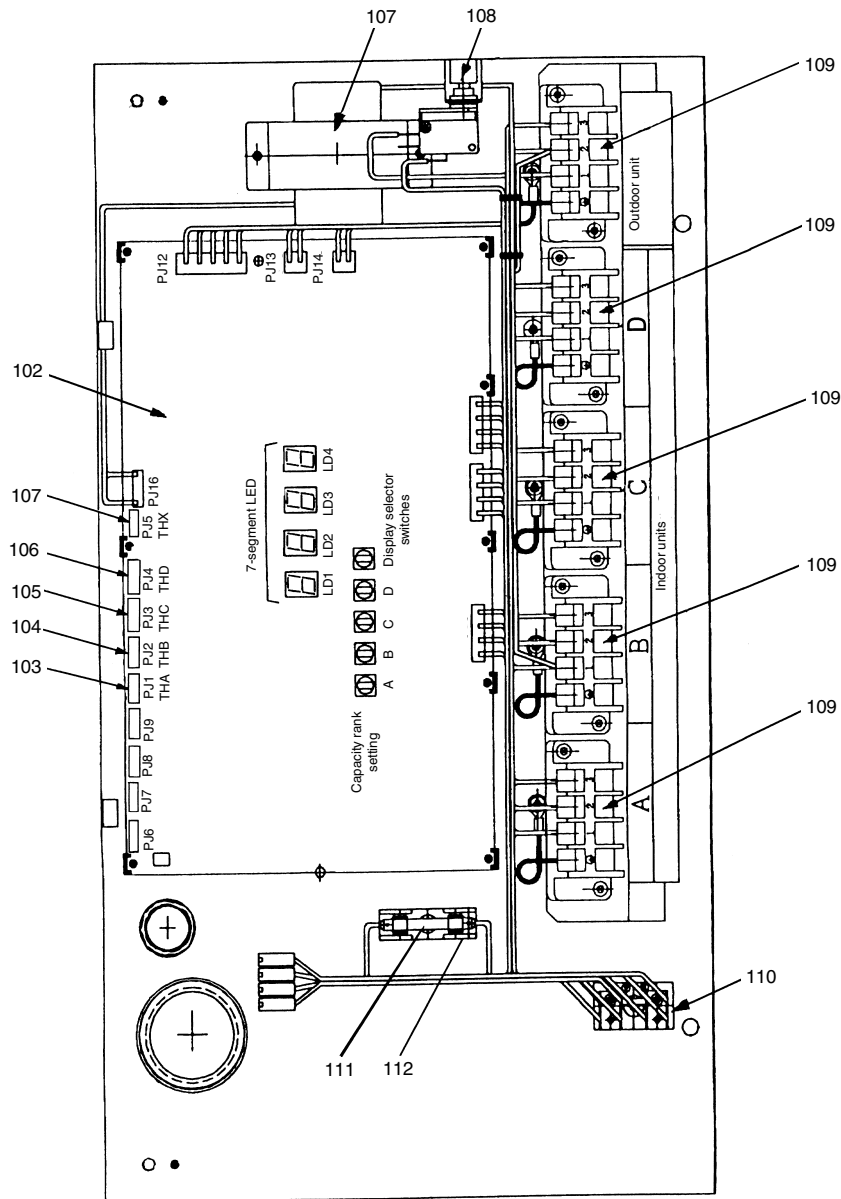
Ref.	Service part No.	Description
1	43107215	Holder, sensor
2	43146443	Solenoid coil
3	43046151	2-way valve
4	43046156	Check valve
5	43046269	Pulse modulating valve
6	43046270	2-way valve
7	43847010	Capillary tube 2.0 diameter
8	43146283	Check valve
9	43146398	Valve body
10	43146506	Solenoid coil
11	43A46011	Expansion valve
12	43146505	2-way valve
13	43151231	Float switch
14	43157252	Panel heater (RBM-Y1044F-PE)

Ref.	Service part No.	Description
15	43157253	Panel heater (RBM-Y1034F-PE)
16	43189026	Insulator, liquid A
17	43189027	Insulator, liquid B
18	43189028	Insulator, header A (RBM-Y1044F-PE)
19	43189029	Insulator, header A (RBM-Y1034F-PE)
20	43189031	Insulator, header B (RBM-Y1044F-PE)
21	43189032	Insulator, header B (RBM-Y1034F-PE)
22	43A47011	Capillary tube 0.8 diameter
23	43A47012	Capillary tube 1.0 diameter

Exploded view and service parts

3-pipe Multi Controller

3-pipe Multi Controller electrical parts assembly (RBM-Y1034F-PE, RBM-Y1044F-PE)



Note: Terminal D is present on RBM-Y1044F-PE unit only.

Ref.	Service part No.	Description
101	43158139	Transformer (FT69)
102	43169627	PC board assembly (MCC-1222)
103	43150162	Sensor (ThA)
104	43150163	Sensor (ThB)
105	43150164	Sensor (ThC)
106	43150165	Sensor (ThD) (RBM-Y1044F-PE)

Ref.	Service part No.	Description
107	43150166	Sensor (ThX)
108	43151242	Micro-switch
109	43A60001	Terminal block 4P
110	43060776	Terminal block 2P
111	43A60008	Fuse T1A
112	43A60007	Fuse holder

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